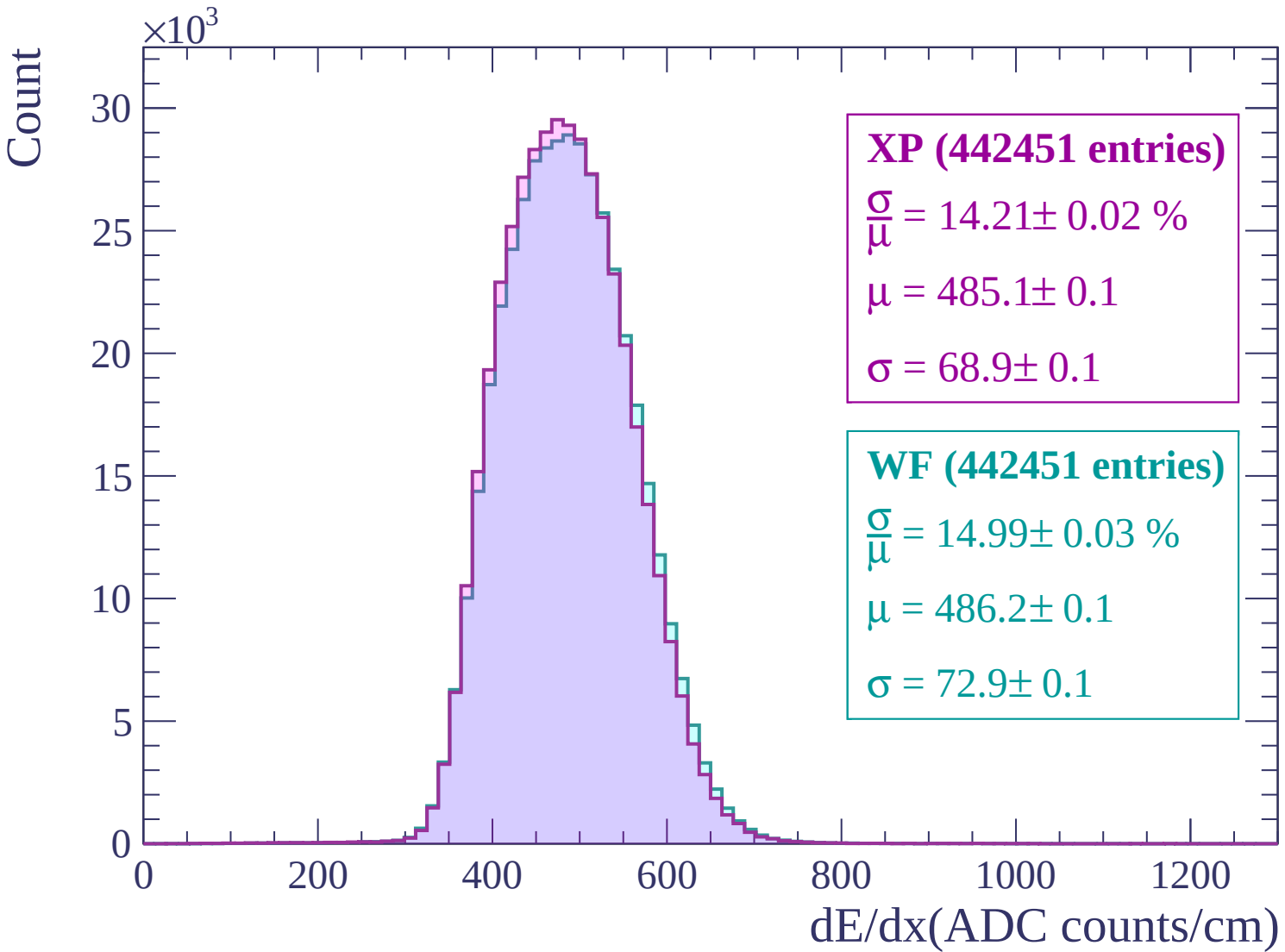
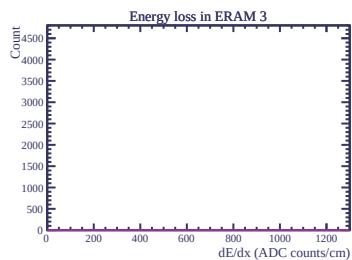
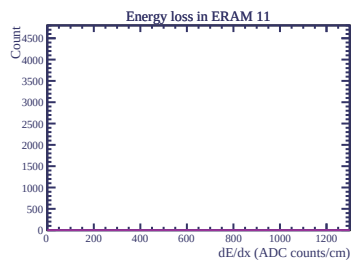
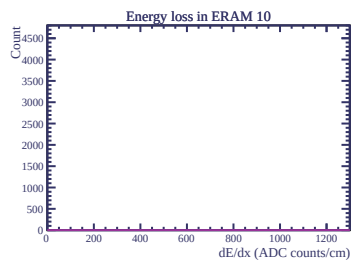
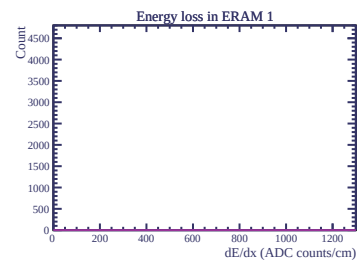
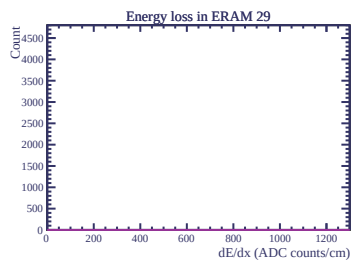
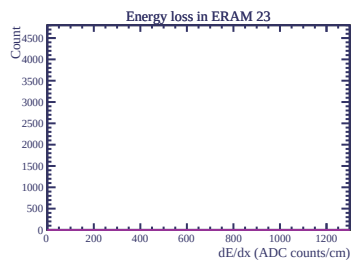
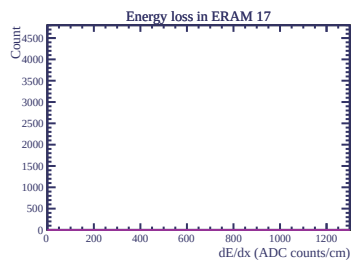
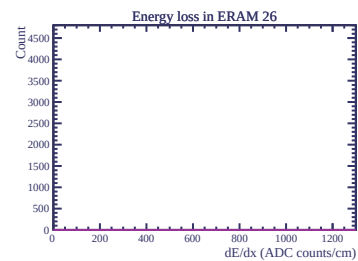
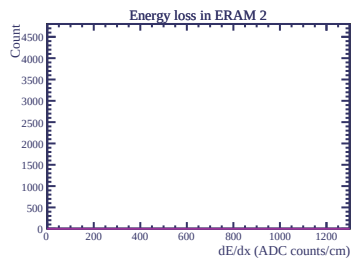
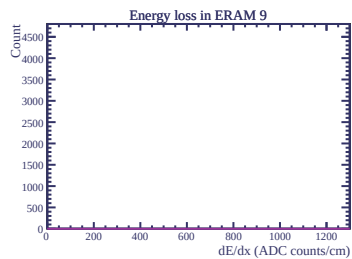
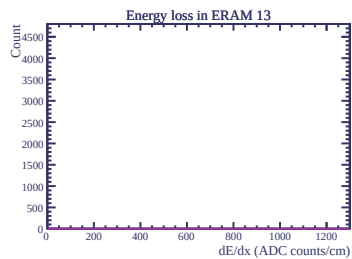
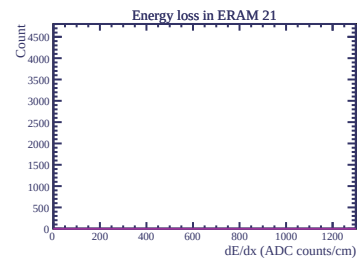
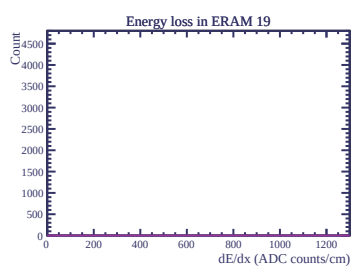
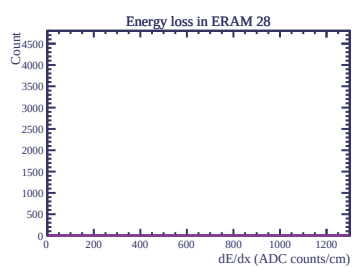
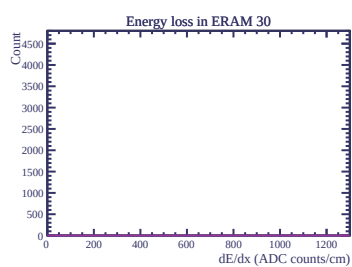
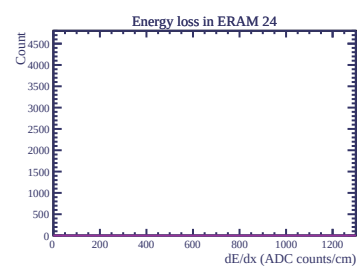
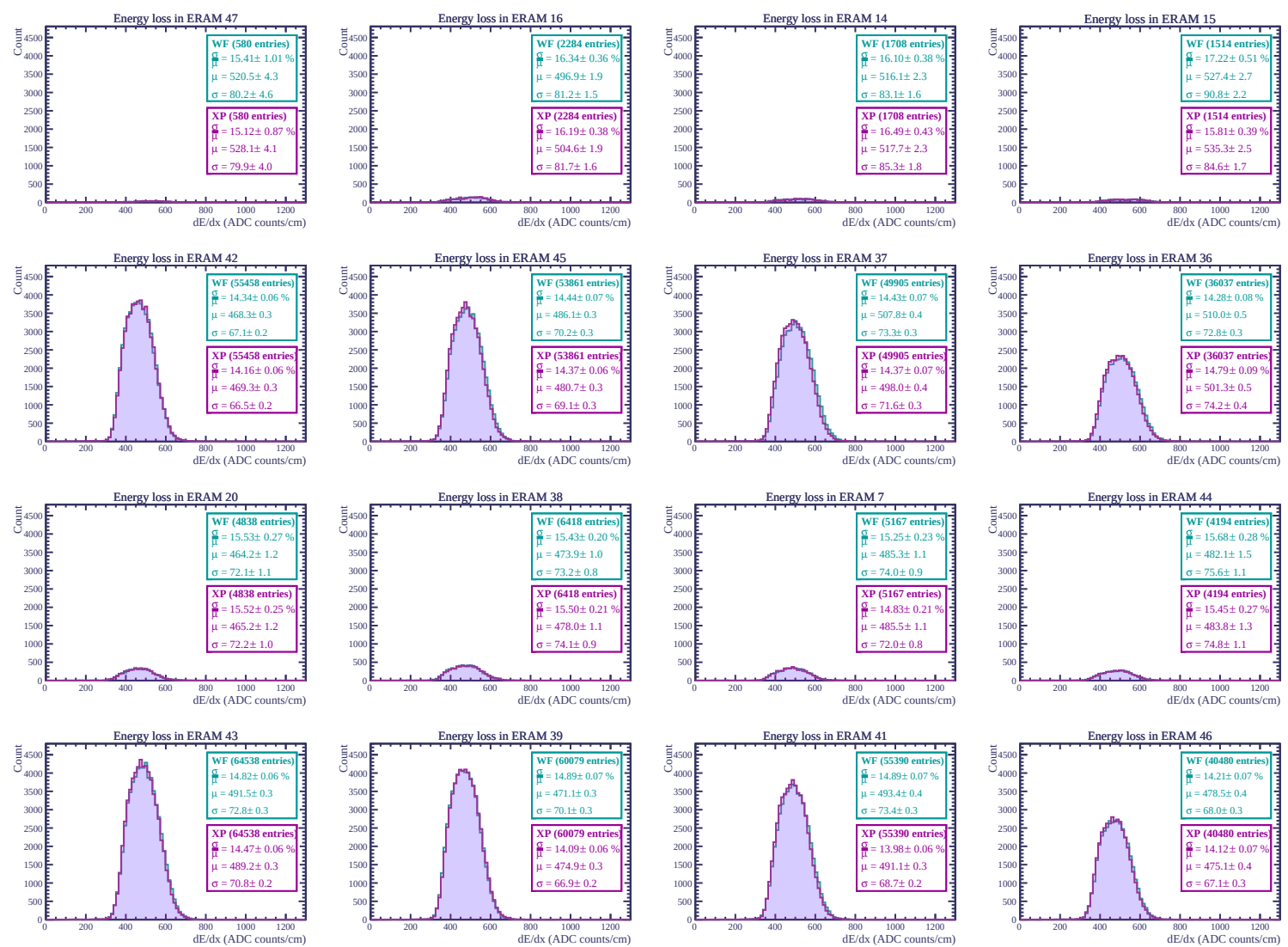
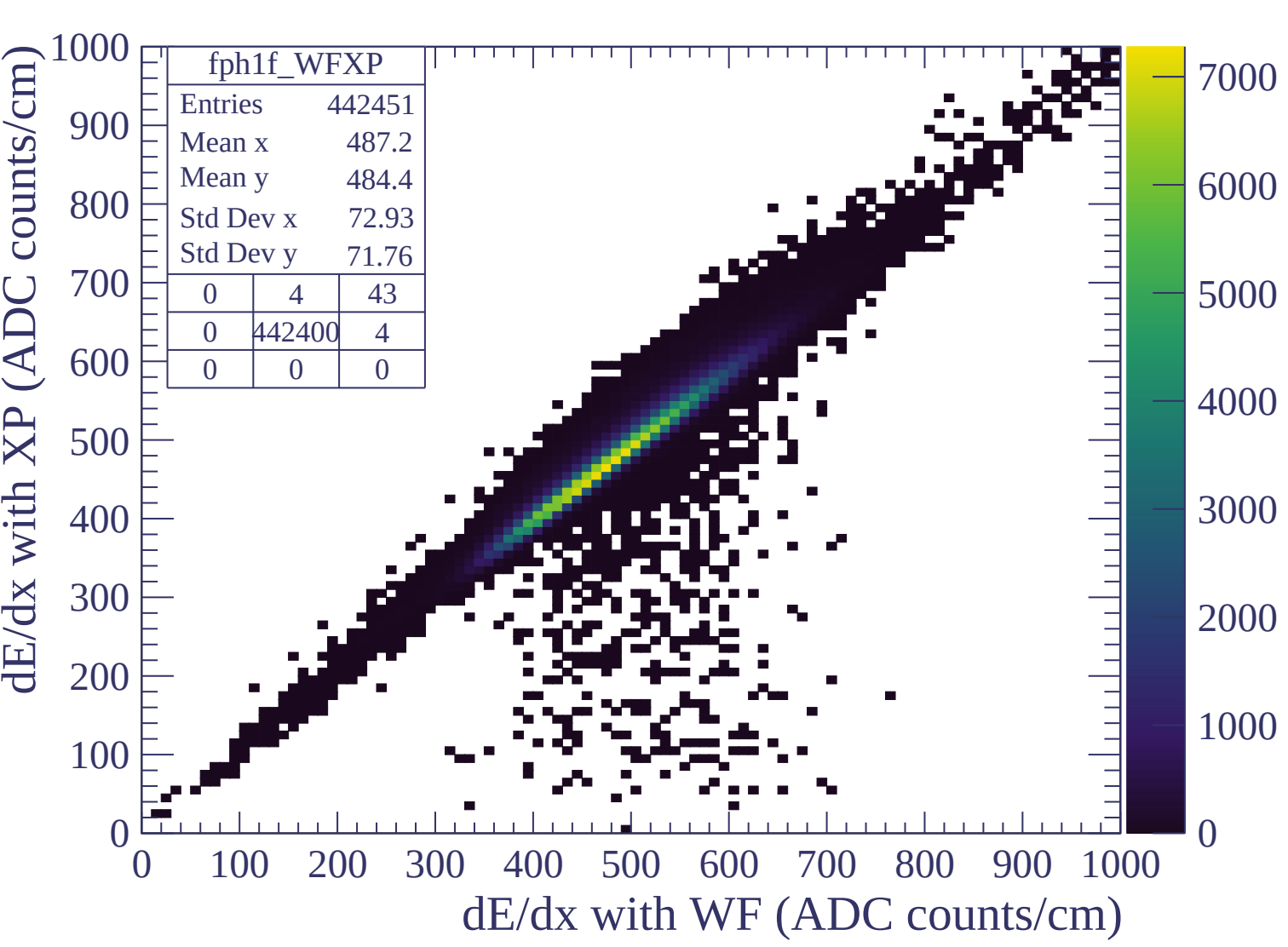
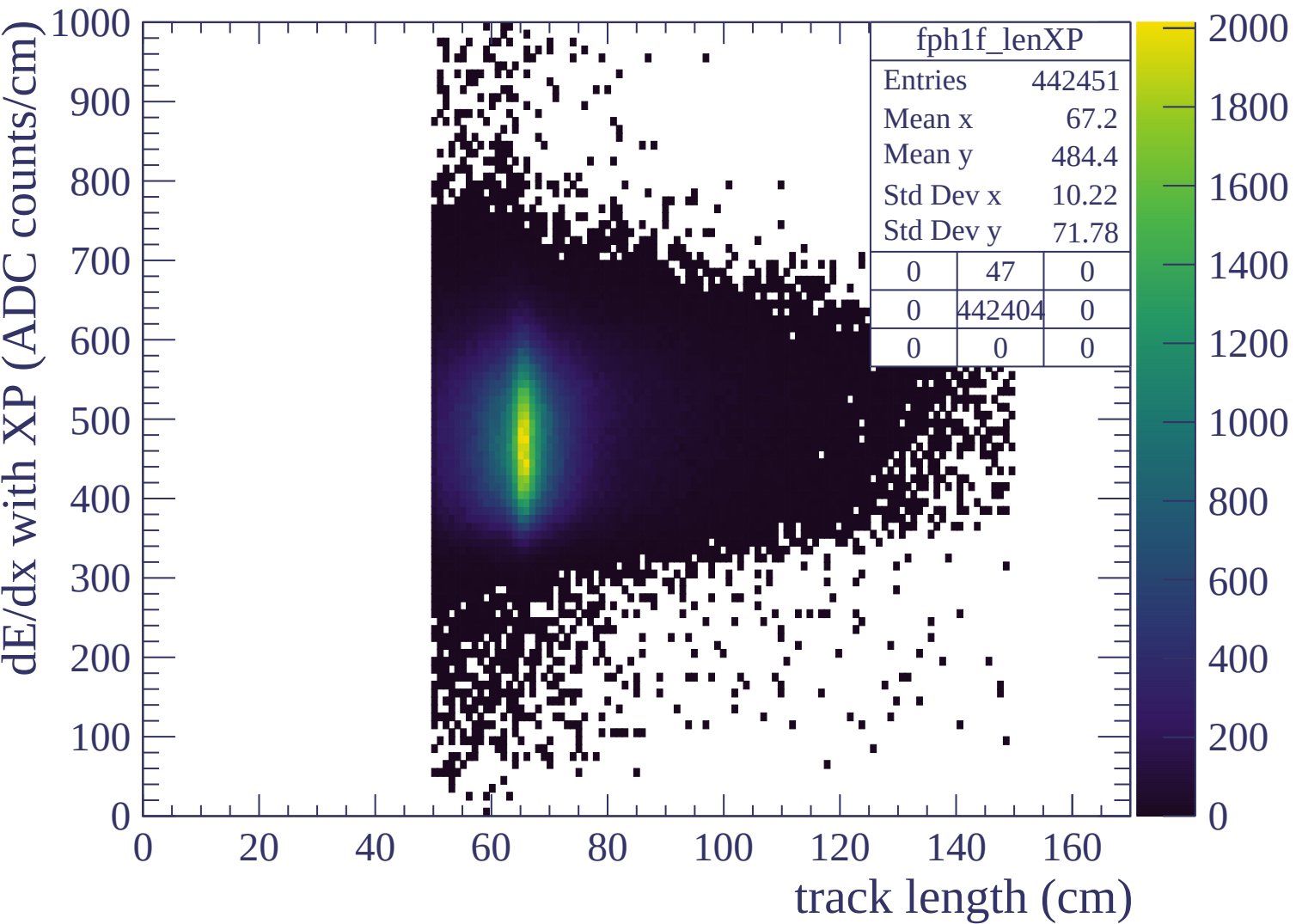


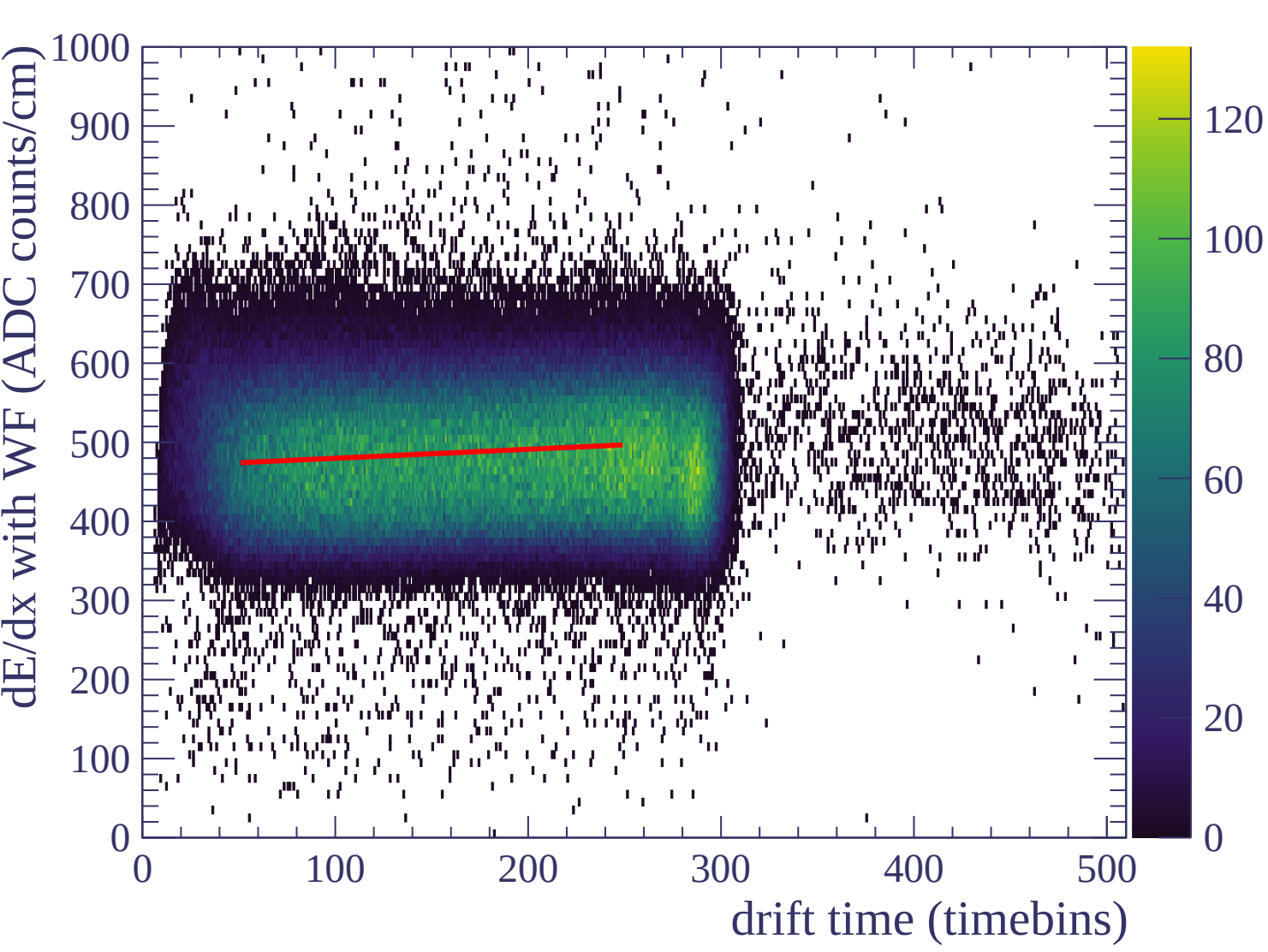


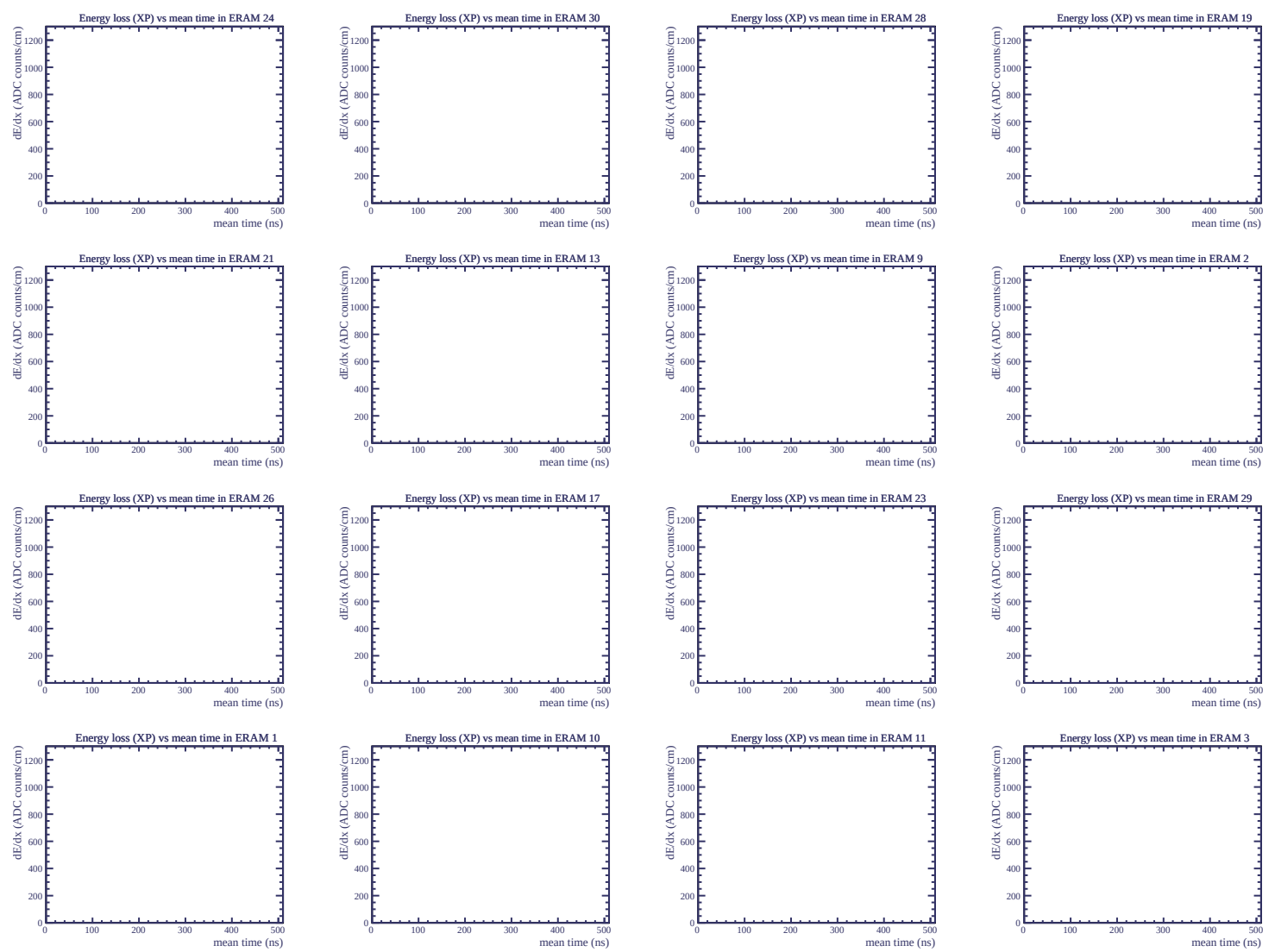


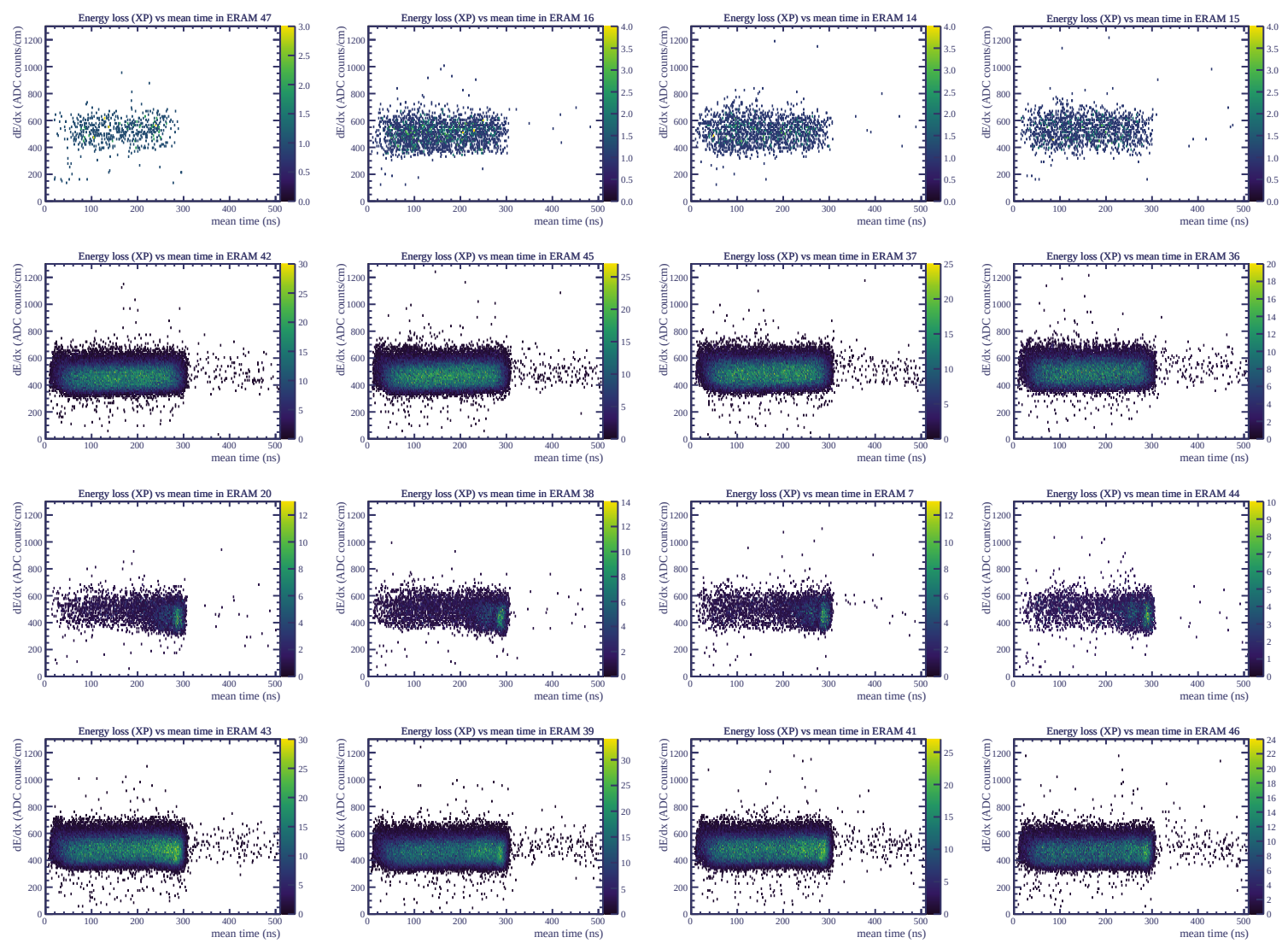




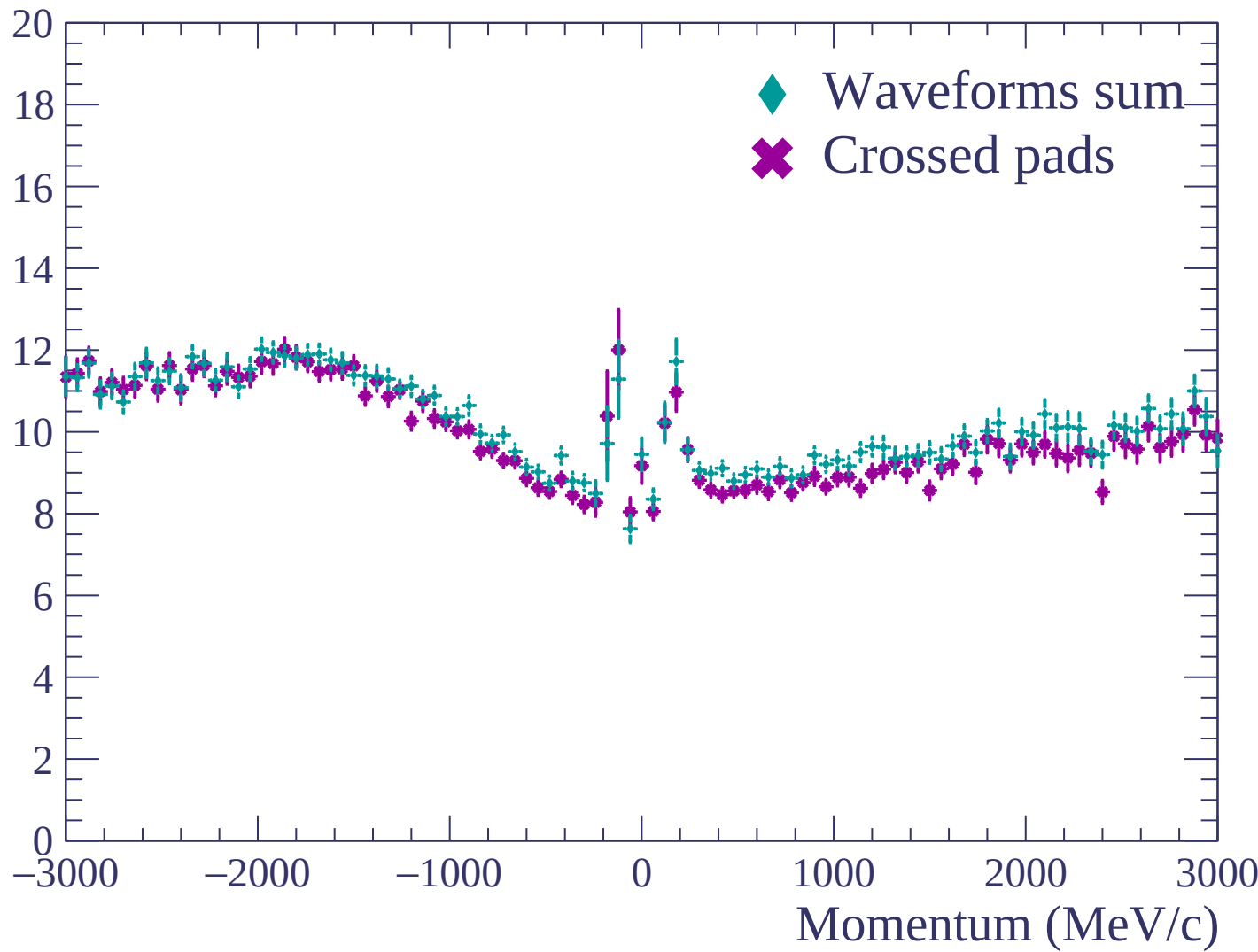


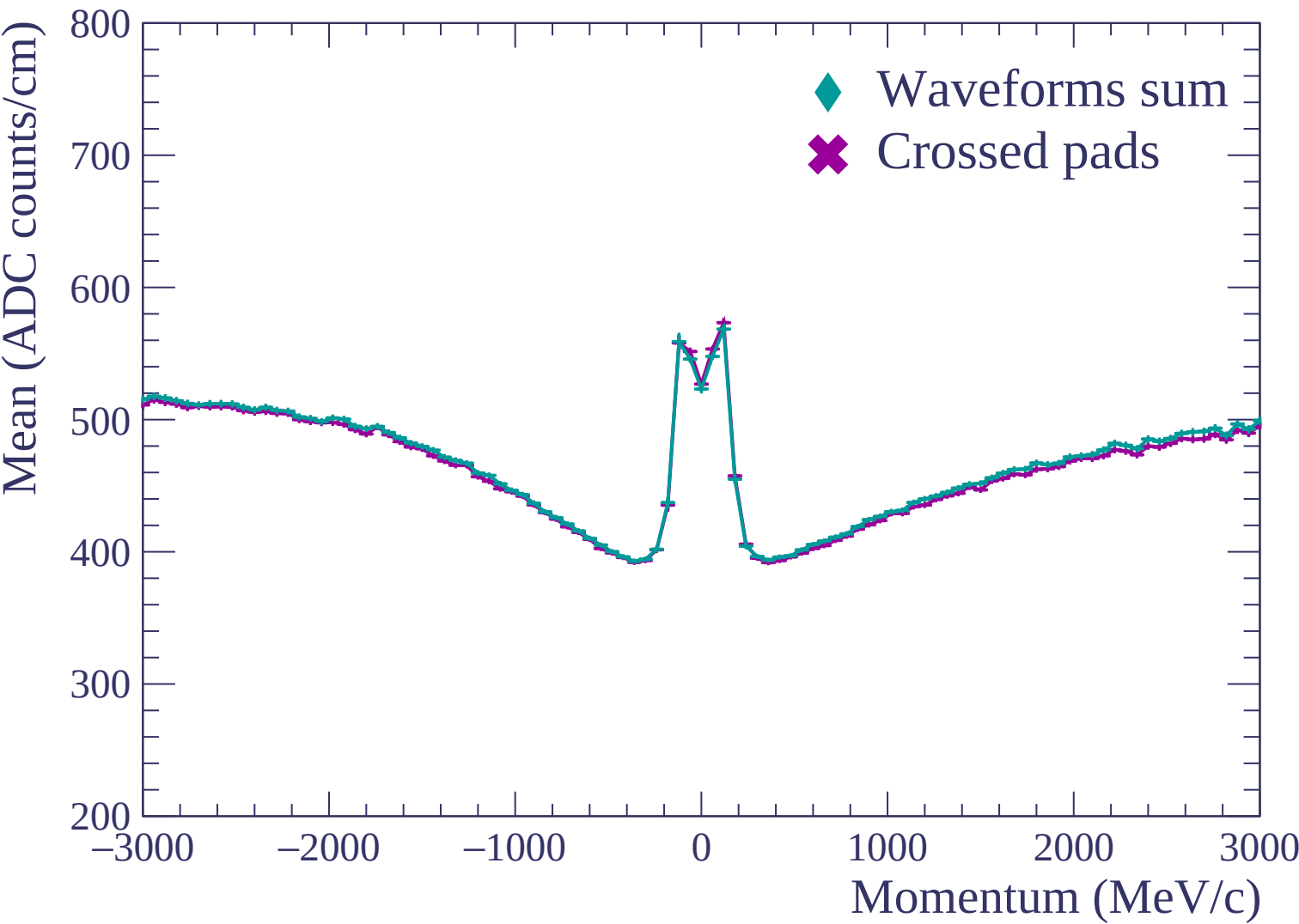


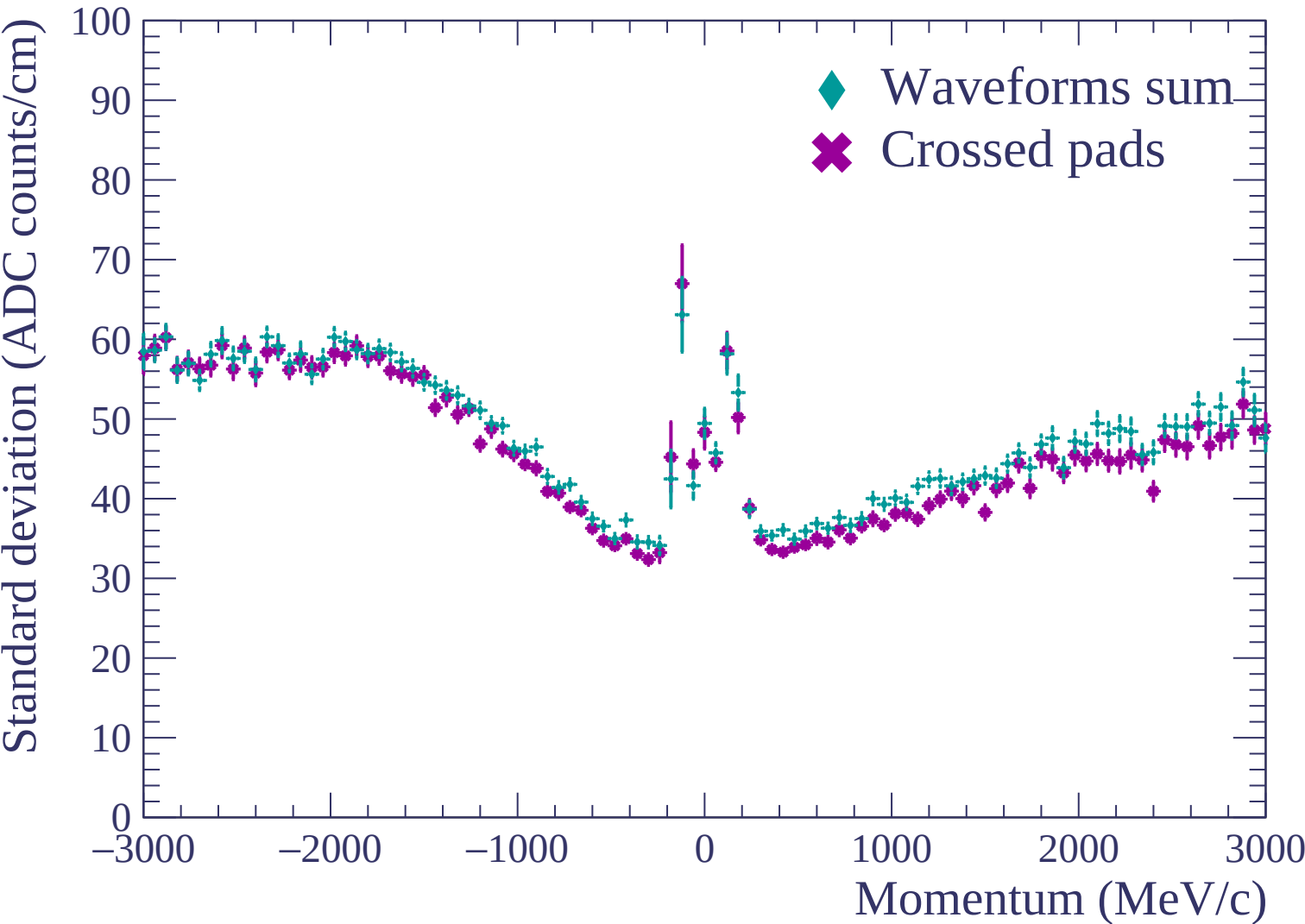


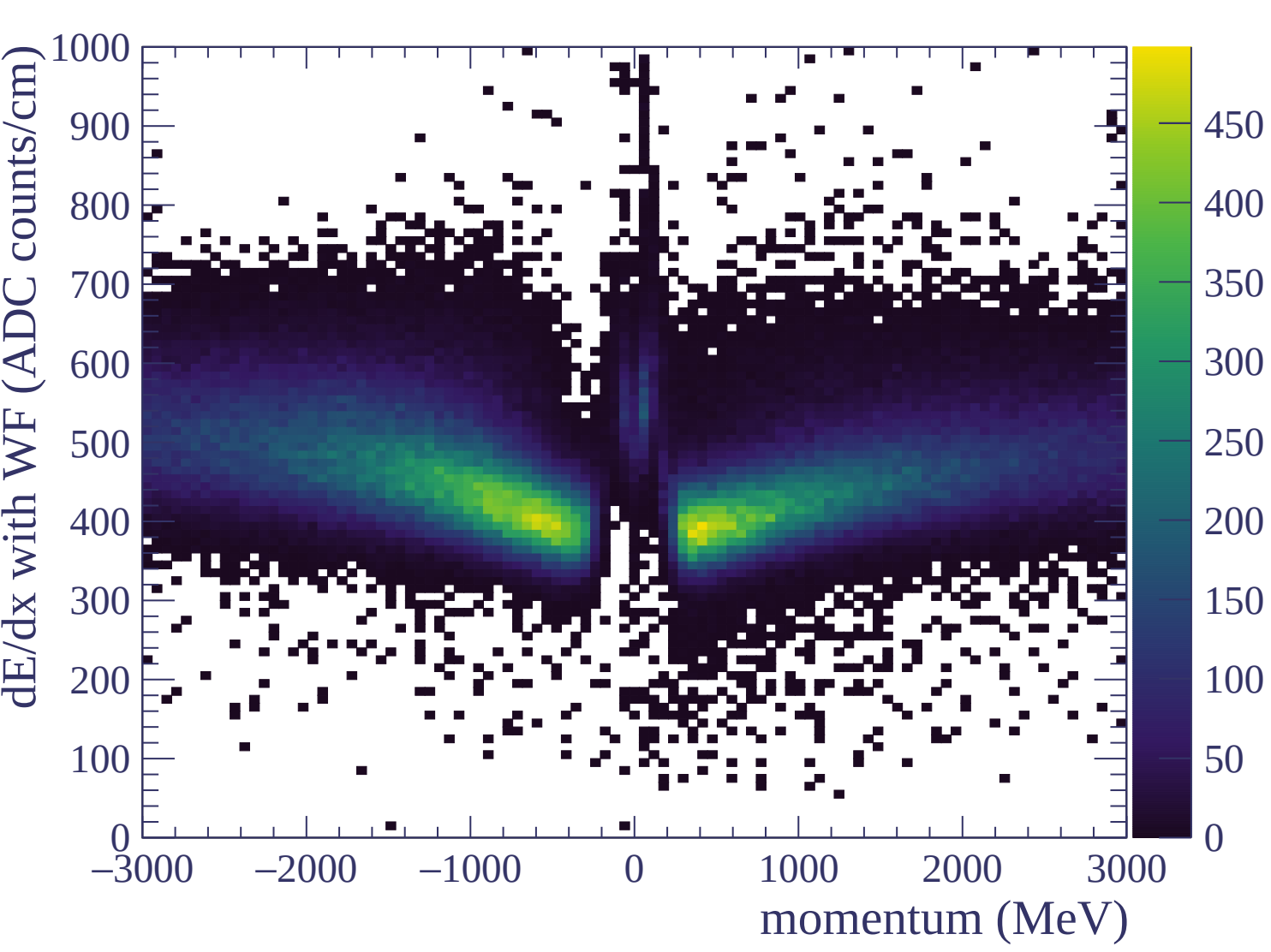


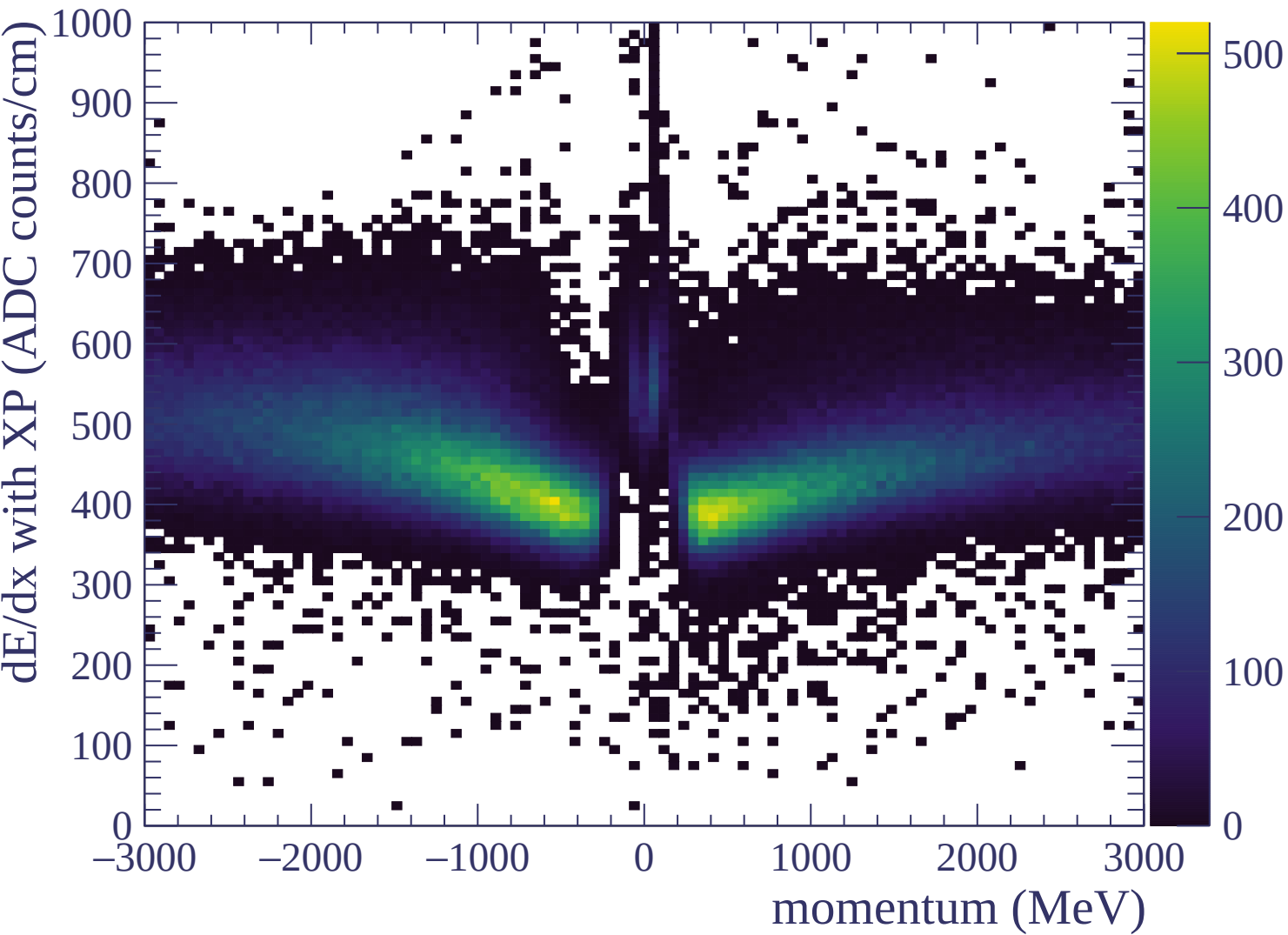
Resolution (%)

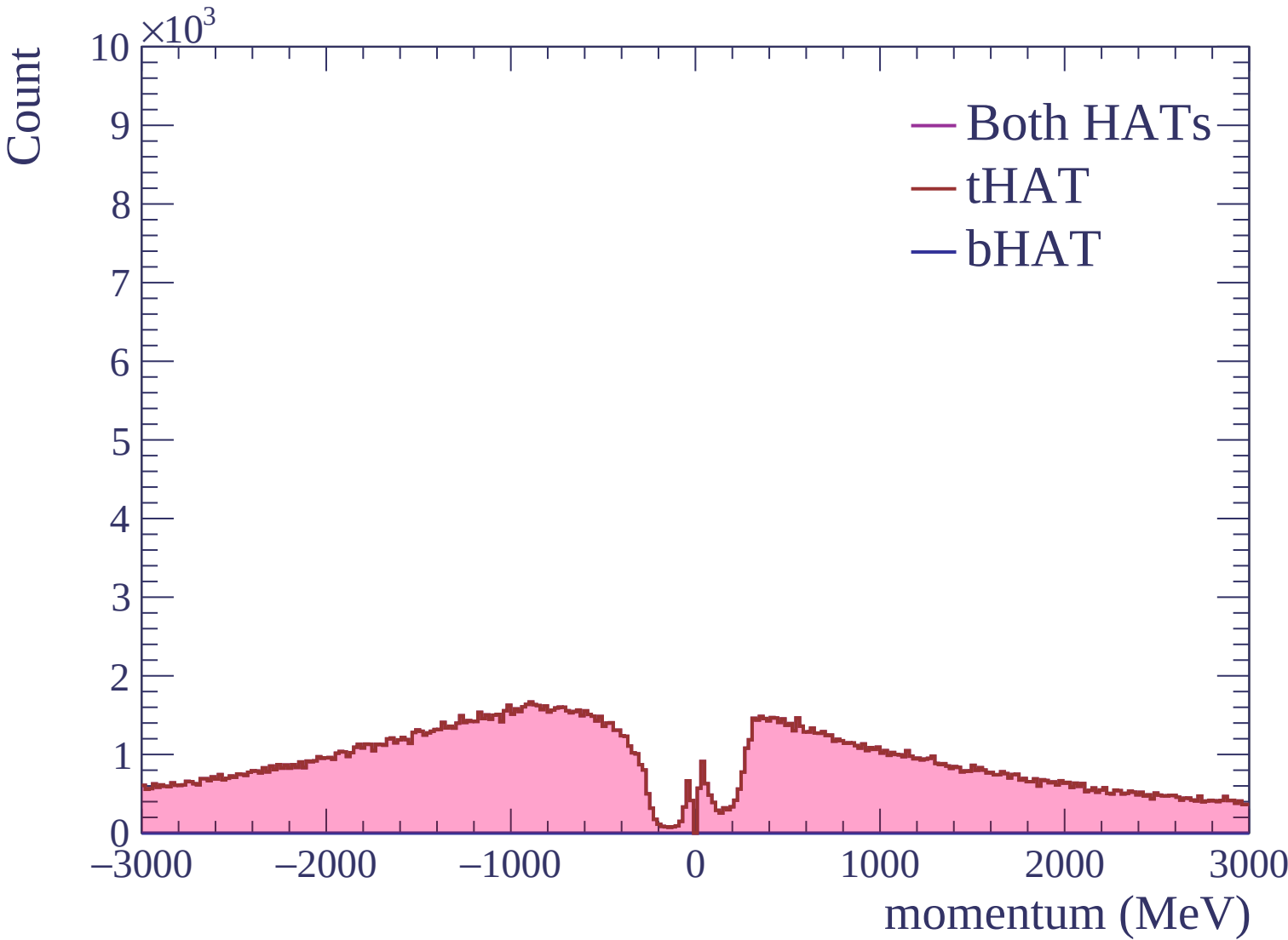






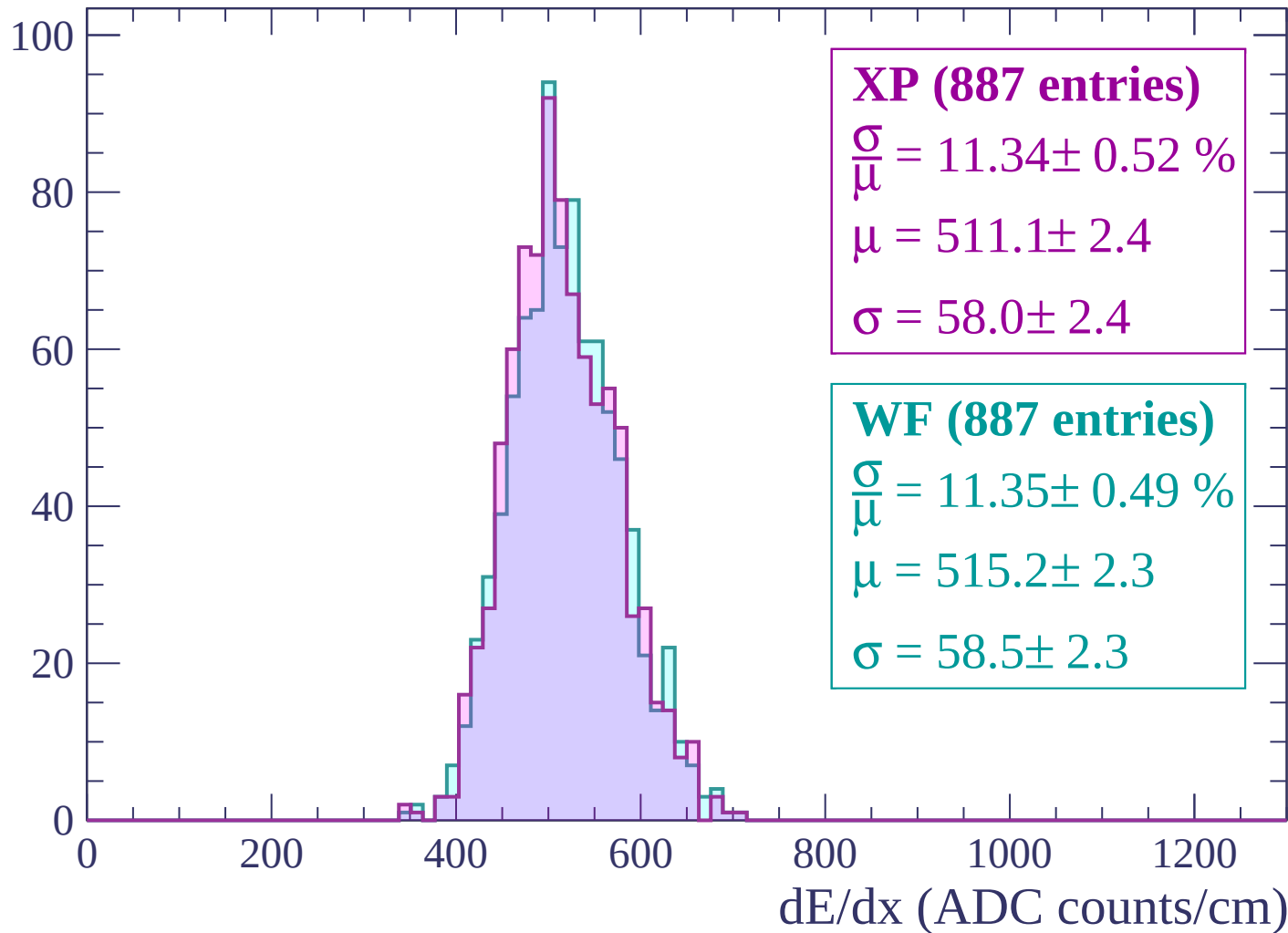






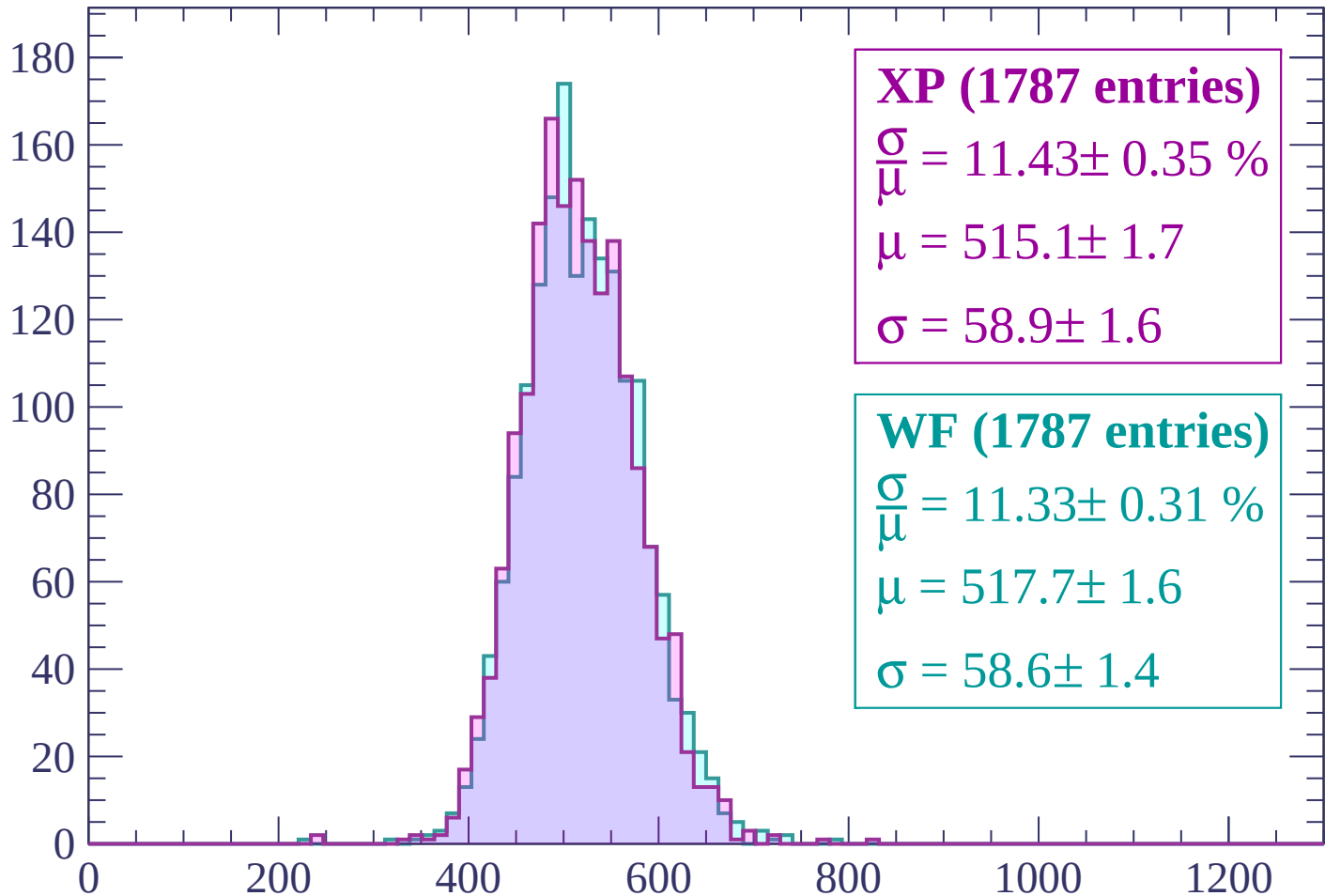
Energy loss | -3000 < p < -2940

Count



Energy loss | $-2940 < p < -2880$

Count



XP (1787 entries)

$$\frac{\sigma}{\mu} = 11.43 \pm 0.35 \%$$

$$\mu = 515.1 \pm 1.7$$

$$\sigma = 58.9 \pm 1.6$$

WF (1787 entries)

$$\frac{\sigma}{\mu} = 11.33 \pm 0.31 \%$$

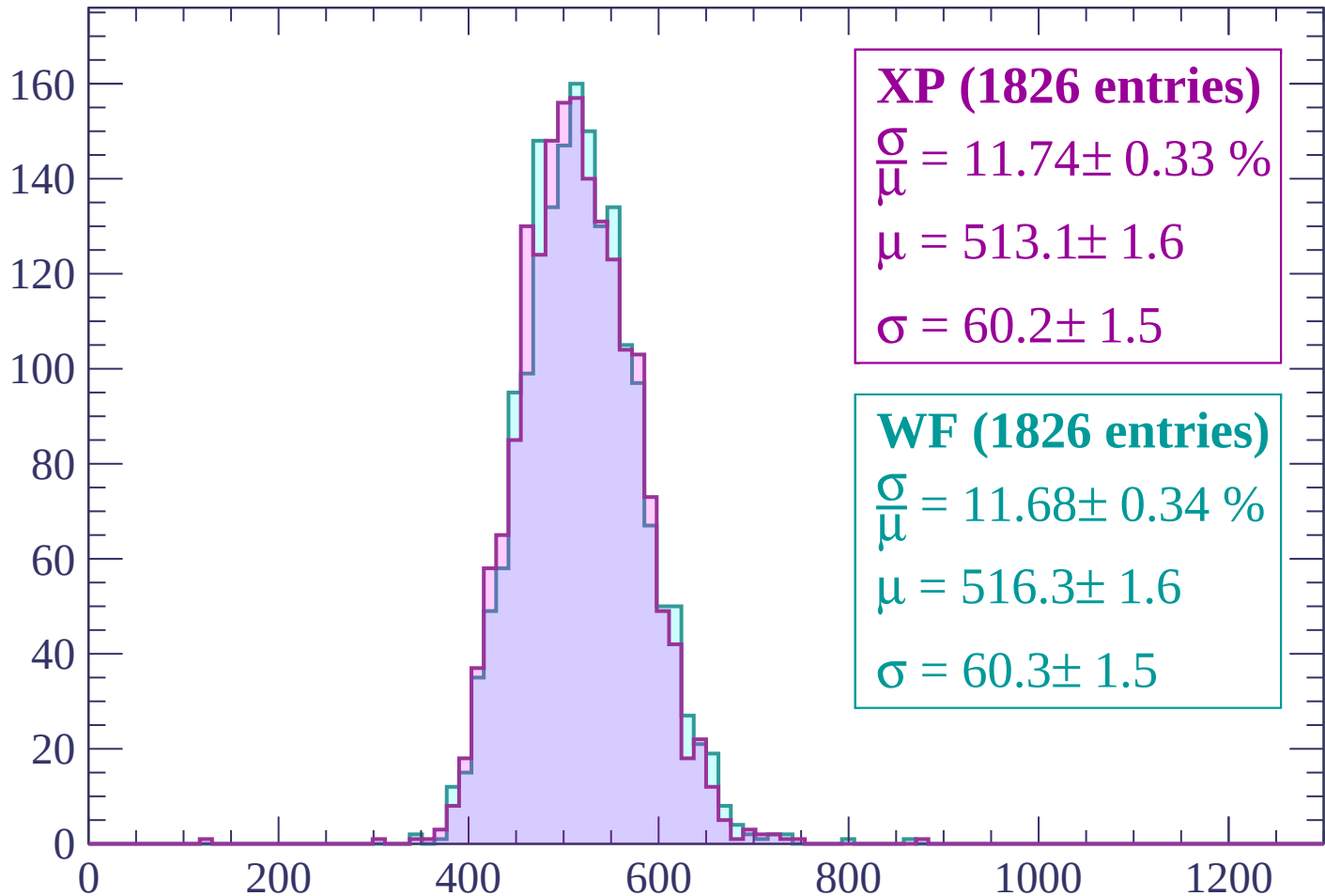
$$\mu = 517.7 \pm 1.6$$

$$\sigma = 58.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP (1826 entries)

$$\frac{\sigma}{\mu} = 11.74 \pm 0.33 \%$$

$$\mu = 513.1 \pm 1.6$$

$$\sigma = 60.2 \pm 1.5$$

WF (1826 entries)

$$\frac{\sigma}{\mu} = 11.68 \pm 0.34 \%$$

$$\mu = 516.3 \pm 1.6$$

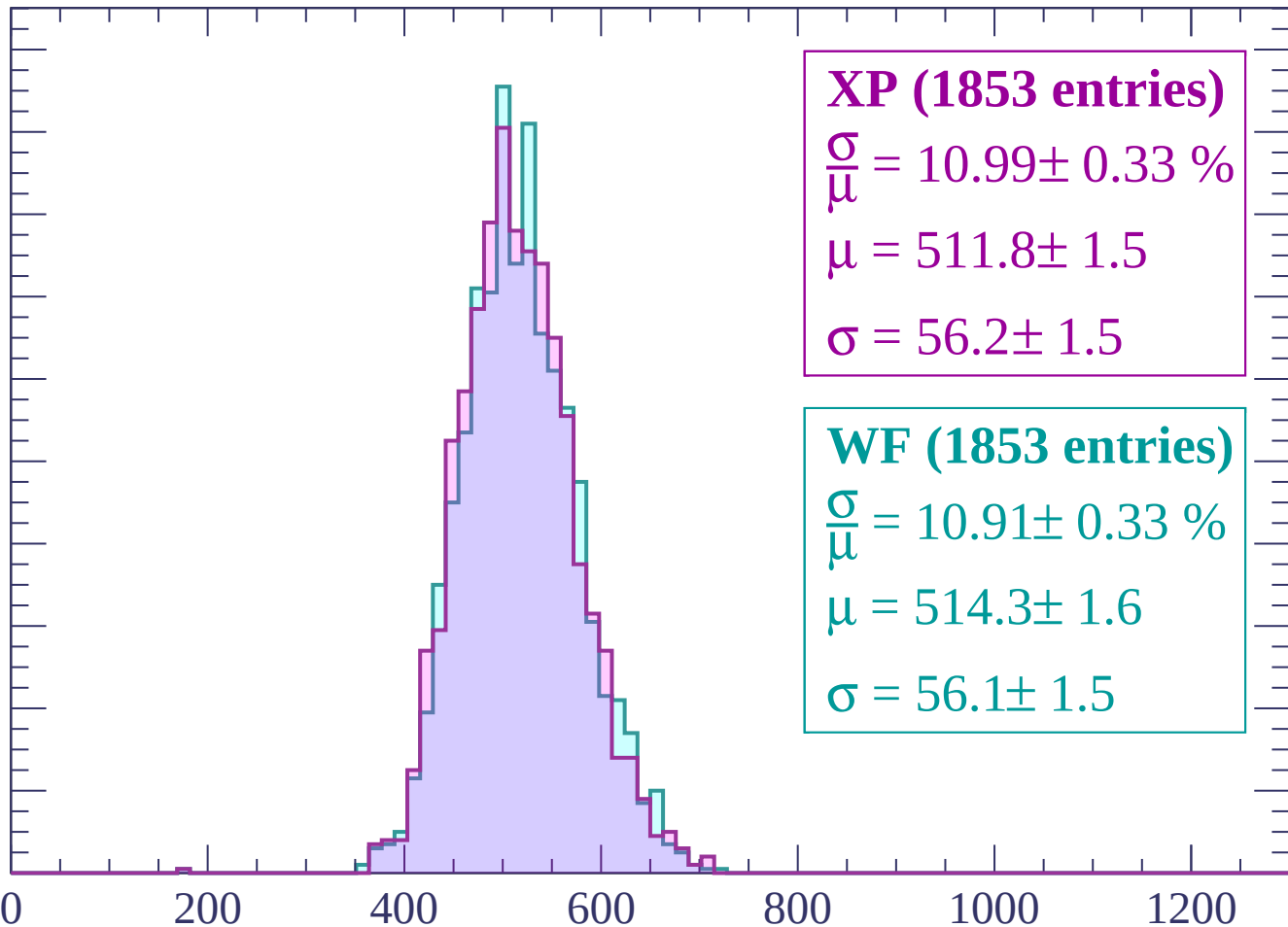
$$\sigma = 60.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-2820 < p < -2760$

Count

200
180
160
140
120
100
80
60
40
20
0



XP (1853 entries)

$$\frac{\sigma}{\mu} = 10.99 \pm 0.33 \%$$

$$\mu = 511.8 \pm 1.5$$

$$\sigma = 56.2 \pm 1.5$$

WF (1853 entries)

$$\frac{\sigma}{\mu} = 10.91 \pm 0.33 \%$$

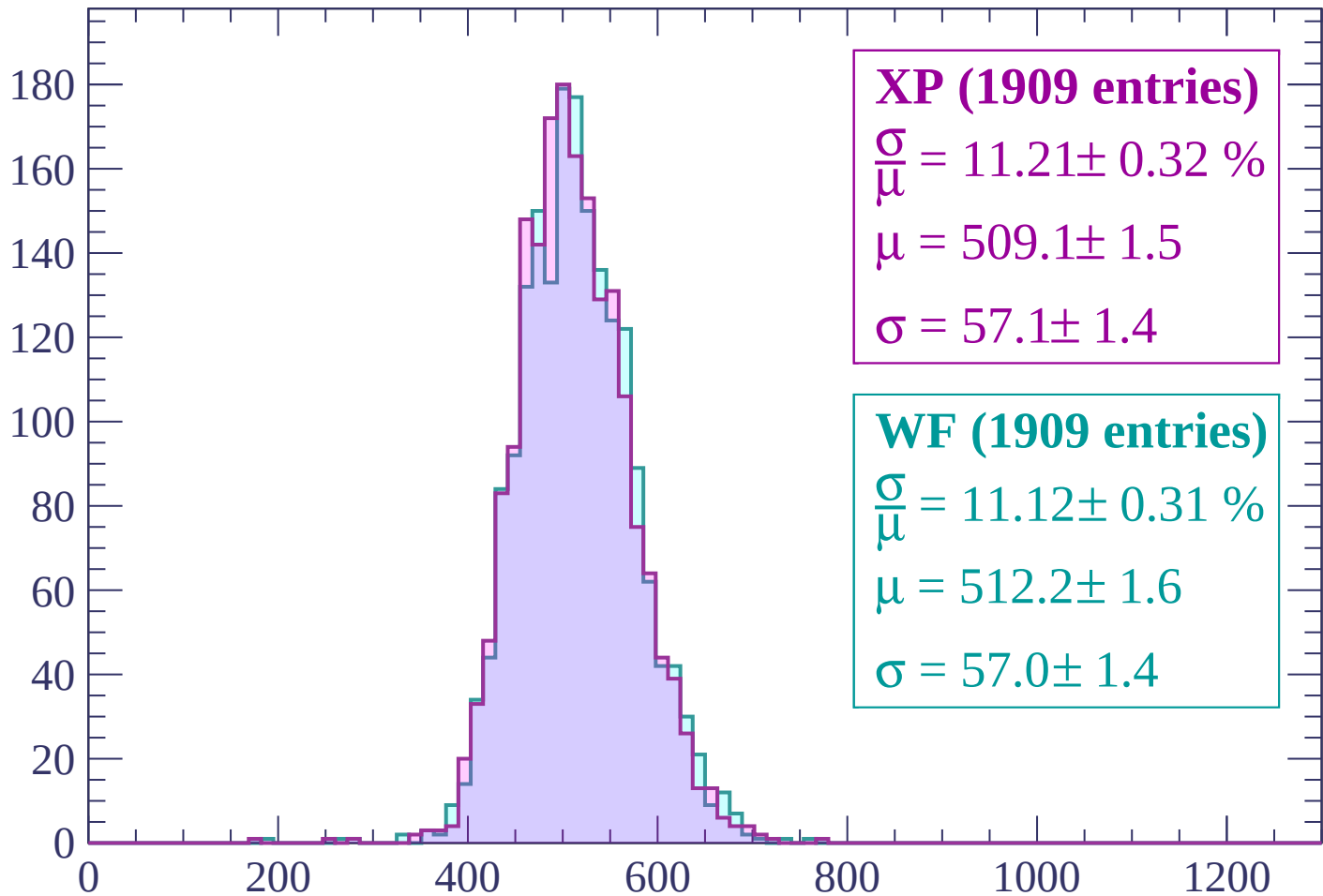
$$\mu = 514.3 \pm 1.6$$

$$\sigma = 56.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700

Count



XP (1909 entries)

$\frac{\sigma}{\mu} = 11.21 \pm 0.32 \%$

$\mu = 509.1 \pm 1.5$

$\sigma = 57.1 \pm 1.4$

WF (1909 entries)

$\frac{\sigma}{\mu} = 11.12 \pm 0.31 \%$

$\mu = 512.2 \pm 1.6$

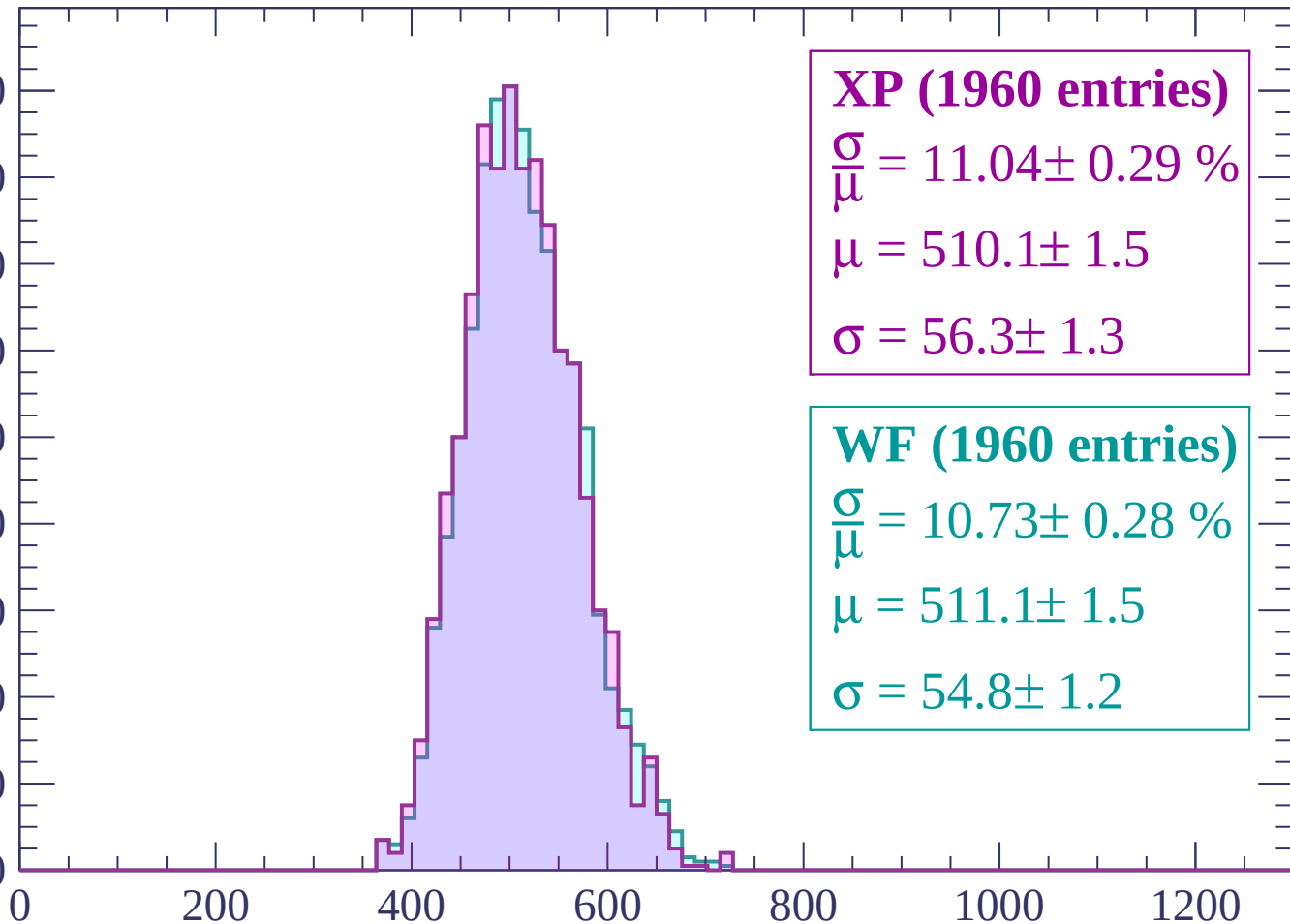
$\sigma = 57.0 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | -2700 < p < -2640

Count

180
160
140
120
100
80
60
40
20
0



XP (1960 entries)

$$\frac{\sigma}{\mu} = 11.04 \pm 0.29 \%$$

$$\mu = 510.1 \pm 1.5$$

$$\sigma = 56.3 \pm 1.3$$

WF (1960 entries)

$$\frac{\sigma}{\mu} = 10.73 \pm 0.28 \%$$

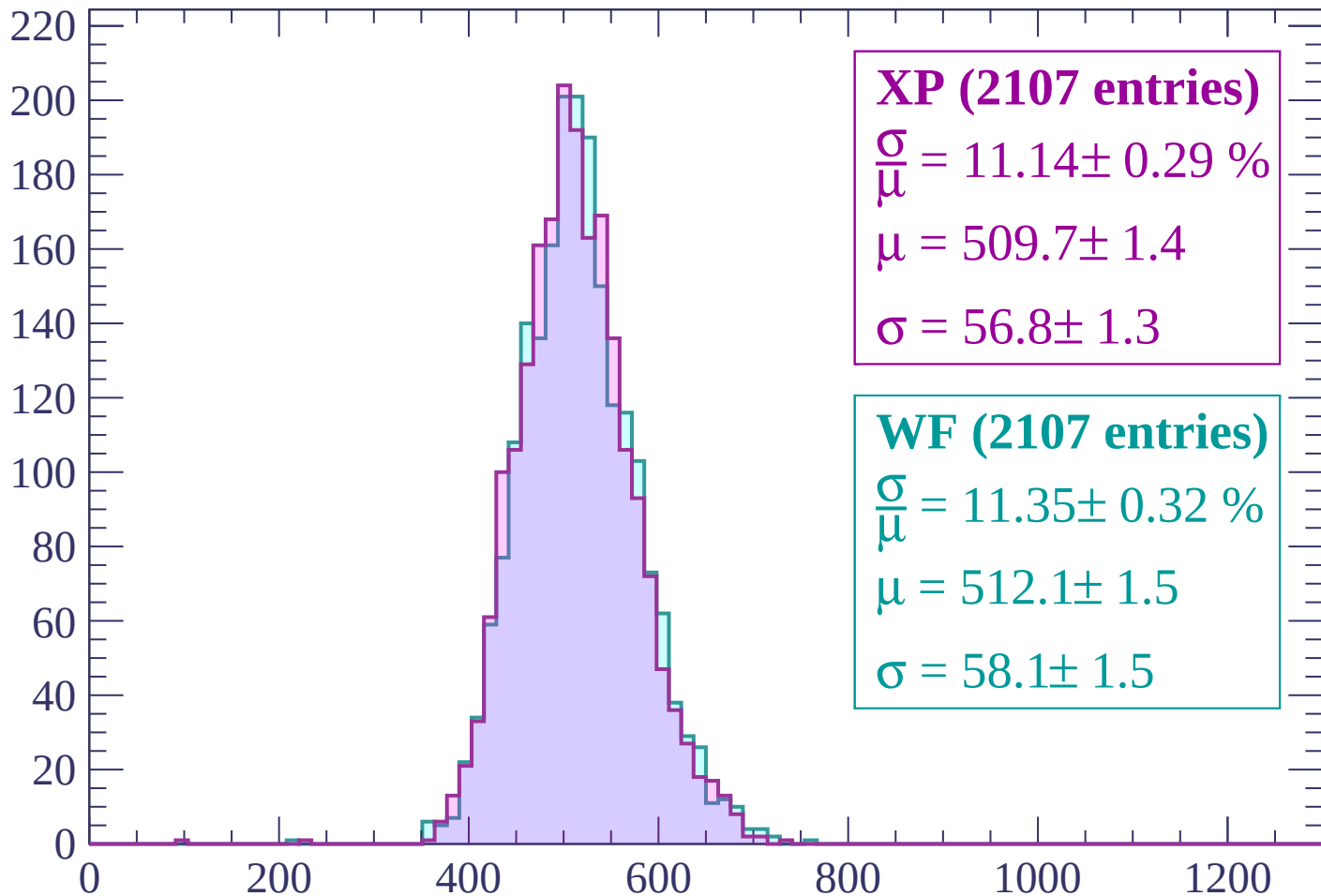
$$\mu = 511.1 \pm 1.5$$

$$\sigma = 54.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (2107 entries)

$$\frac{\sigma}{\mu} = 11.14 \pm 0.29 \%$$

$$\mu = 509.7 \pm 1.4$$

$$\sigma = 56.8 \pm 1.3$$

WF (2107 entries)

$$\frac{\sigma}{\mu} = 11.35 \pm 0.32 \%$$

$$\mu = 512.1 \pm 1.5$$

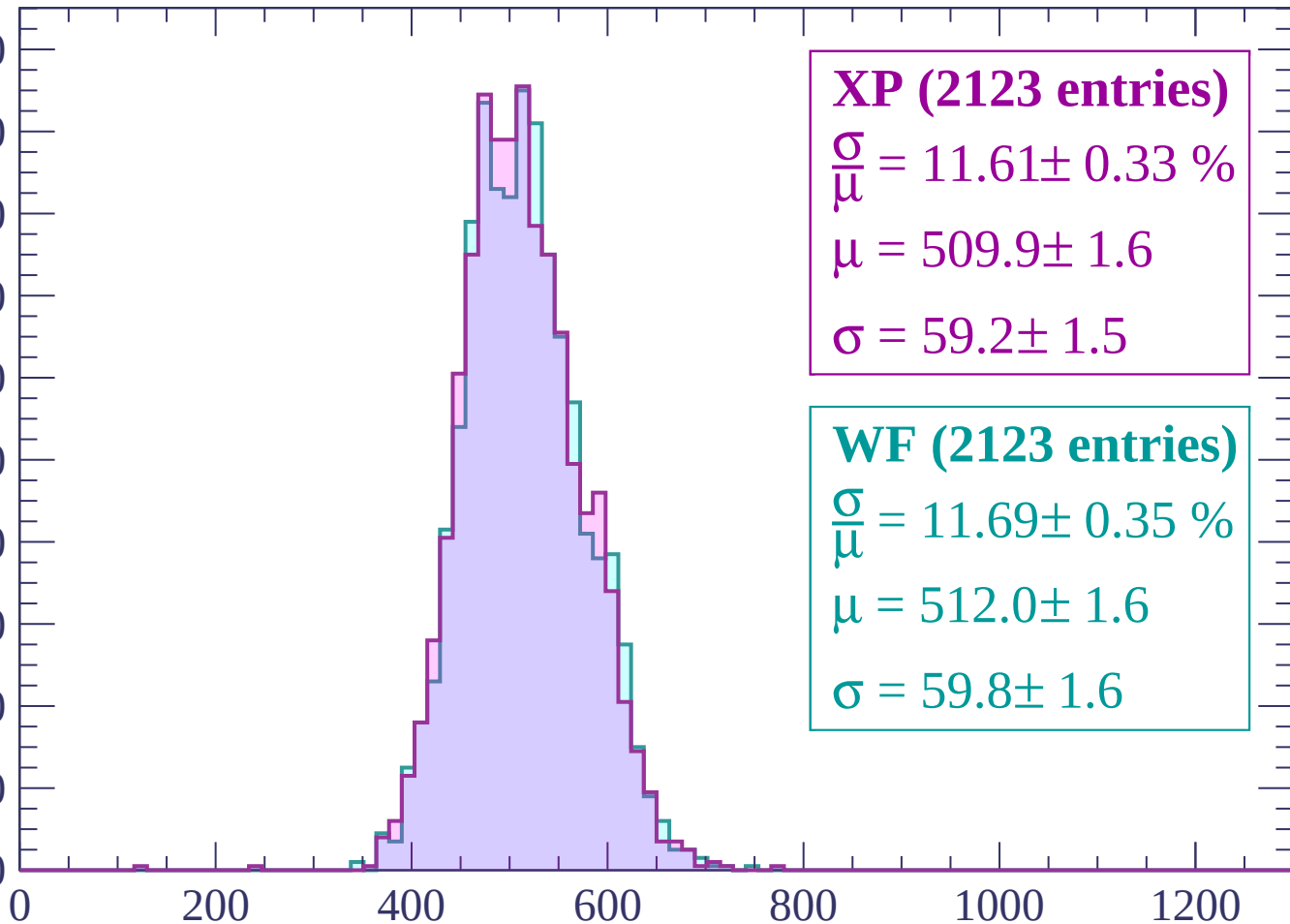
$$\sigma = 58.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-2580 < p < -2520$

Count

200
180
160
140
120
100
80
60
40
20
0



XP (2123 entries)

$$\frac{\sigma}{\mu} = 11.61 \pm 0.33 \%$$

$$\mu = 509.9 \pm 1.6$$

$$\sigma = 59.2 \pm 1.5$$

WF (2123 entries)

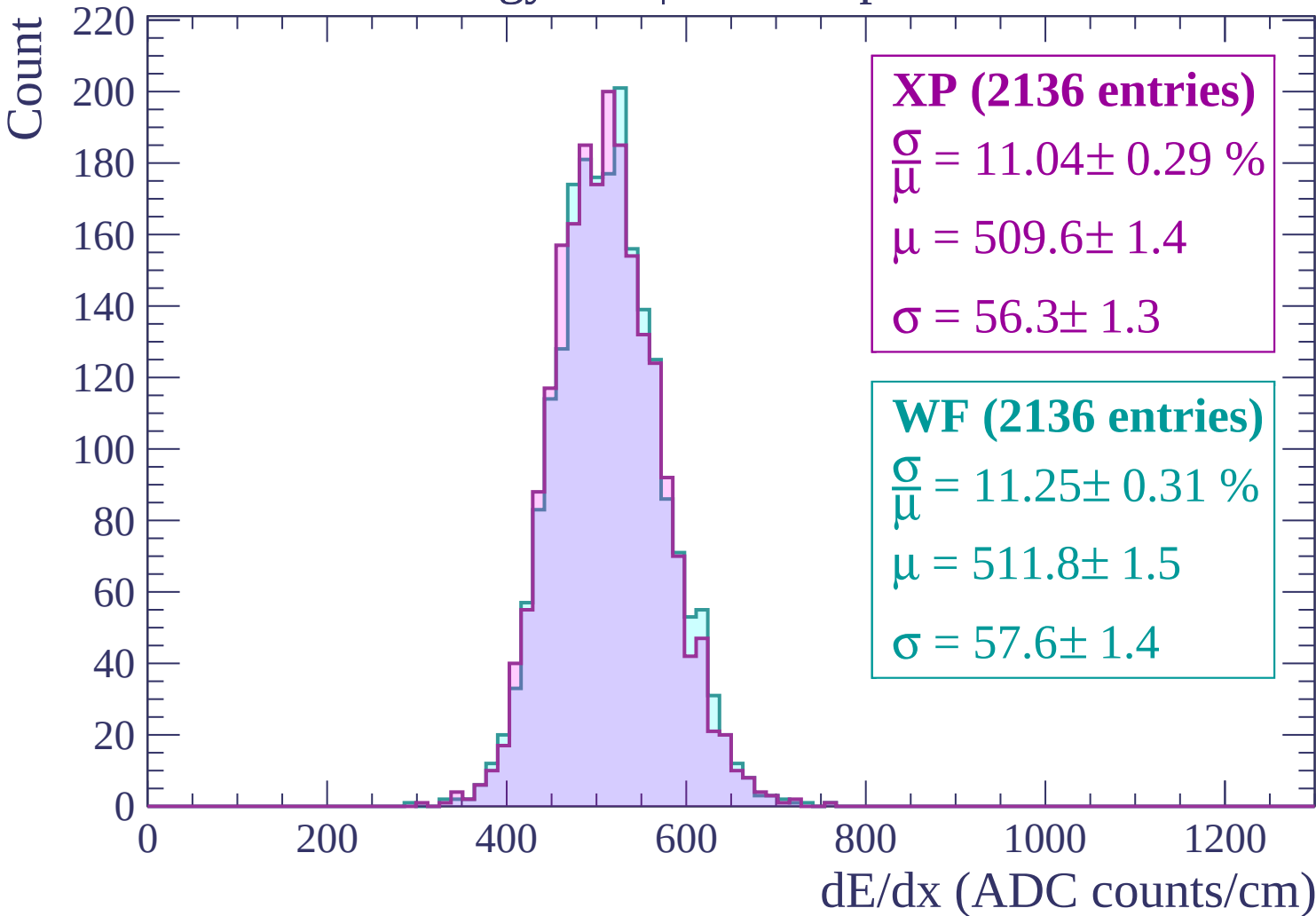
$$\frac{\sigma}{\mu} = 11.69 \pm 0.35 \%$$

$$\mu = 512.0 \pm 1.6$$

$$\sigma = 59.8 \pm 1.6$$

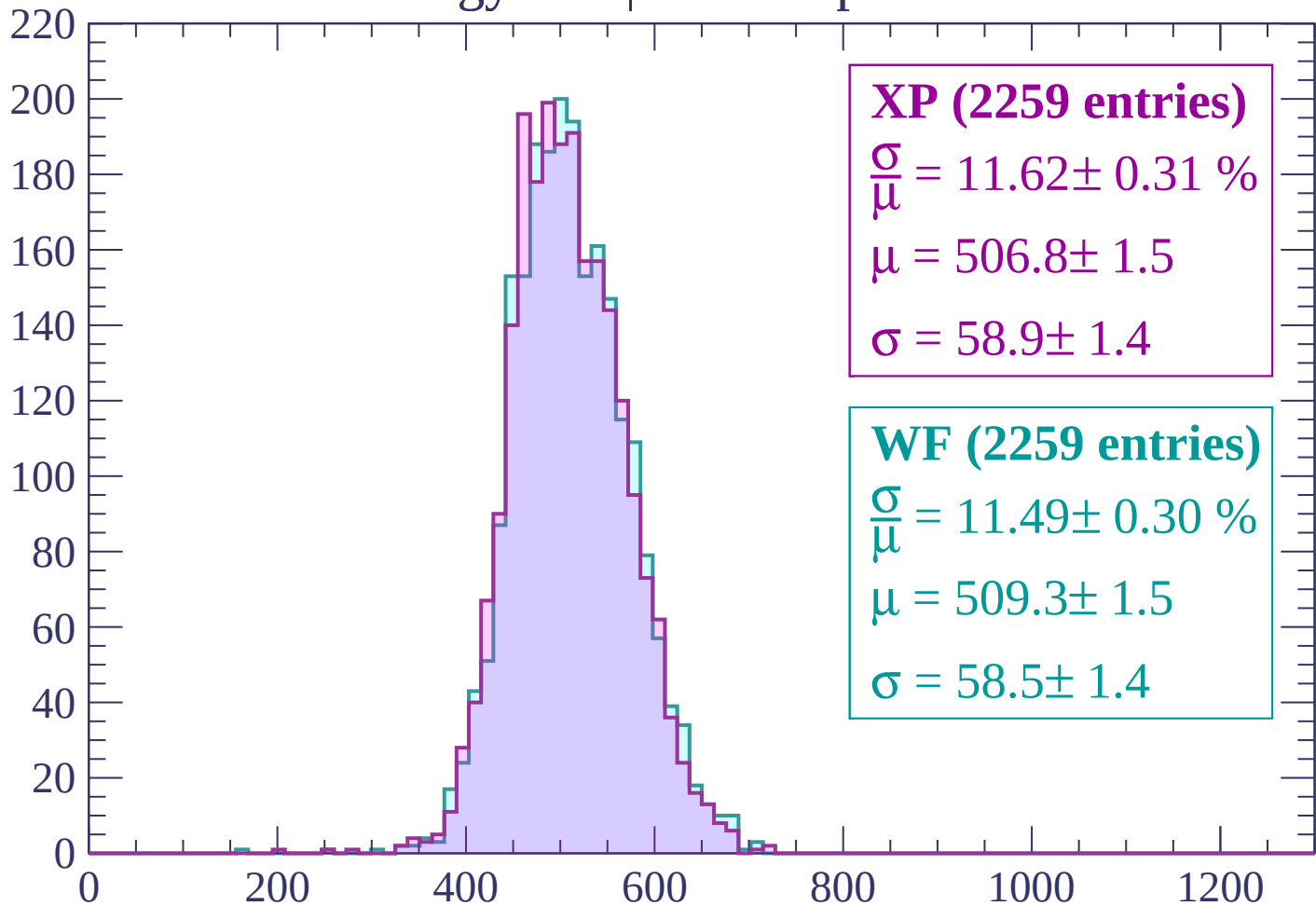
dE/dx (ADC counts/cm)

Energy loss | $-2520 < p < -2460$



Energy loss | $-2460 < p < -2400$

Count



XP (2259 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.31 \%$$

$$\mu = 506.8 \pm 1.5$$

$$\sigma = 58.9 \pm 1.4$$

WF (2259 entries)

$$\frac{\sigma}{\mu} = 11.49 \pm 0.30 \%$$

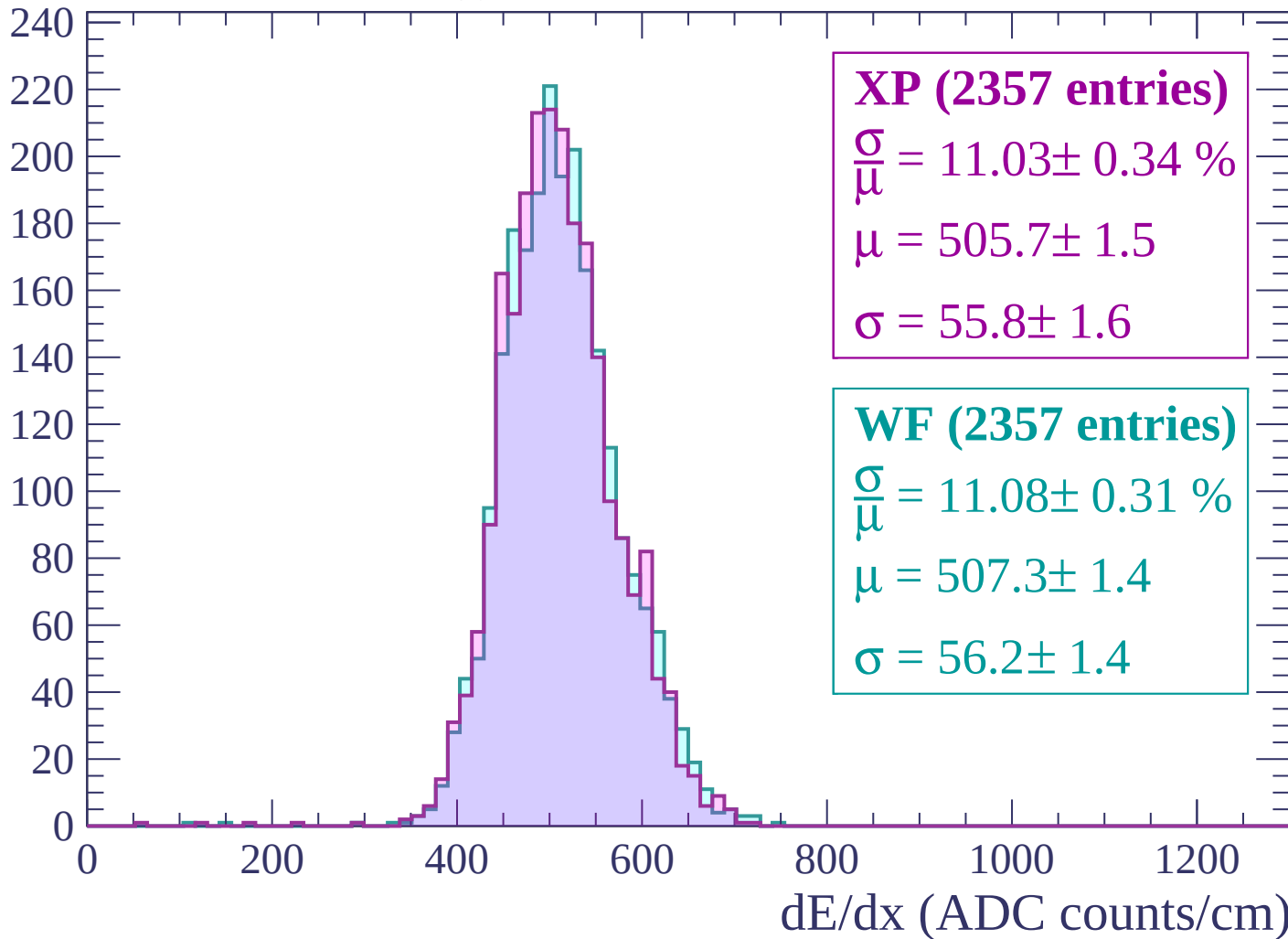
$$\mu = 509.3 \pm 1.5$$

$$\sigma = 58.5 \pm 1.4$$

dE/dx (ADC counts/cm)

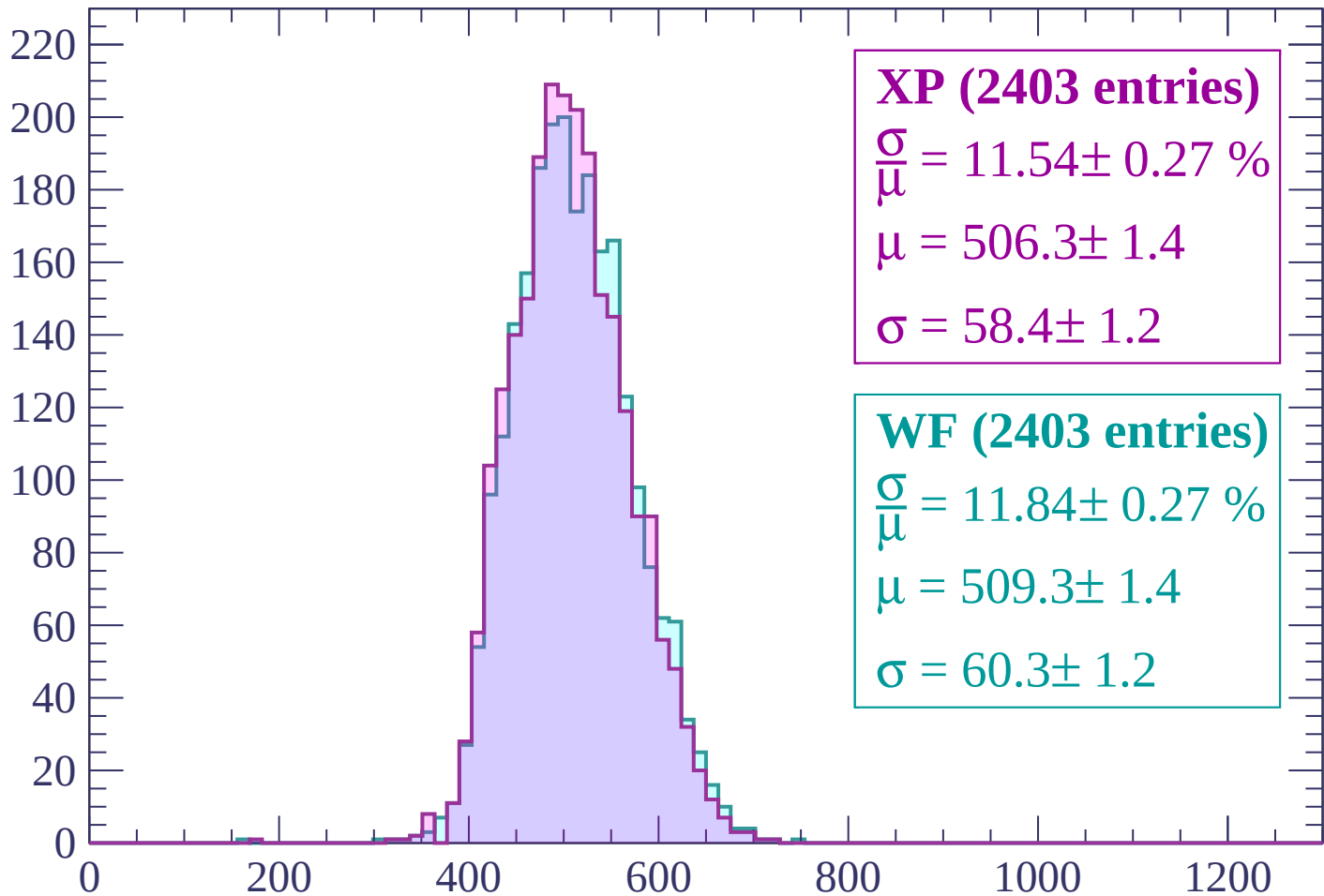
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (2403 entries)

$$\frac{\sigma}{\mu} = 11.54 \pm 0.27 \%$$

$$\mu = 506.3 \pm 1.4$$

$$\sigma = 58.4 \pm 1.2$$

WF (2403 entries)

$$\frac{\sigma}{\mu} = 11.84 \pm 0.27 \%$$

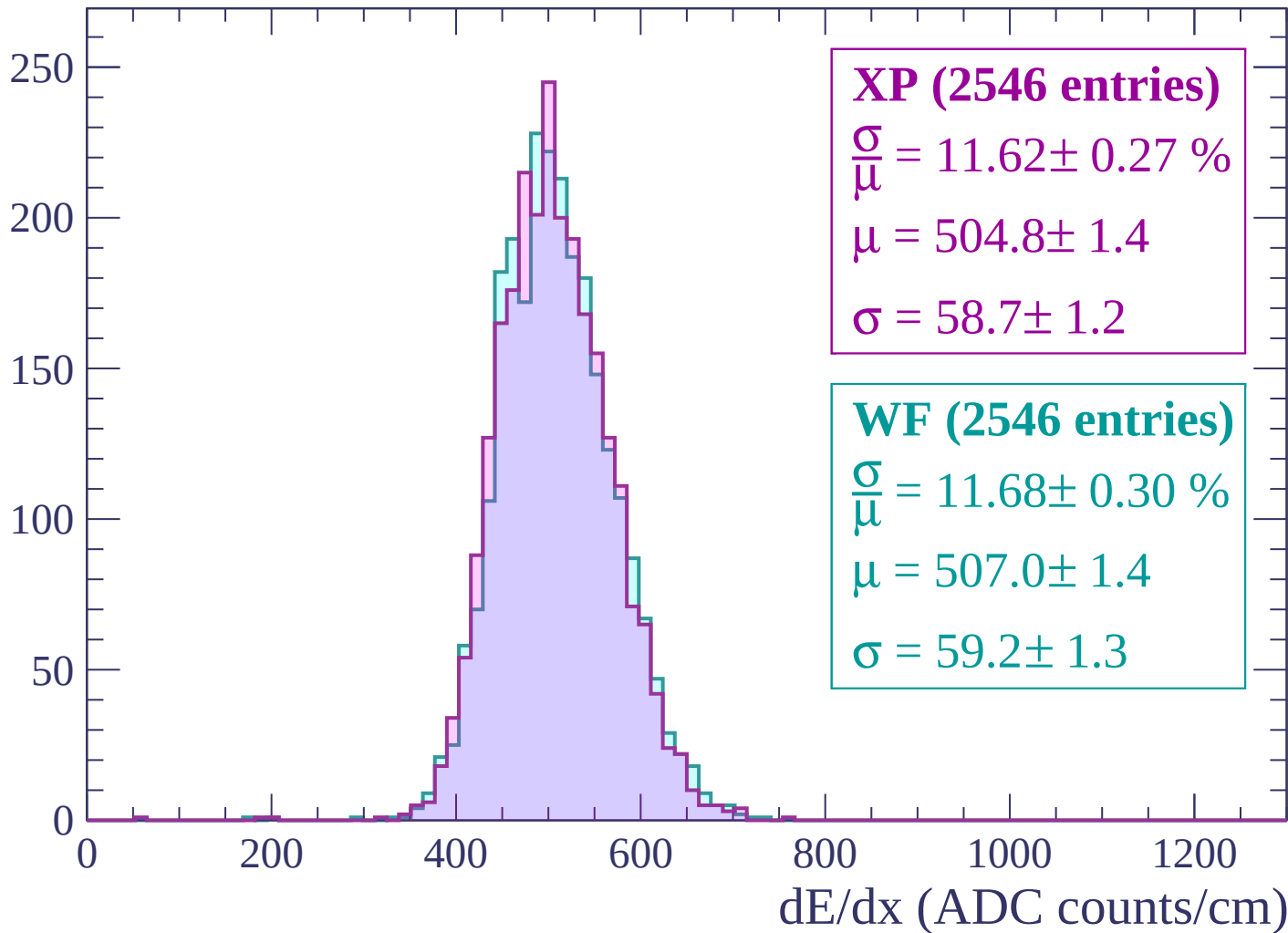
$$\mu = 509.3 \pm 1.4$$

$$\sigma = 60.3 \pm 1.2$$

dE/dx (ADC counts/cm)

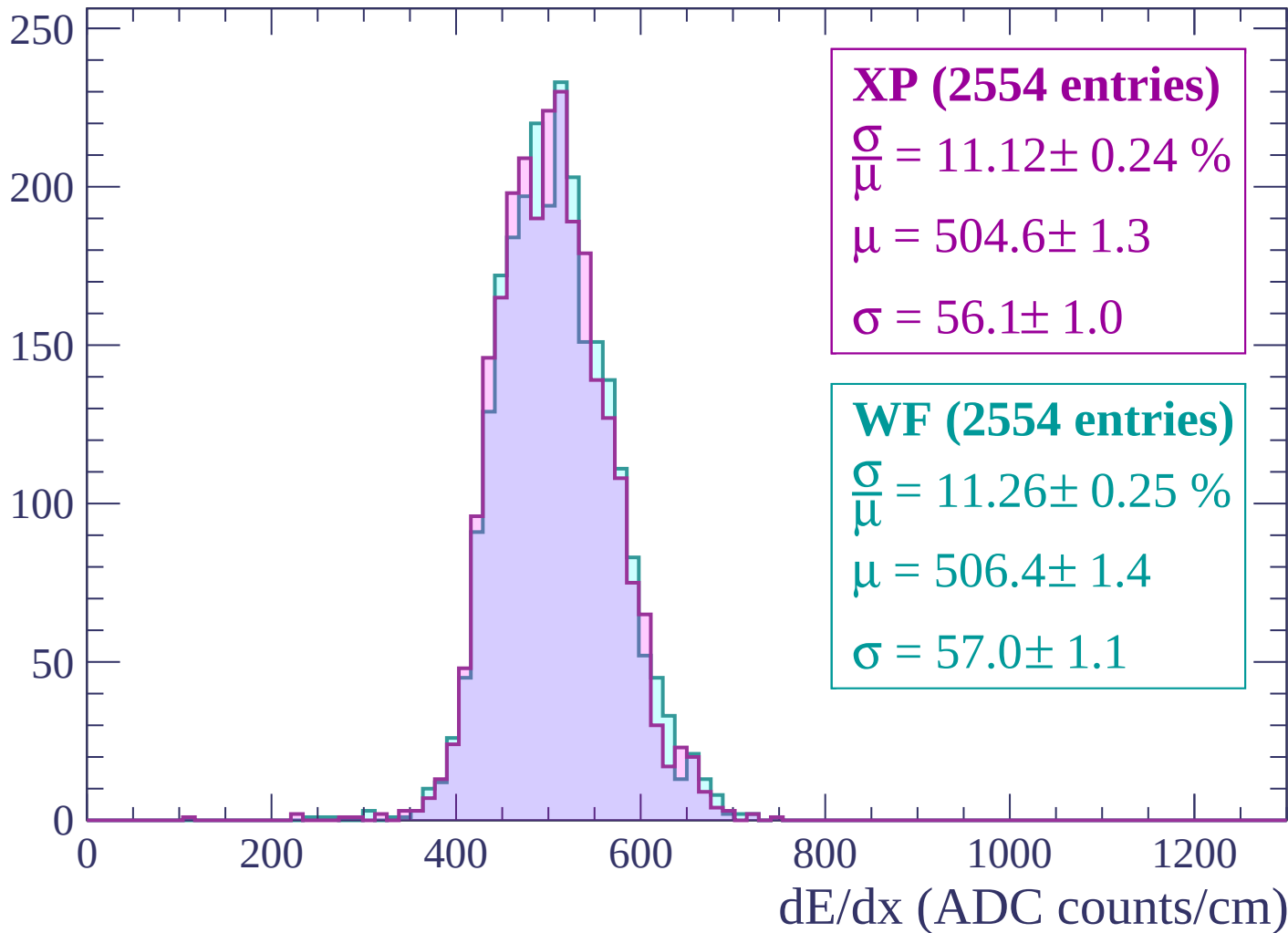
Energy loss | -2280 < p < -2220

Count



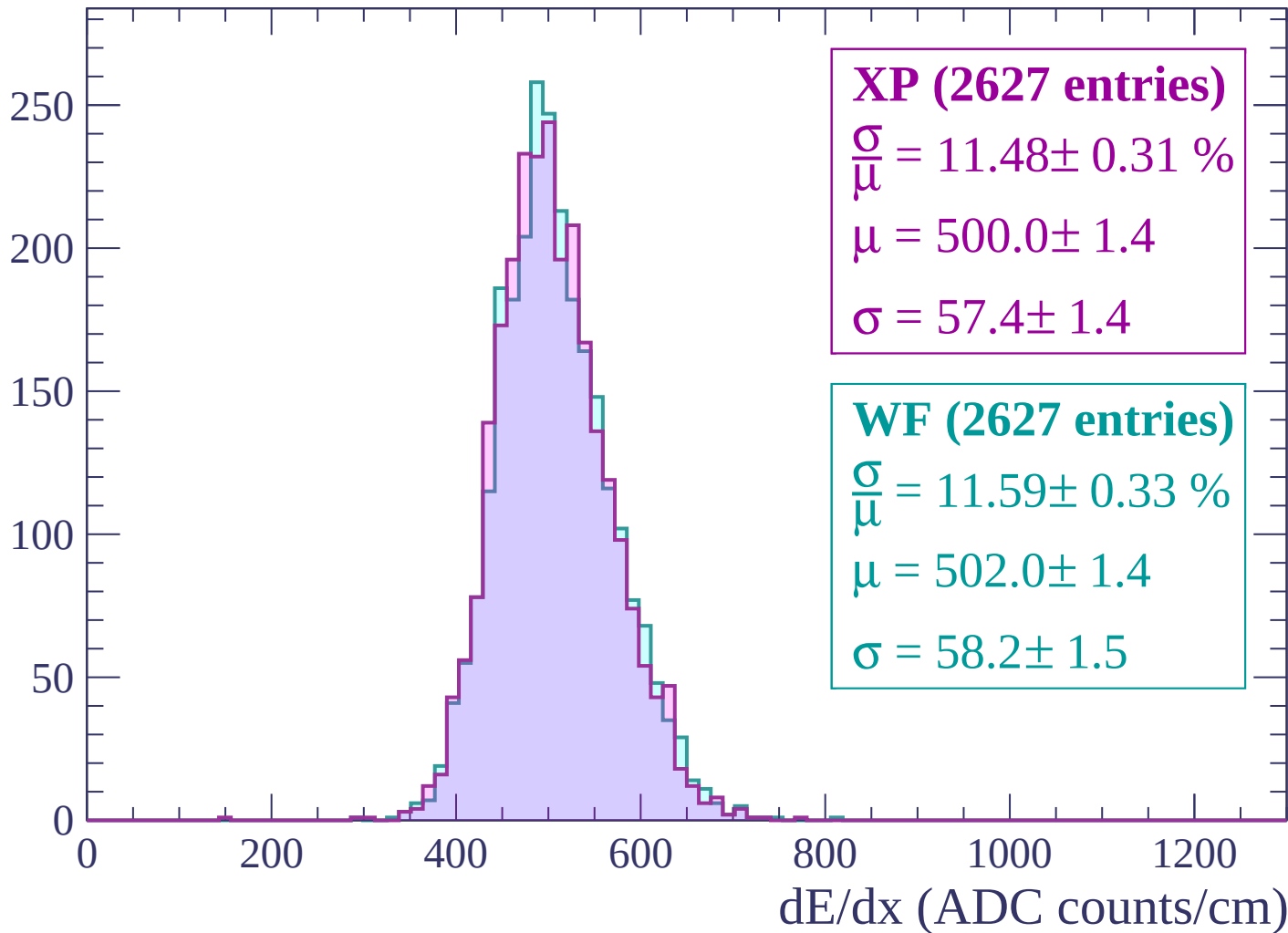
Energy loss | -2220 < p < -2160

Count



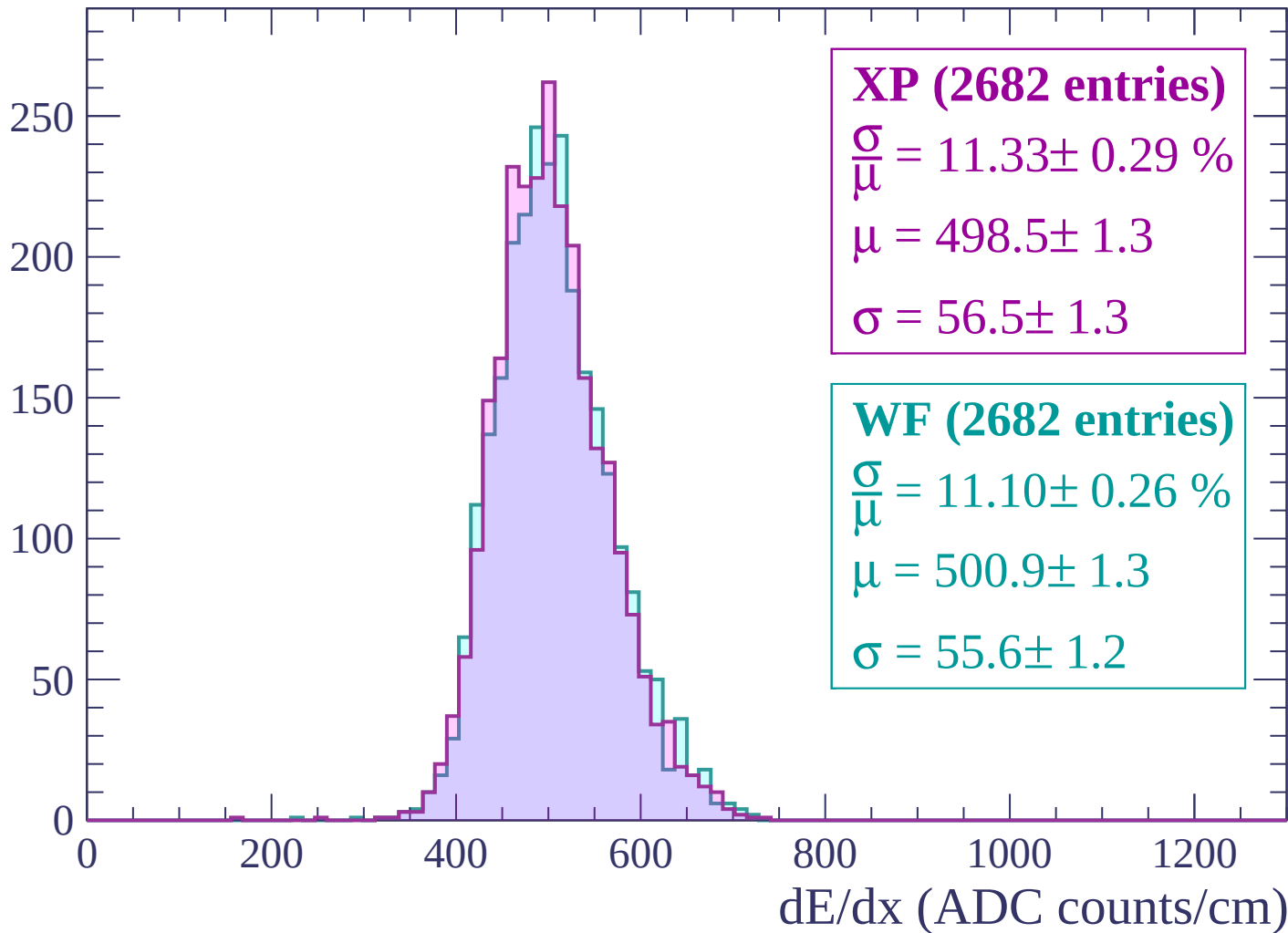
Energy loss | -2160 < p < -2100

Count



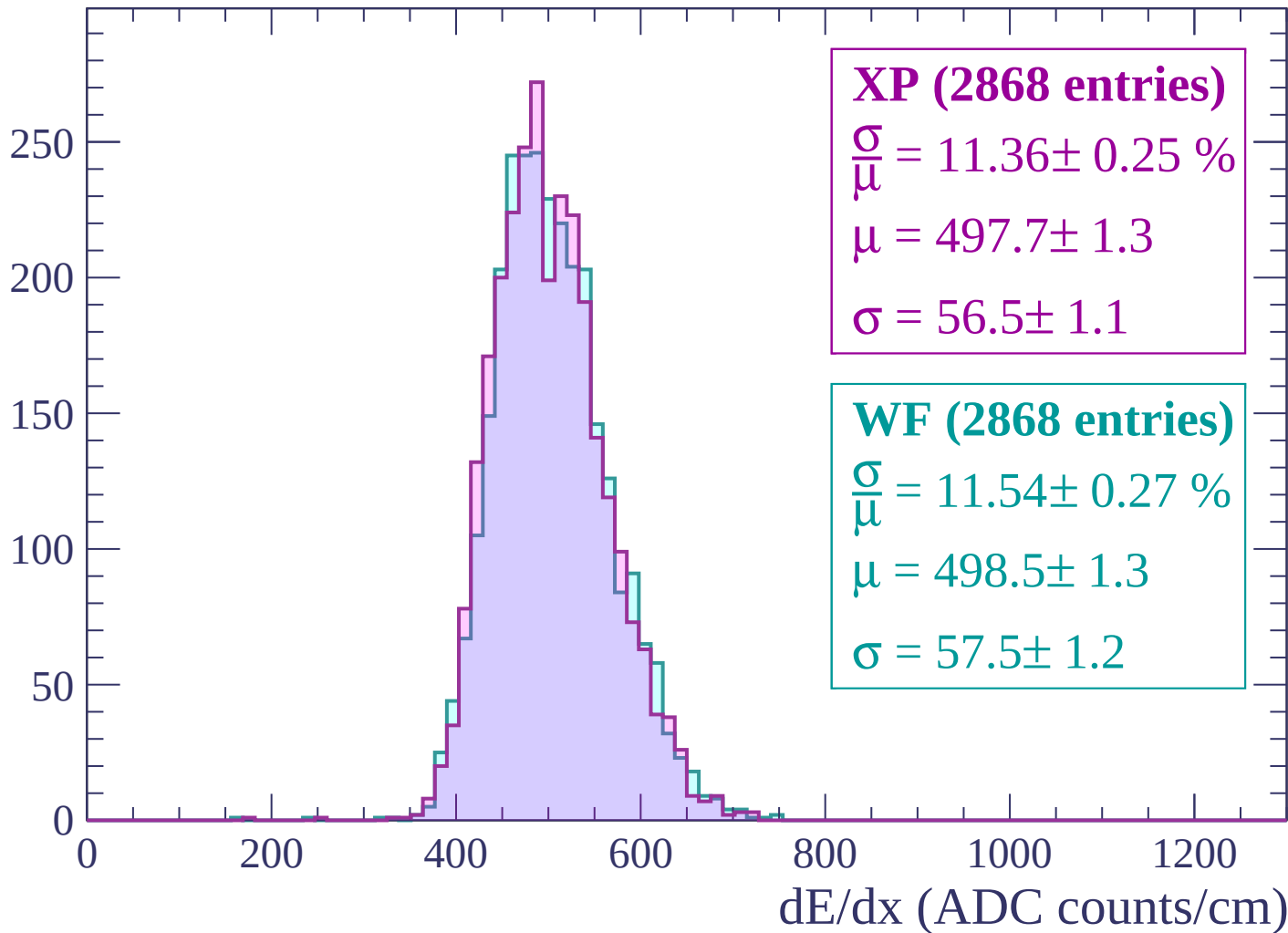
Energy loss | -2100 < p < -2040

Count



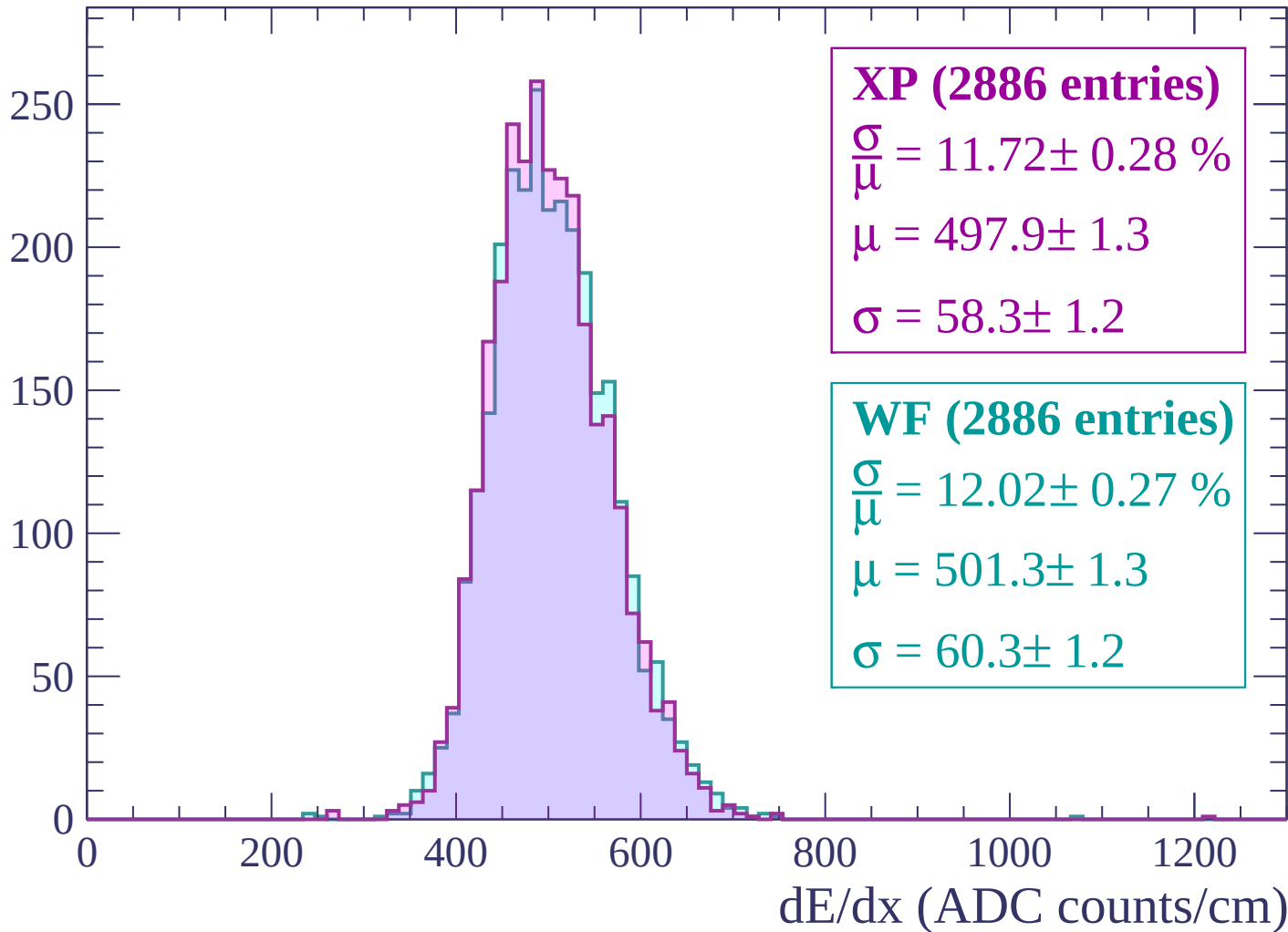
Energy loss | -2040 < p < -1980

Count



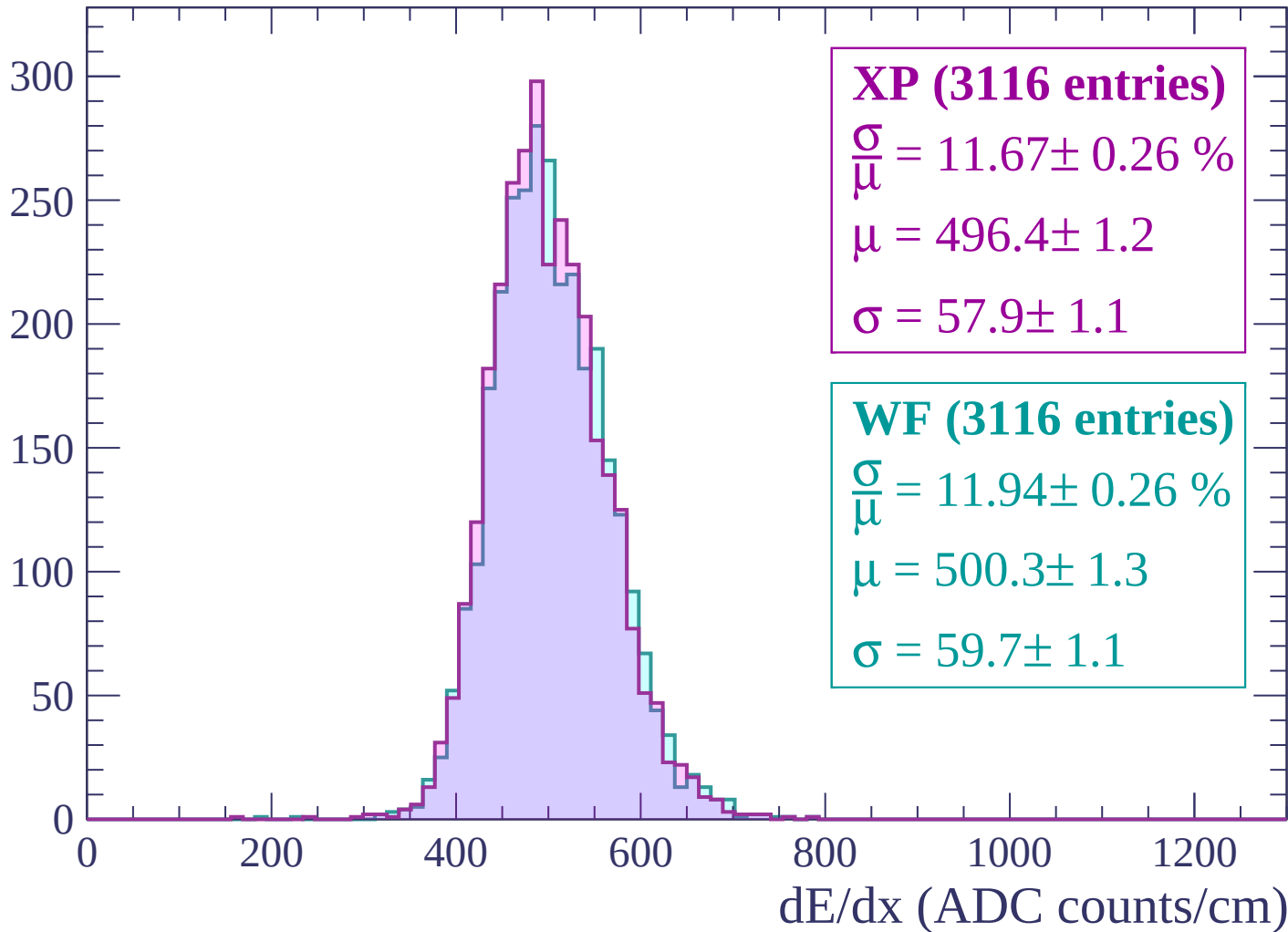
Energy loss | -1980 < p < -1920

Count



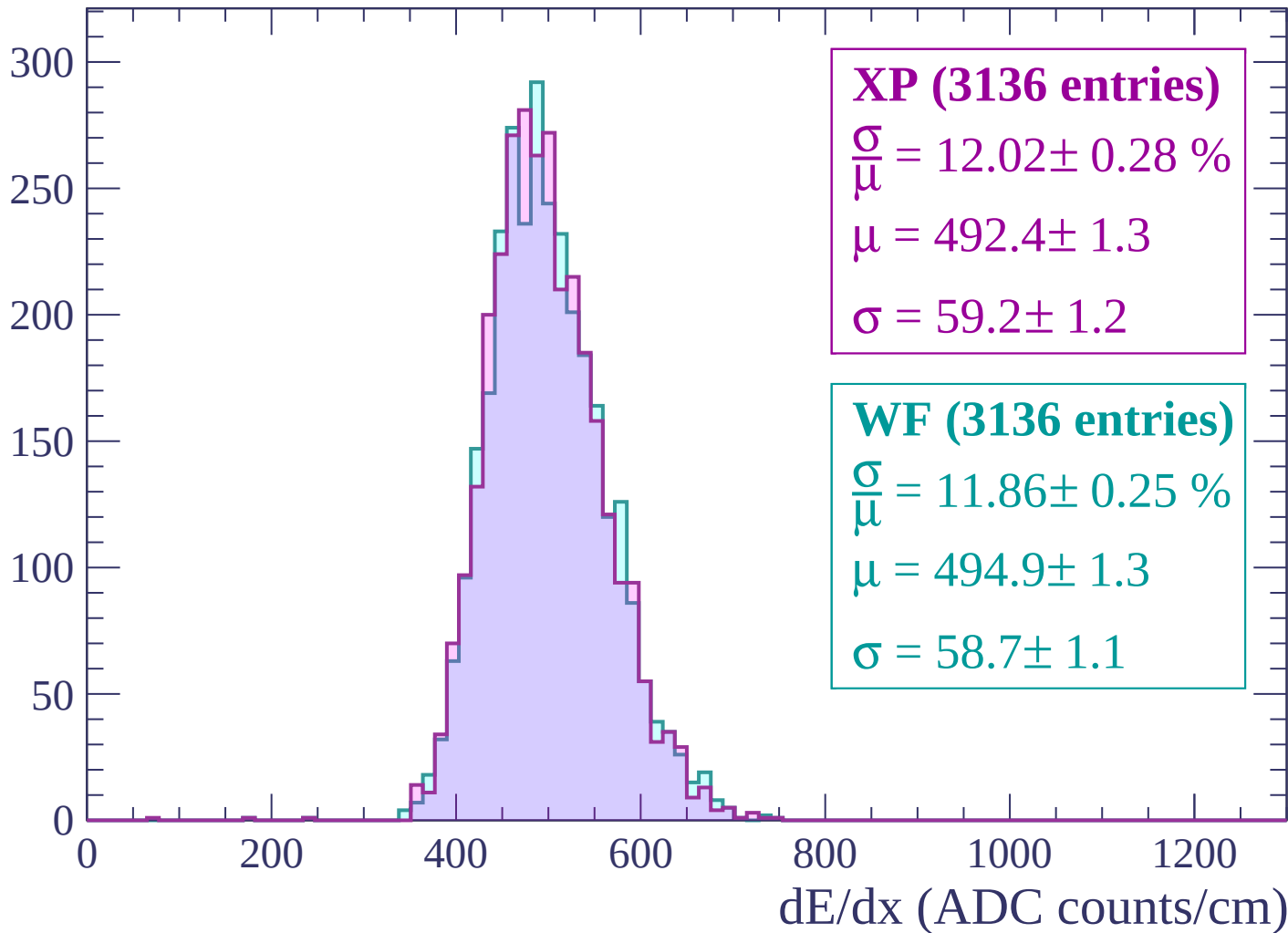
Energy loss | -1920 < p < -1860

Count



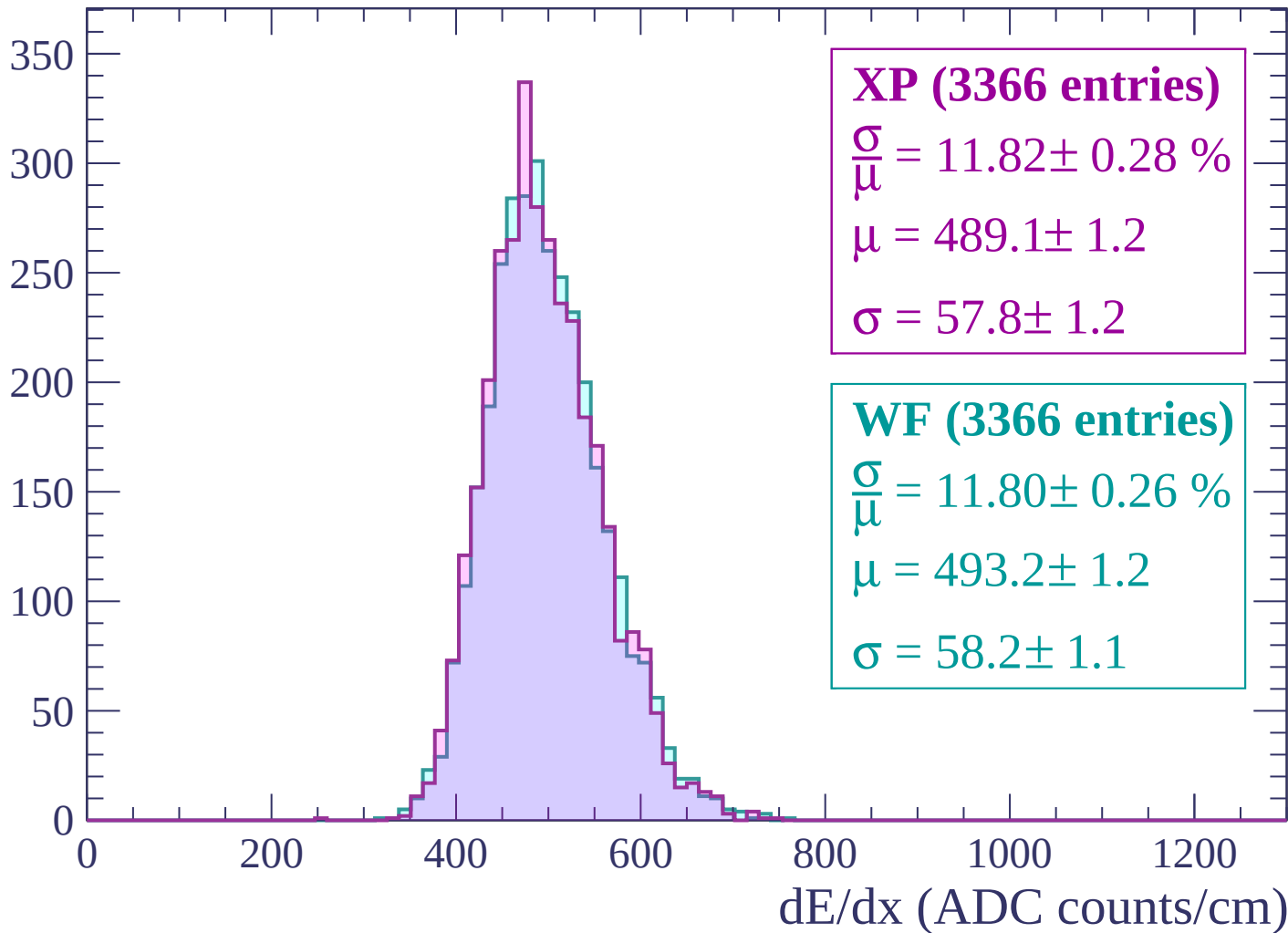
Energy loss | -1860 < p < -1800

Count



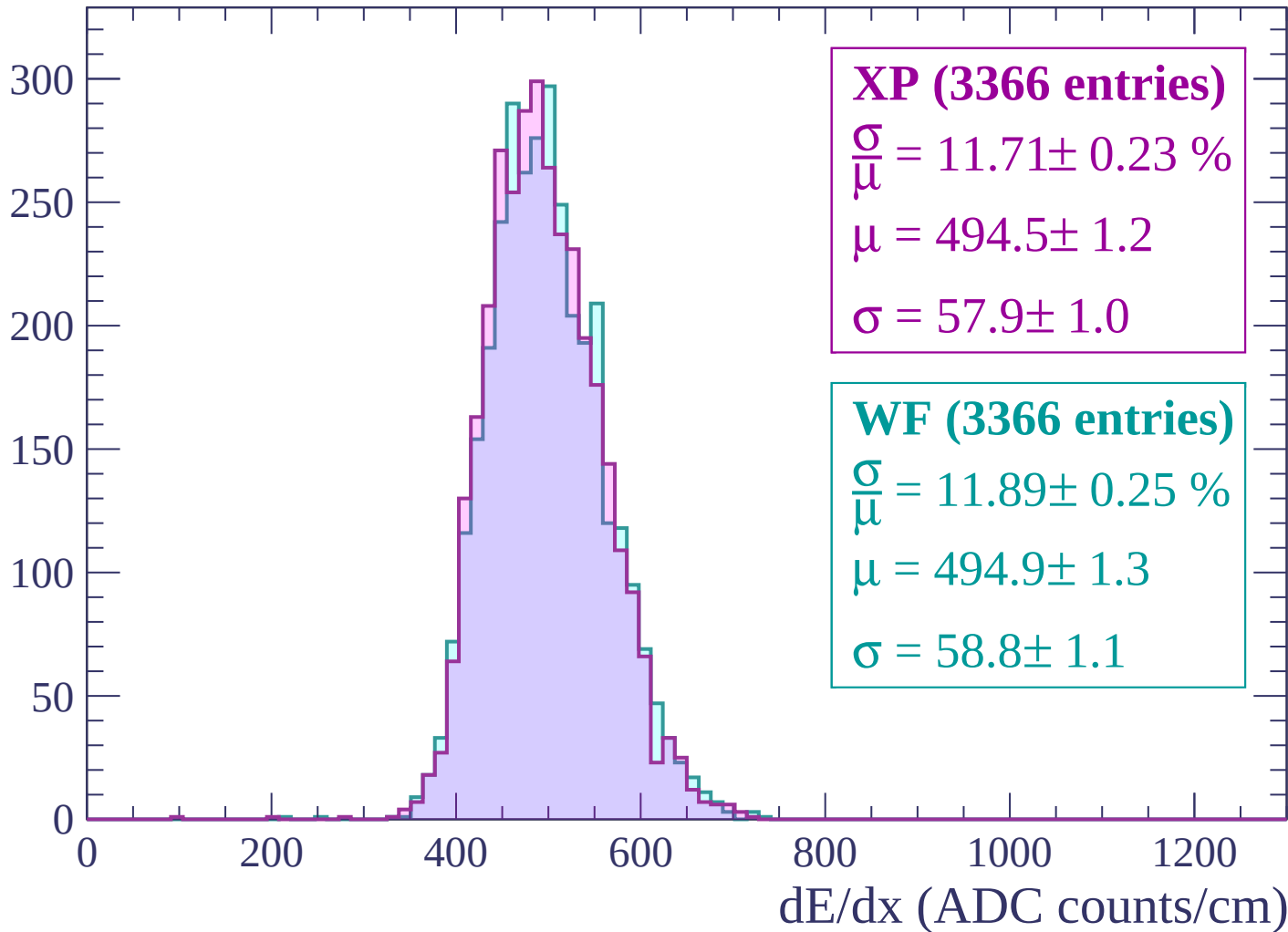
Energy loss | $-1800 < p < -1740$

Count



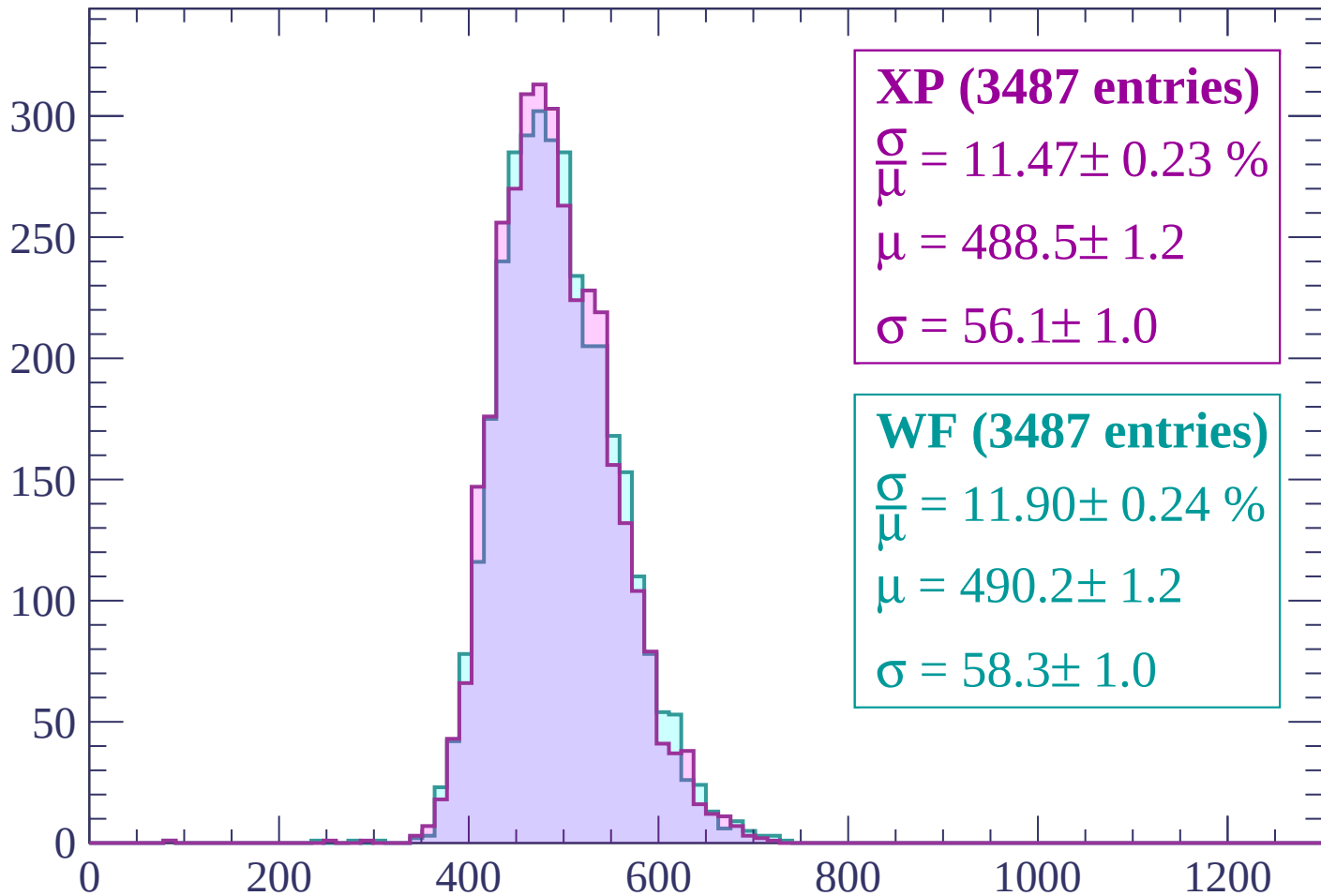
Energy loss | $-1740 < p < -1680$

Count



Energy loss | -1680 < p < -1620

Count



XP (3487 entries)

$$\frac{\sigma}{\mu} = 11.47 \pm 0.23 \%$$

$$\mu = 488.5 \pm 1.2$$

$$\sigma = 56.1 \pm 1.0$$

WF (3487 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.24 \%$$

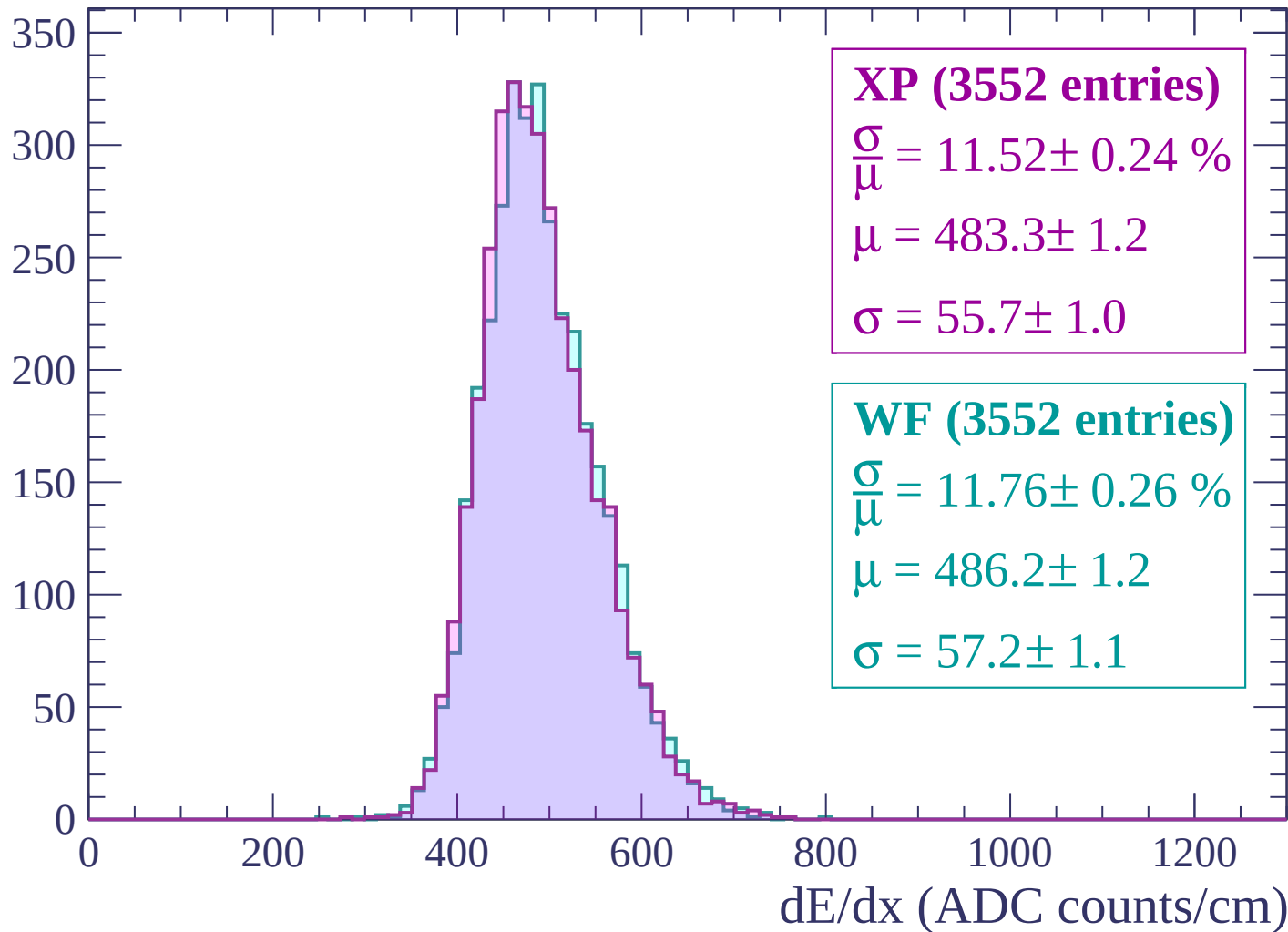
$$\mu = 490.2 \pm 1.2$$

$$\sigma = 58.3 \pm 1.0$$

dE/dx (ADC counts/cm)

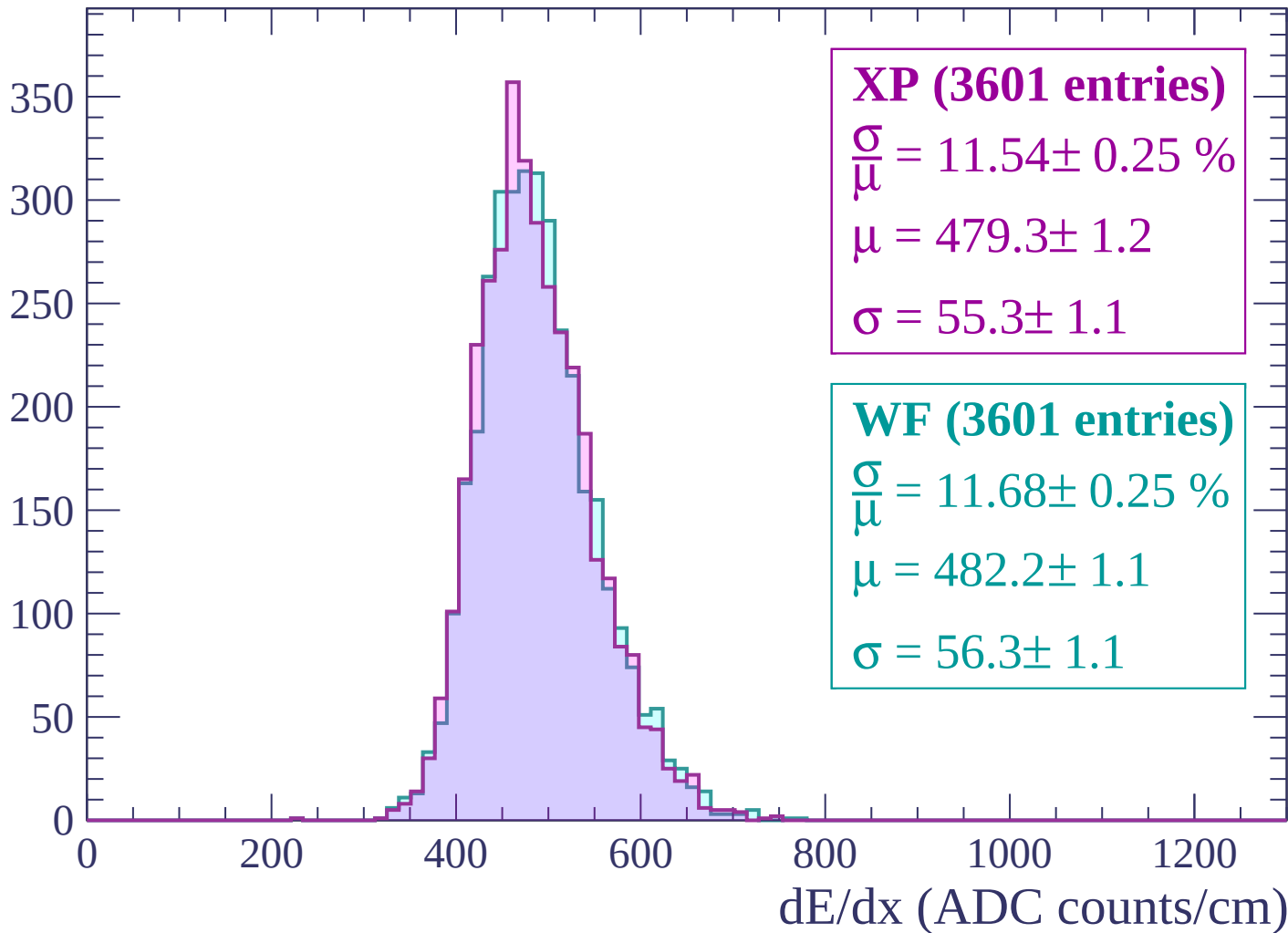
Energy loss | -1620 < p < -1560

Count



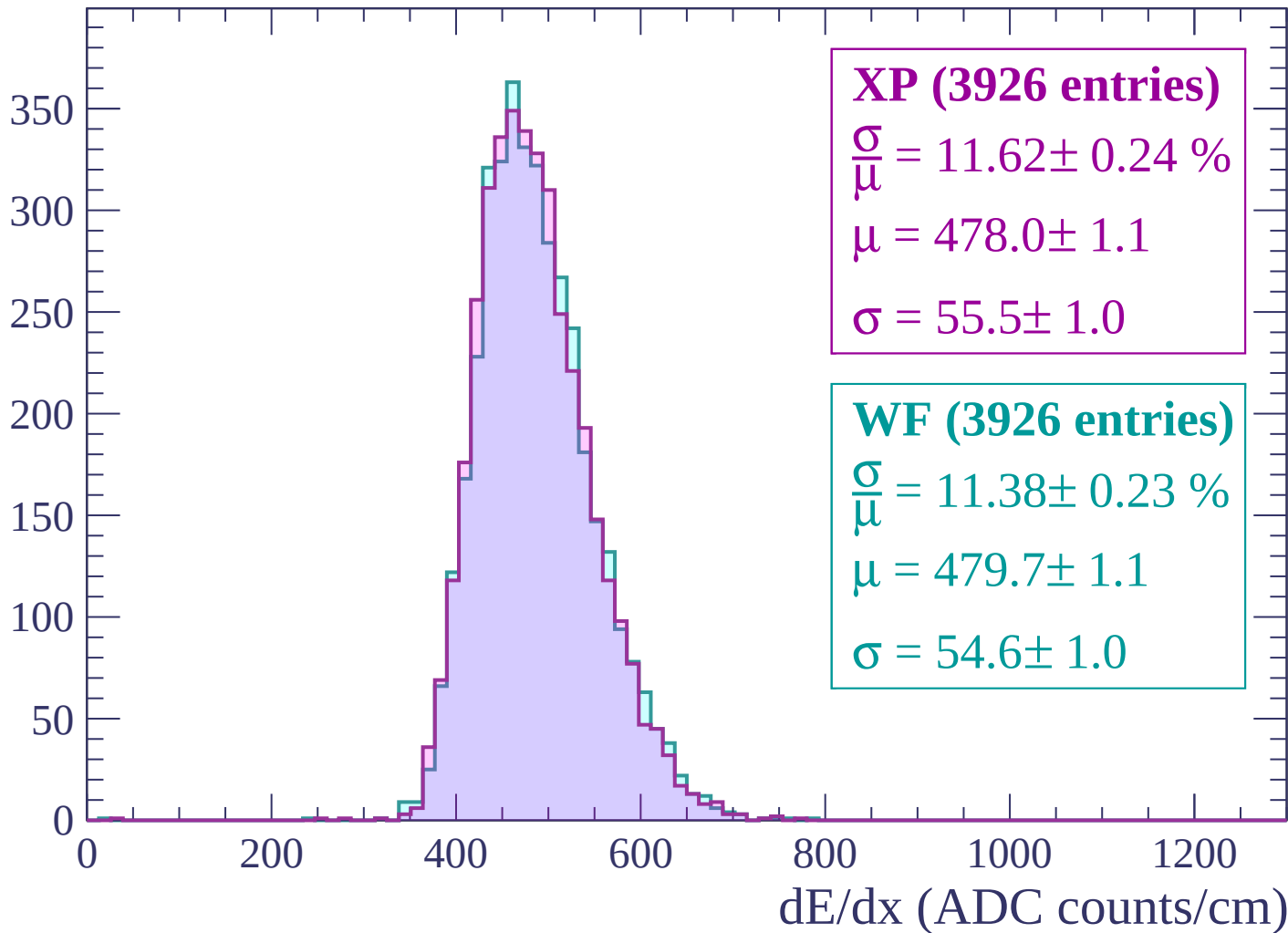
Energy loss | $-1560 < p < -1500$

Count



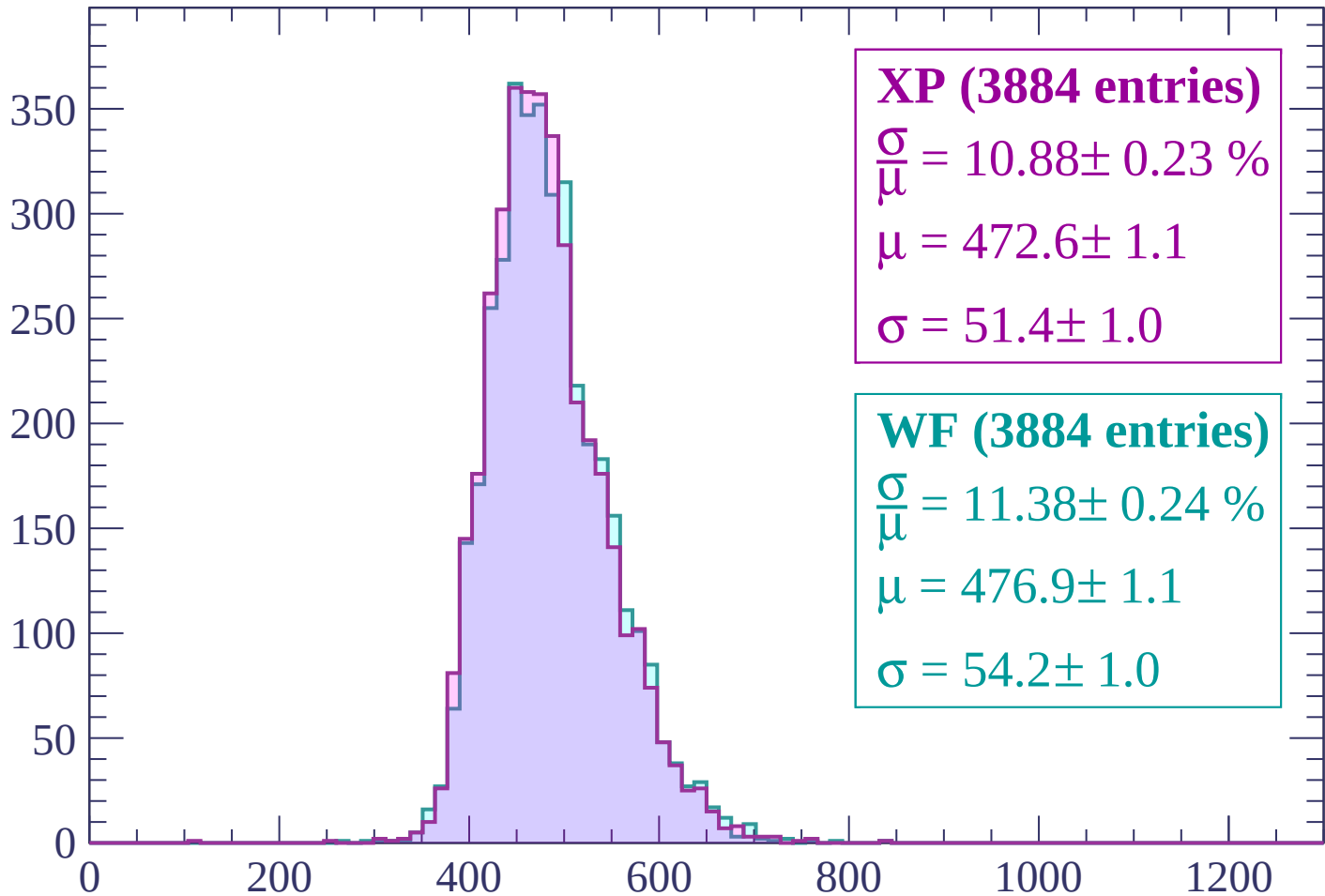
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

Count



XP (3884 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.23 \%$$

$$\mu = 472.6 \pm 1.1$$

$$\sigma = 51.4 \pm 1.0$$

WF (3884 entries)

$$\frac{\sigma}{\mu} = 11.38 \pm 0.24 \%$$

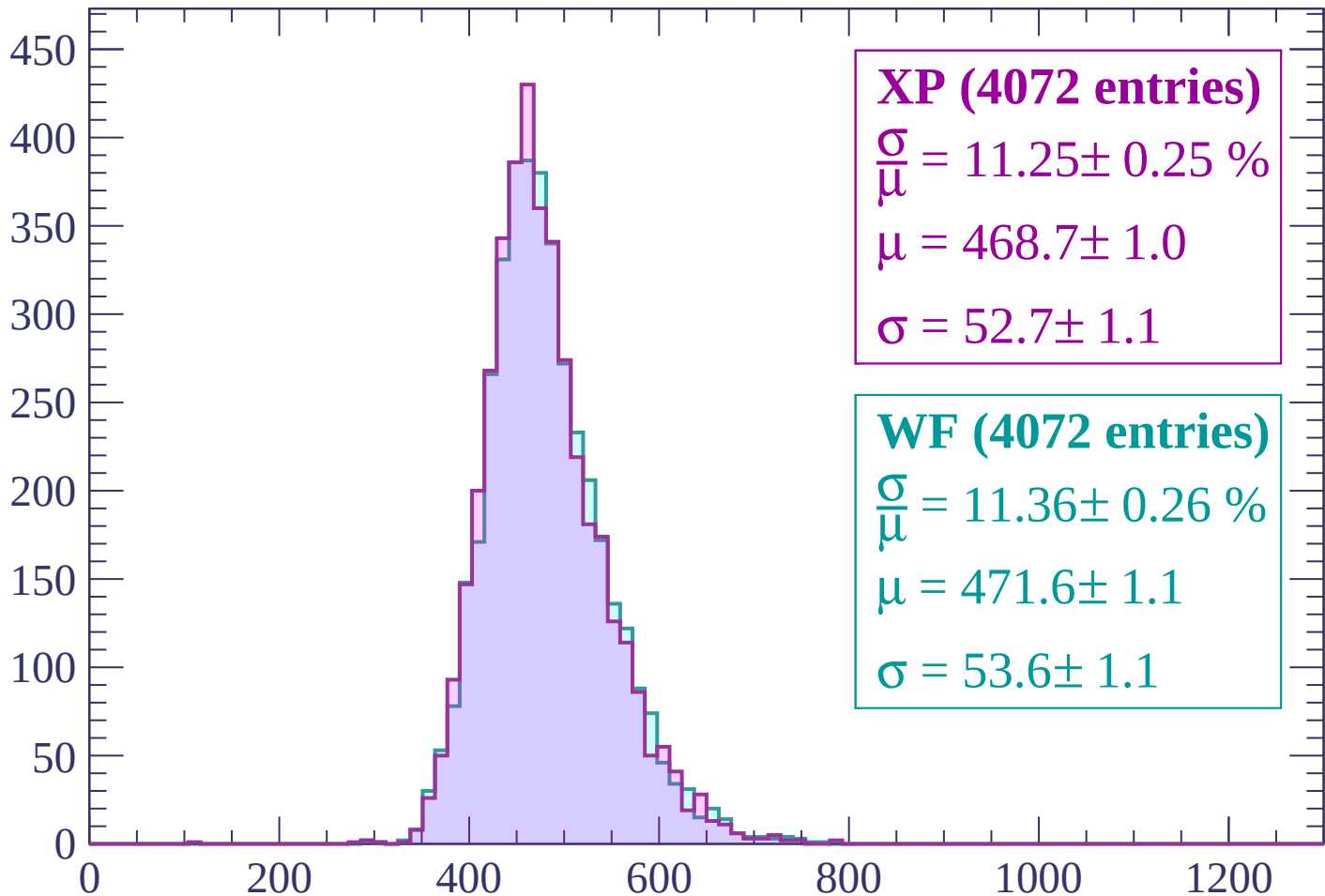
$$\mu = 476.9 \pm 1.1$$

$$\sigma = 54.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP (4072 entries)

$\frac{\sigma}{\bar{\mu}} = 11.25 \pm 0.25 \%$

$\mu = 468.7 \pm 1.0$

$\sigma = 52.7 \pm 1.1$

WF (4072 entries)

$\frac{\sigma}{\bar{\mu}} = 11.36 \pm 0.26 \%$

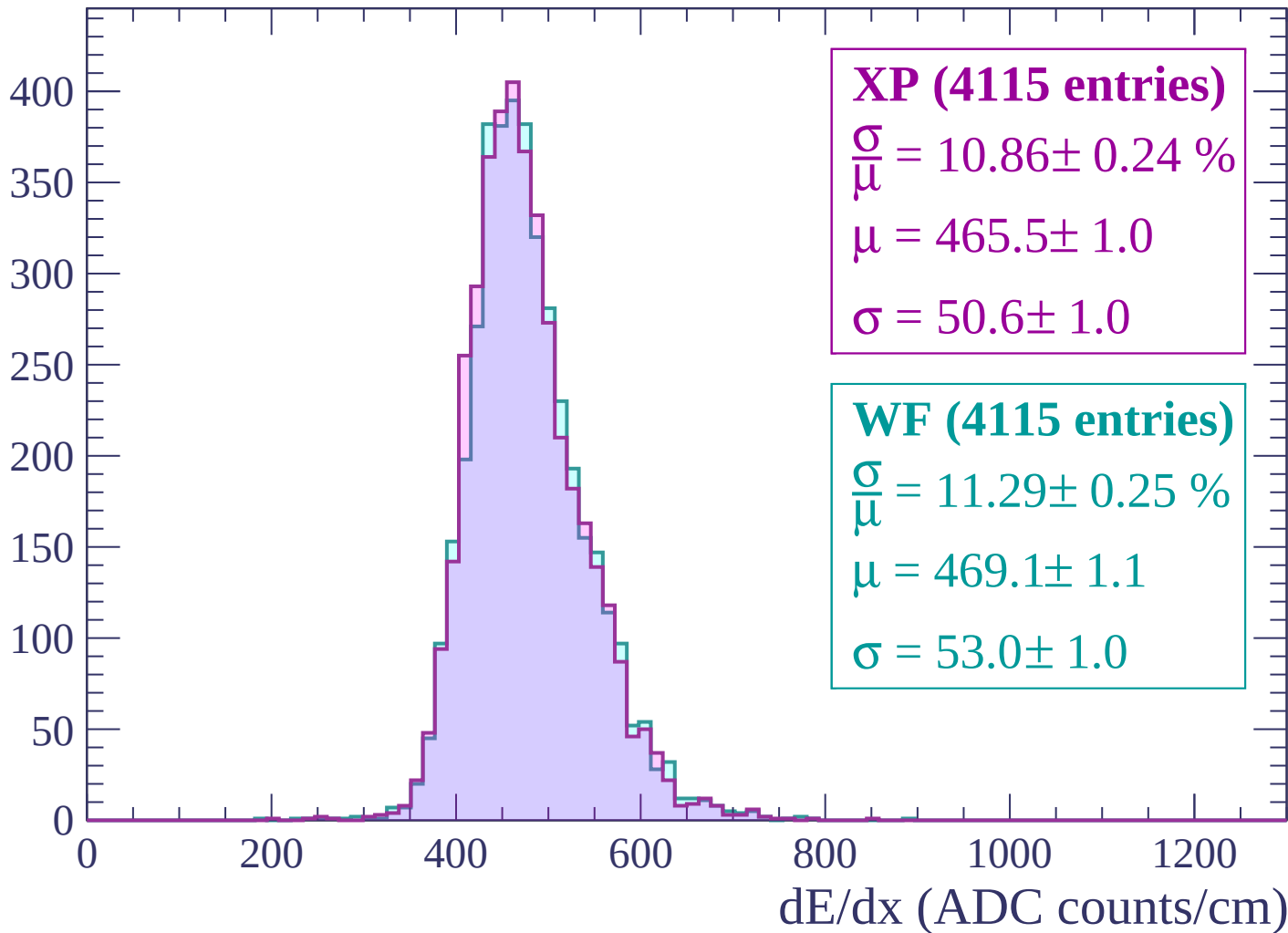
$\mu = 471.6 \pm 1.1$

$\sigma = 53.6 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

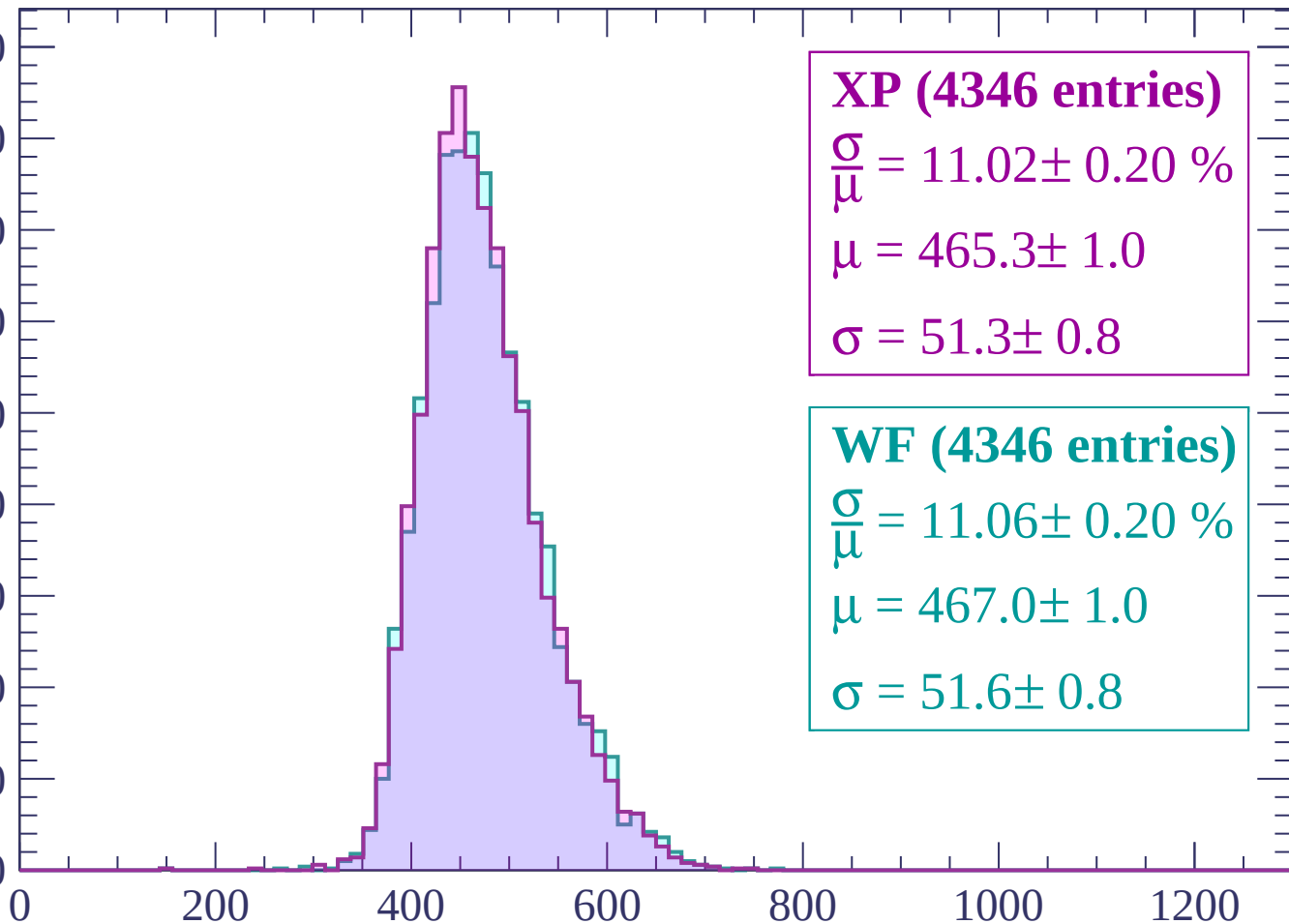
Count



Energy loss | -1260 < p < -1200

Count

450
400
350
300
250
200
150
100
50
0



XP (4346 entries)

$\frac{\sigma}{\mu} = 11.02 \pm 0.20 \%$

$\mu = 465.3 \pm 1.0$

$\sigma = 51.3 \pm 0.8$

WF (4346 entries)

$\frac{\sigma}{\mu} = 11.06 \pm 0.20 \%$

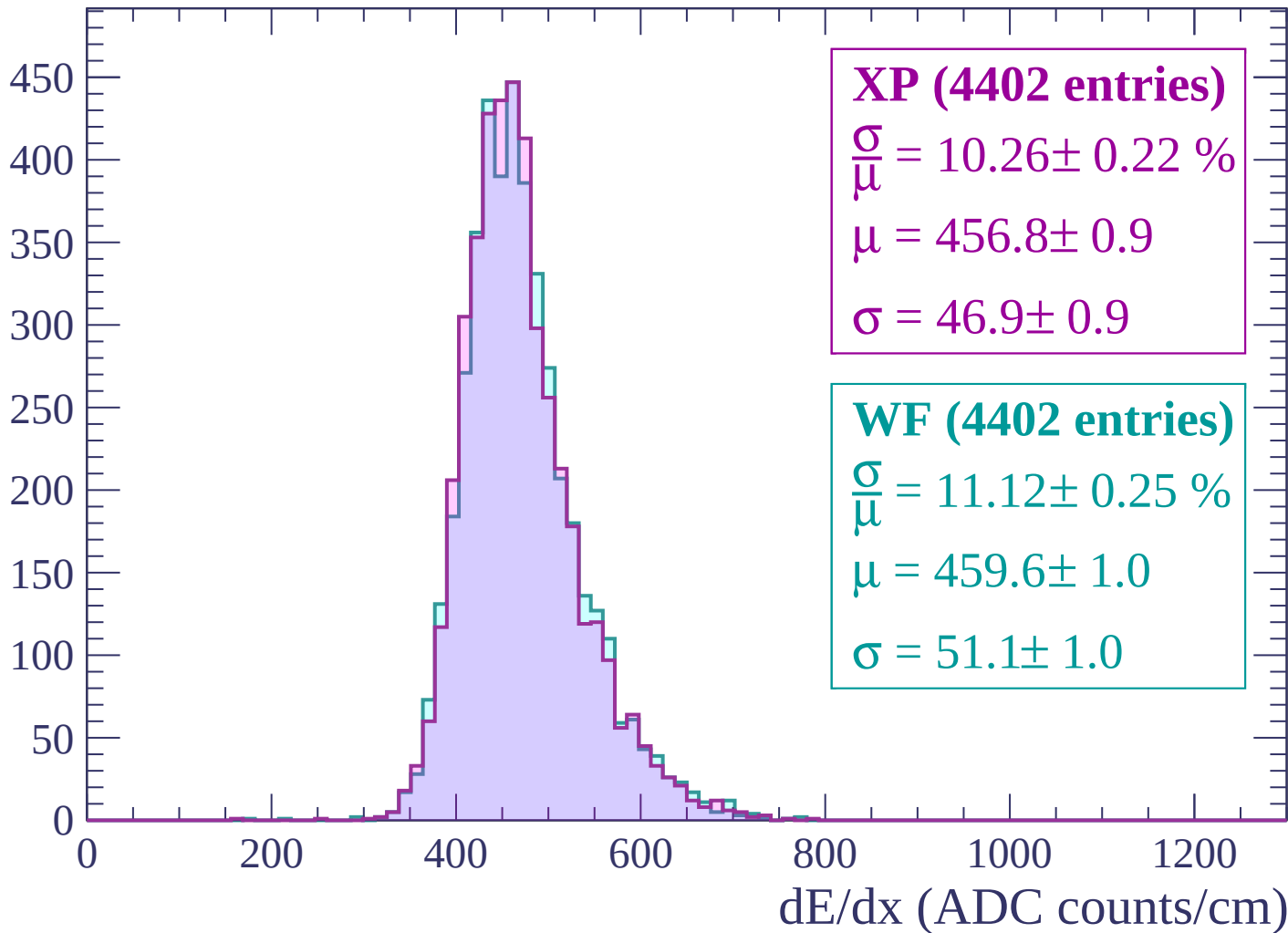
$\mu = 467.0 \pm 1.0$

$\sigma = 51.6 \pm 0.8$

dE/dx (ADC counts/cm)

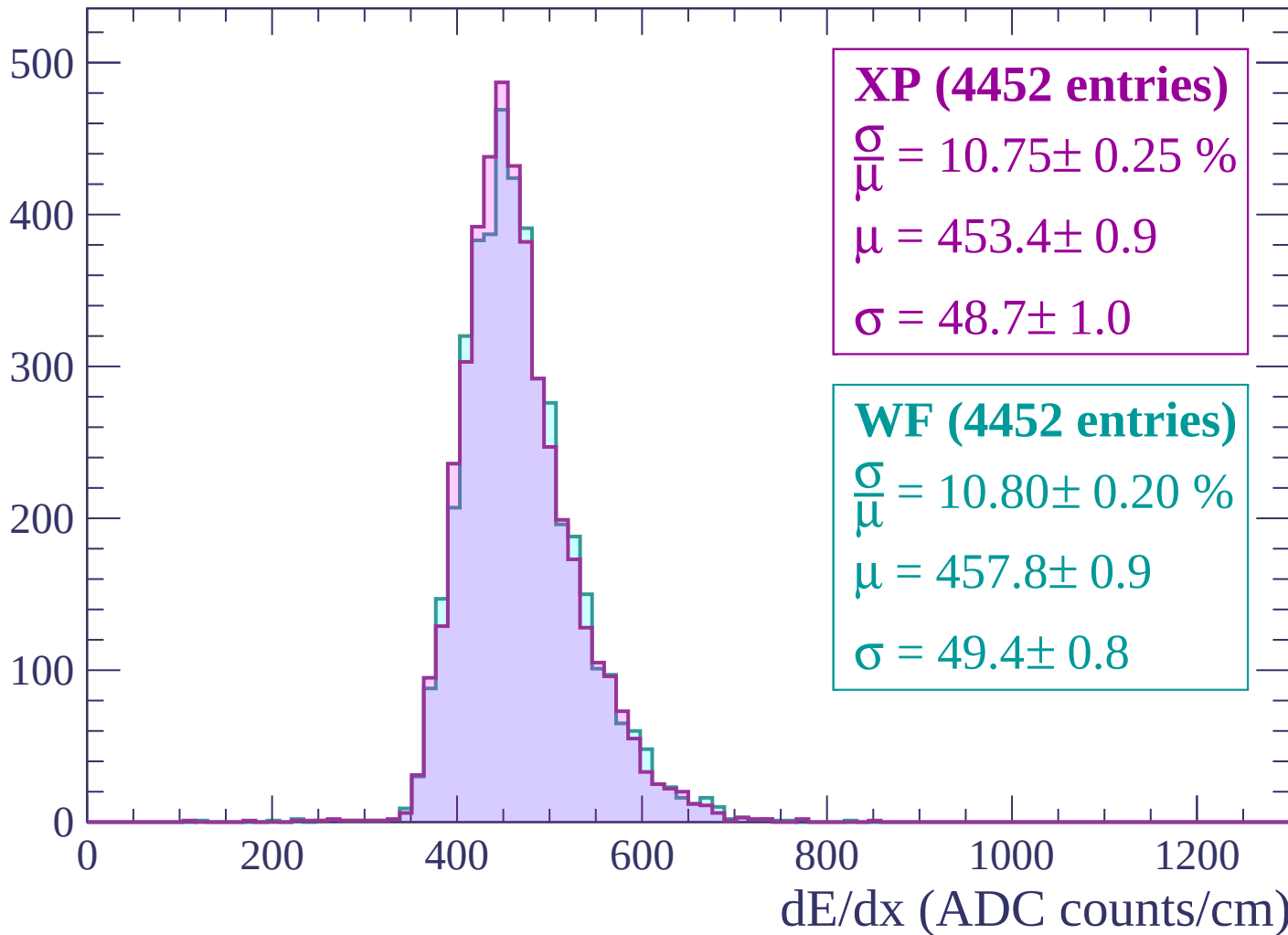
Energy loss | $-1200 < p < -1140$

Count



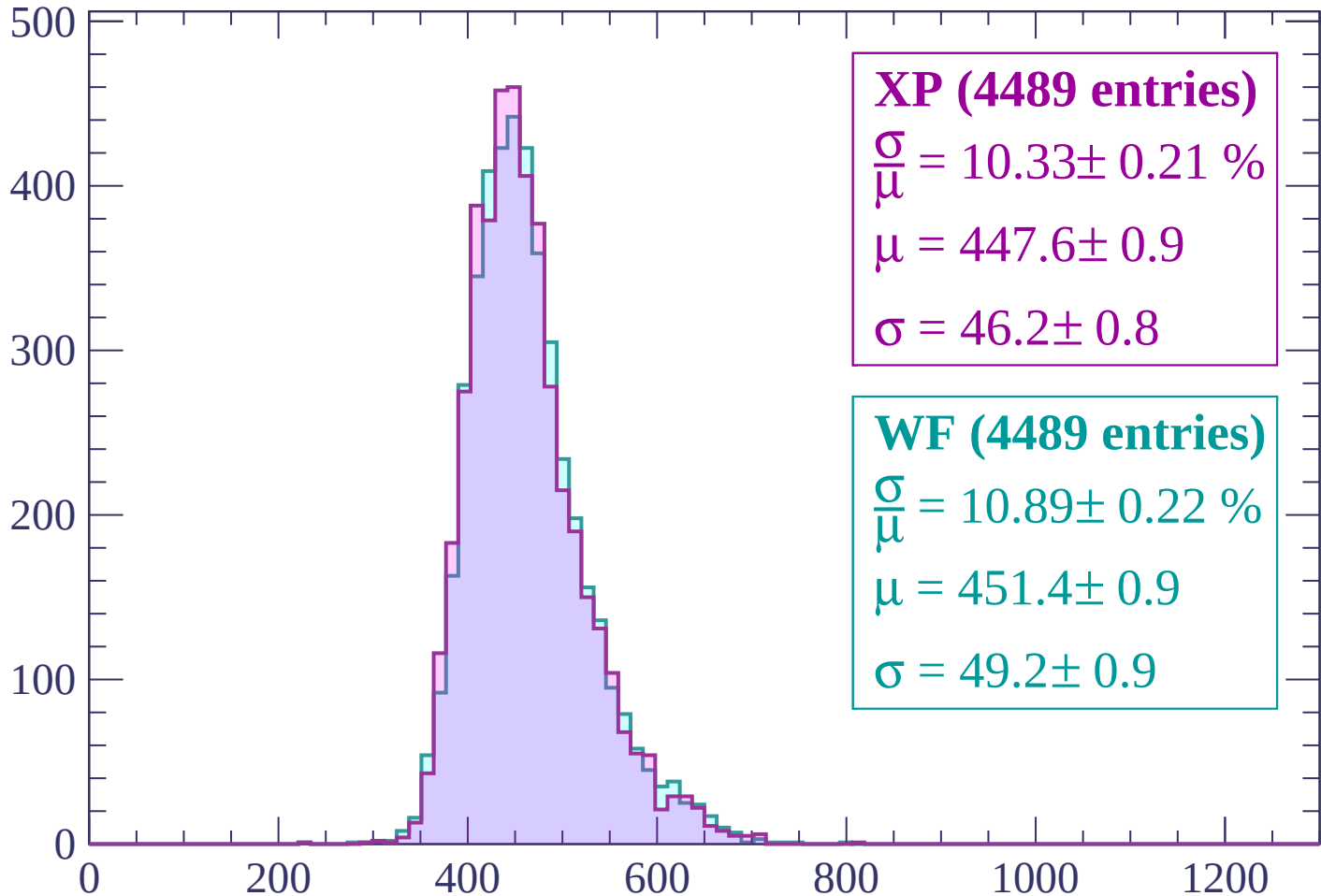
Energy loss | $-1140 < p < -1080$

Count



Energy loss | $-1080 < p < -1020$

Count



XP (4489 entries)

$\frac{\sigma}{\mu} = 10.33 \pm 0.21 \%$

$\mu = 447.6 \pm 0.9$

$\sigma = 46.2 \pm 0.8$

WF (4489 entries)

$\frac{\sigma}{\mu} = 10.89 \pm 0.22 \%$

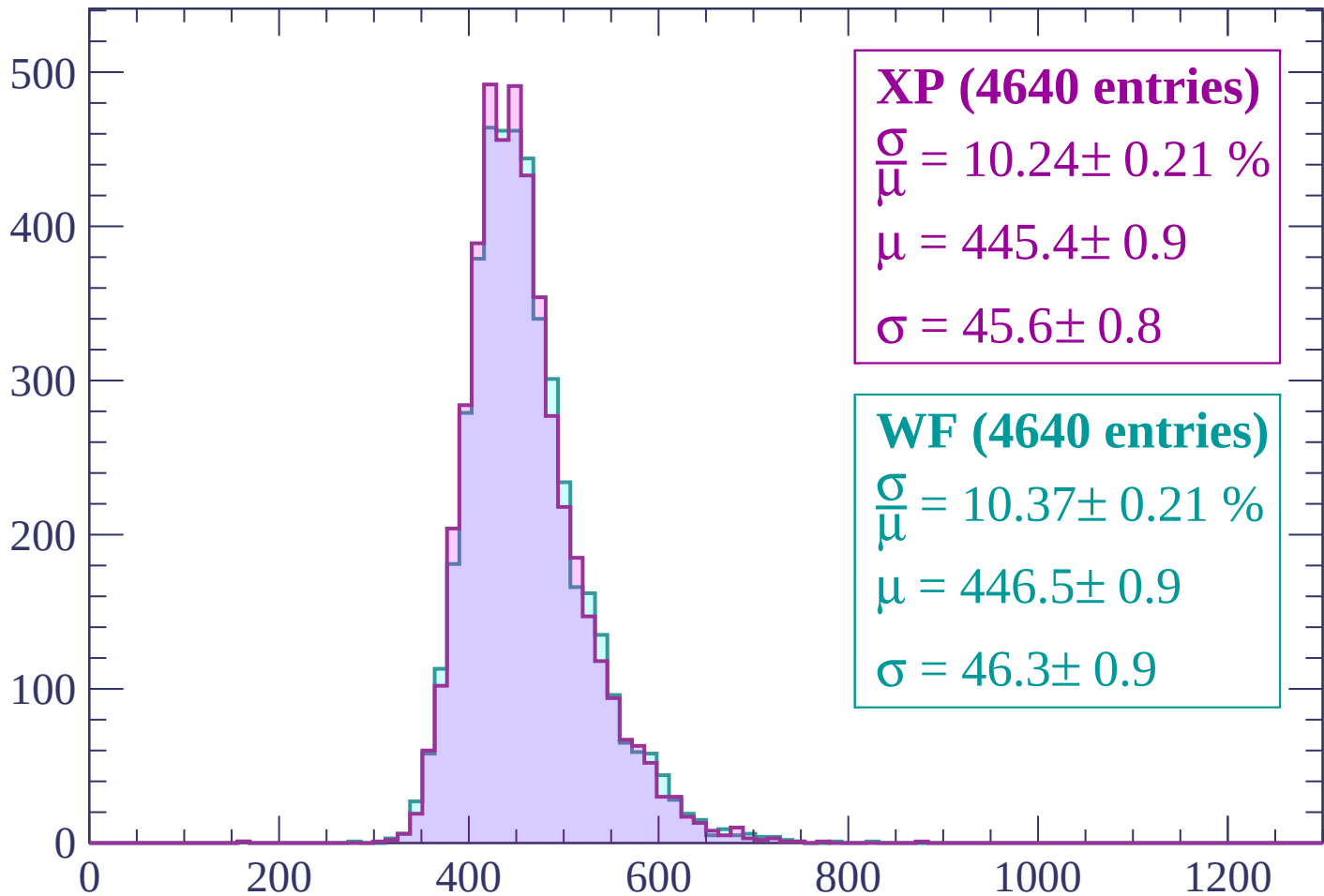
$\mu = 451.4 \pm 0.9$

$\sigma = 49.2 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (4640 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.21 \%$$

$$\mu = 445.4 \pm 0.9$$

$$\sigma = 45.6 \pm 0.8$$

WF (4640 entries)

$$\frac{\sigma}{\mu} = 10.37 \pm 0.21 \%$$

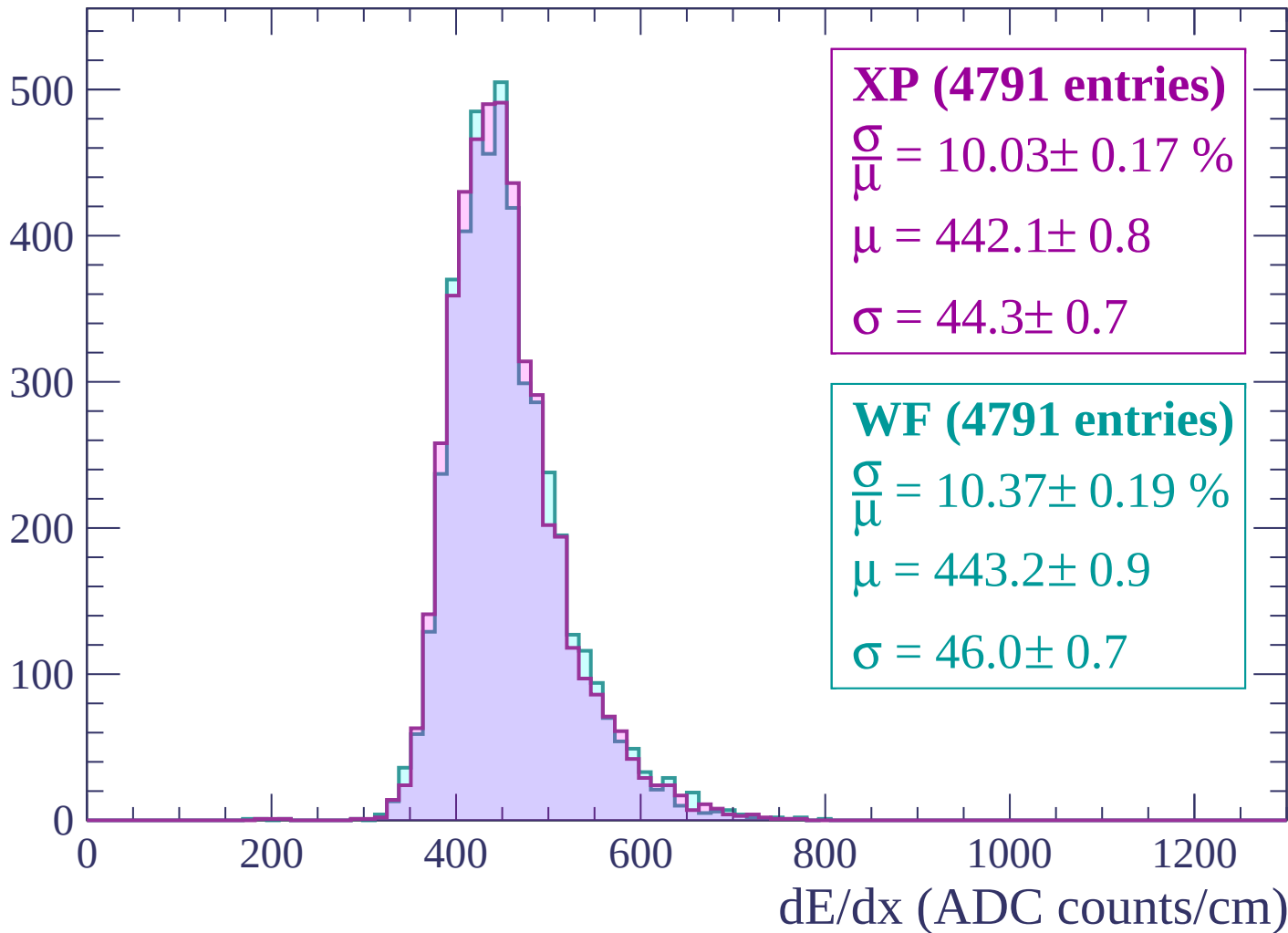
$$\mu = 446.5 \pm 0.9$$

$$\sigma = 46.3 \pm 0.9$$

dE/dx (ADC counts/cm)

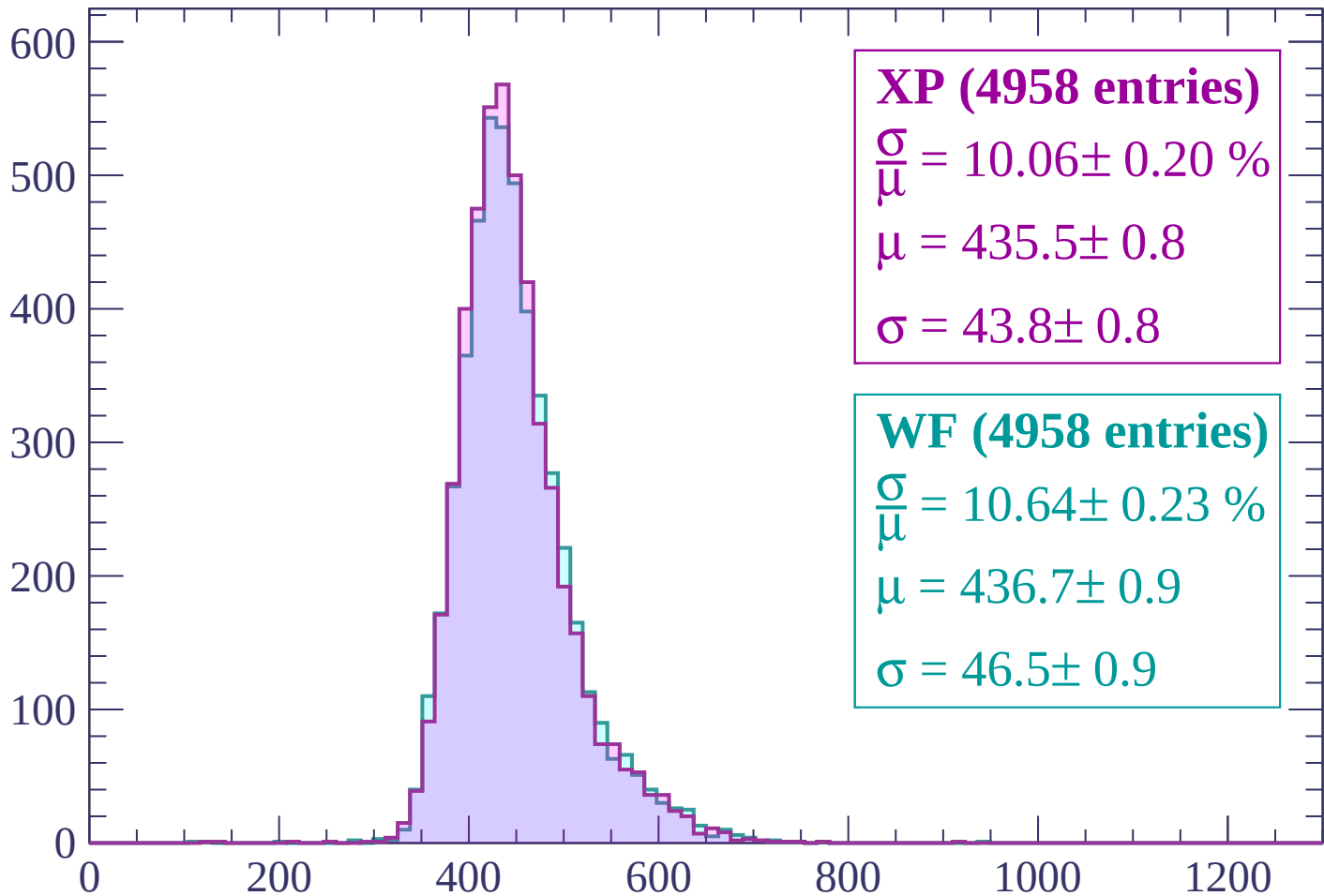
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP (4958 entries)

$\frac{\sigma}{\mu} = 10.06 \pm 0.20 \%$

$\mu = 435.5 \pm 0.8$

$\sigma = 43.8 \pm 0.8$

WF (4958 entries)

$\frac{\sigma}{\mu} = 10.64 \pm 0.23 \%$

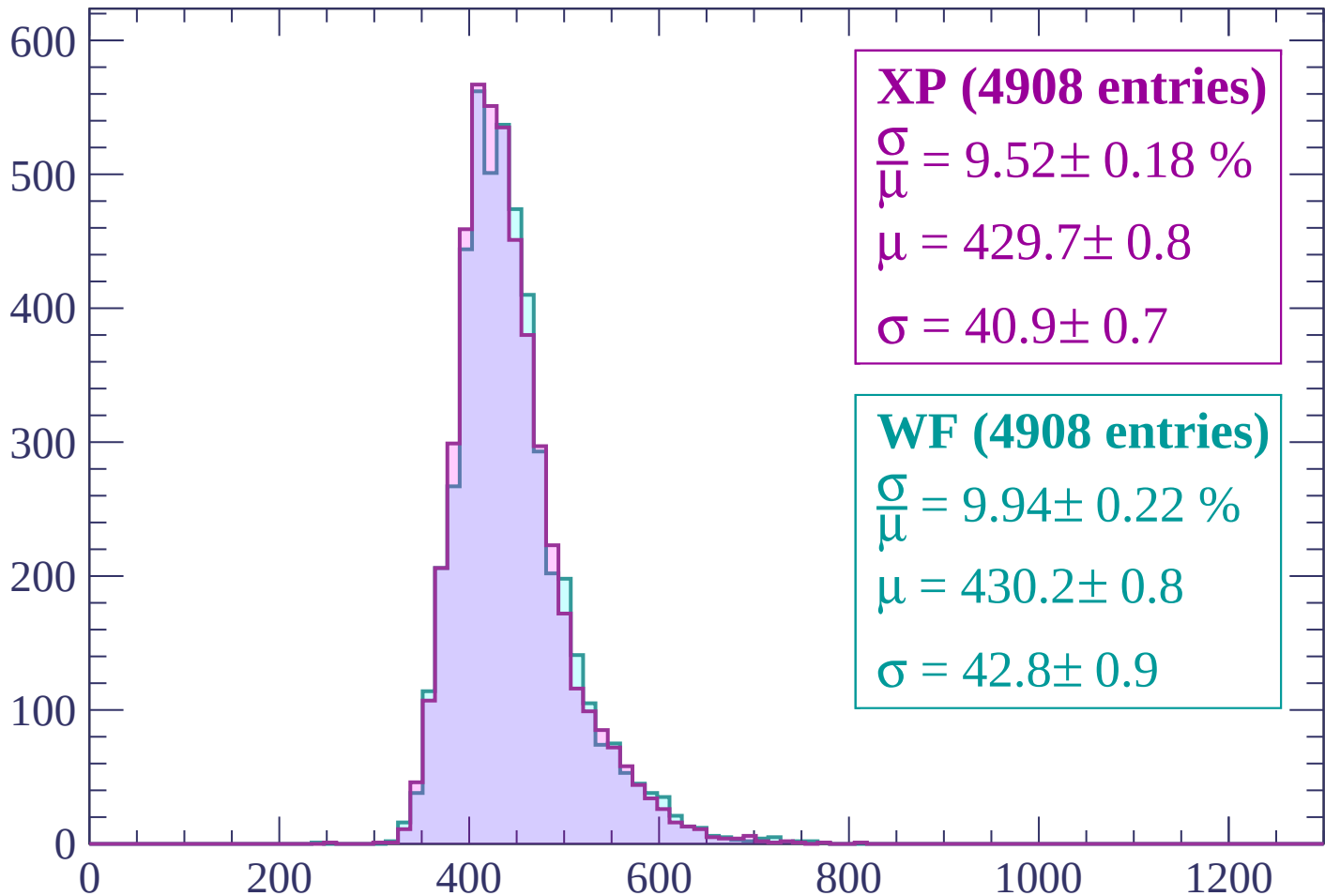
$\mu = 436.7 \pm 0.9$

$\sigma = 46.5 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (4908 entries)

$\frac{\sigma}{\mu} = 9.52 \pm 0.18 \%$

$\mu = 429.7 \pm 0.8$

$\sigma = 40.9 \pm 0.7$

WF (4908 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.22 \%$

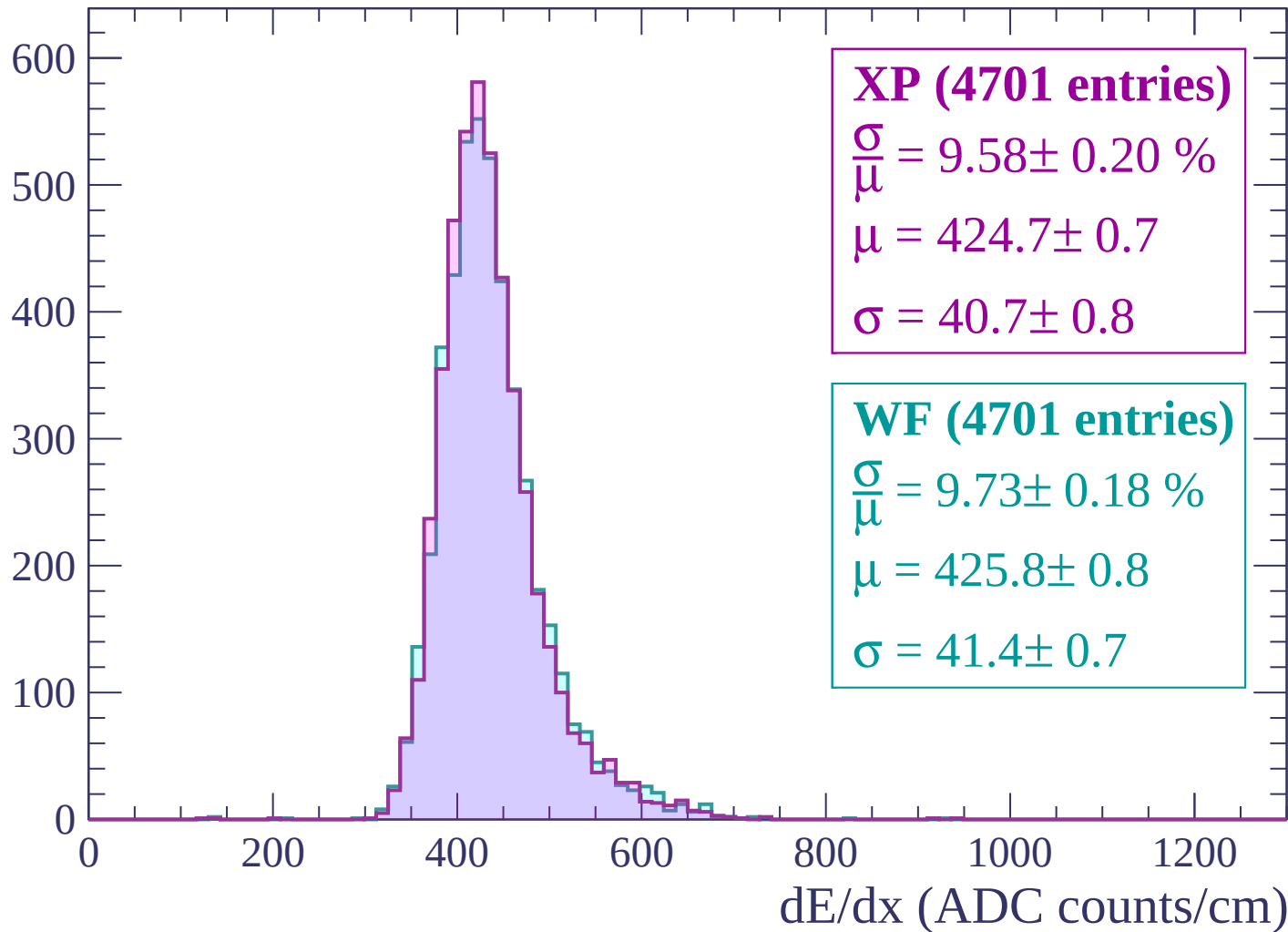
$\mu = 430.2 \pm 0.8$

$\sigma = 42.8 \pm 0.9$

dE/dx (ADC counts/cm)

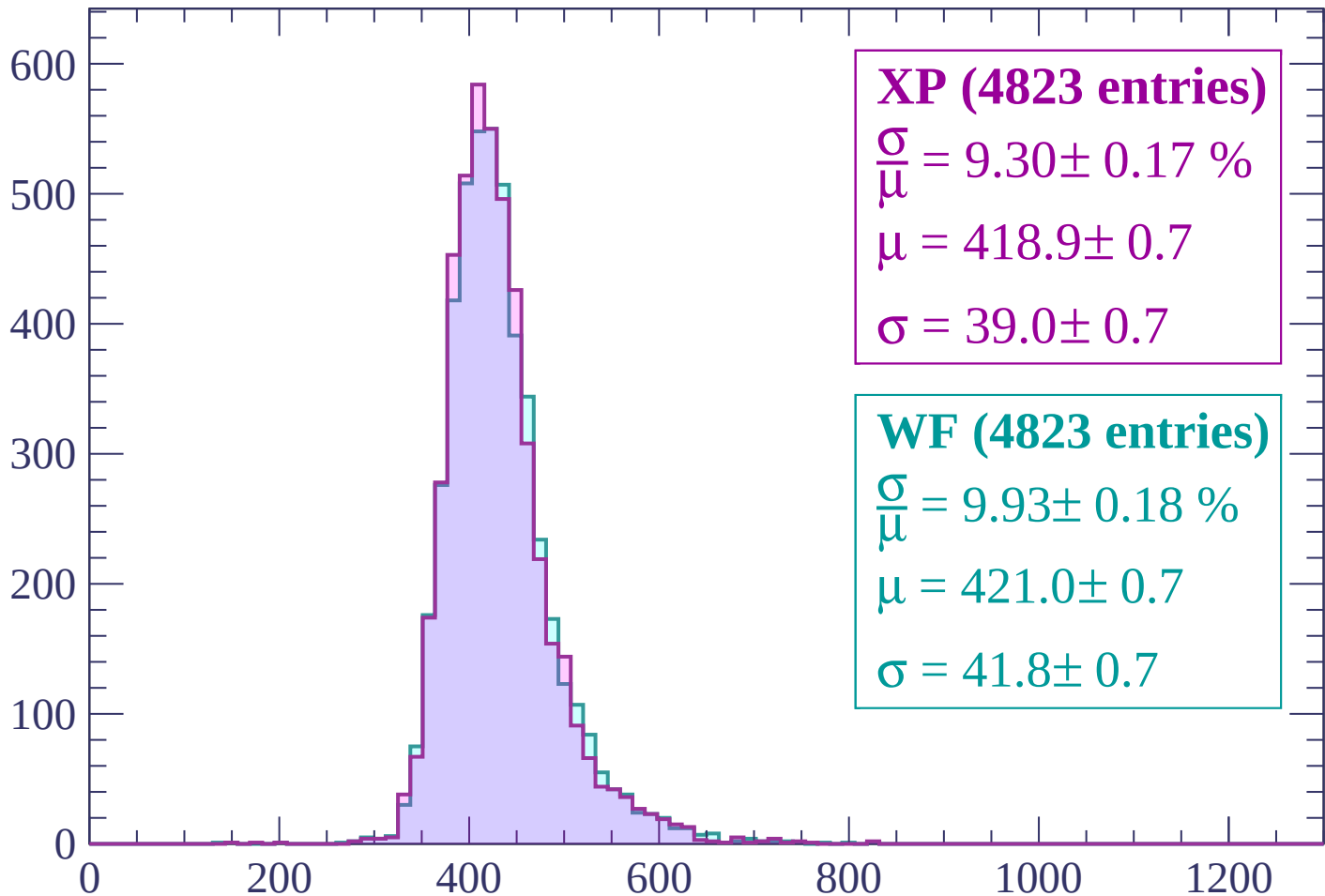
Energy loss | $-780 < p < -720$

Count



Energy loss | $-720 < p < -660$

Count



XP (4823 entries)

$\frac{\sigma}{\mu} = 9.30 \pm 0.17 \%$

$\mu = 418.9 \pm 0.7$

$\sigma = 39.0 \pm 0.7$

WF (4823 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.18 \%$

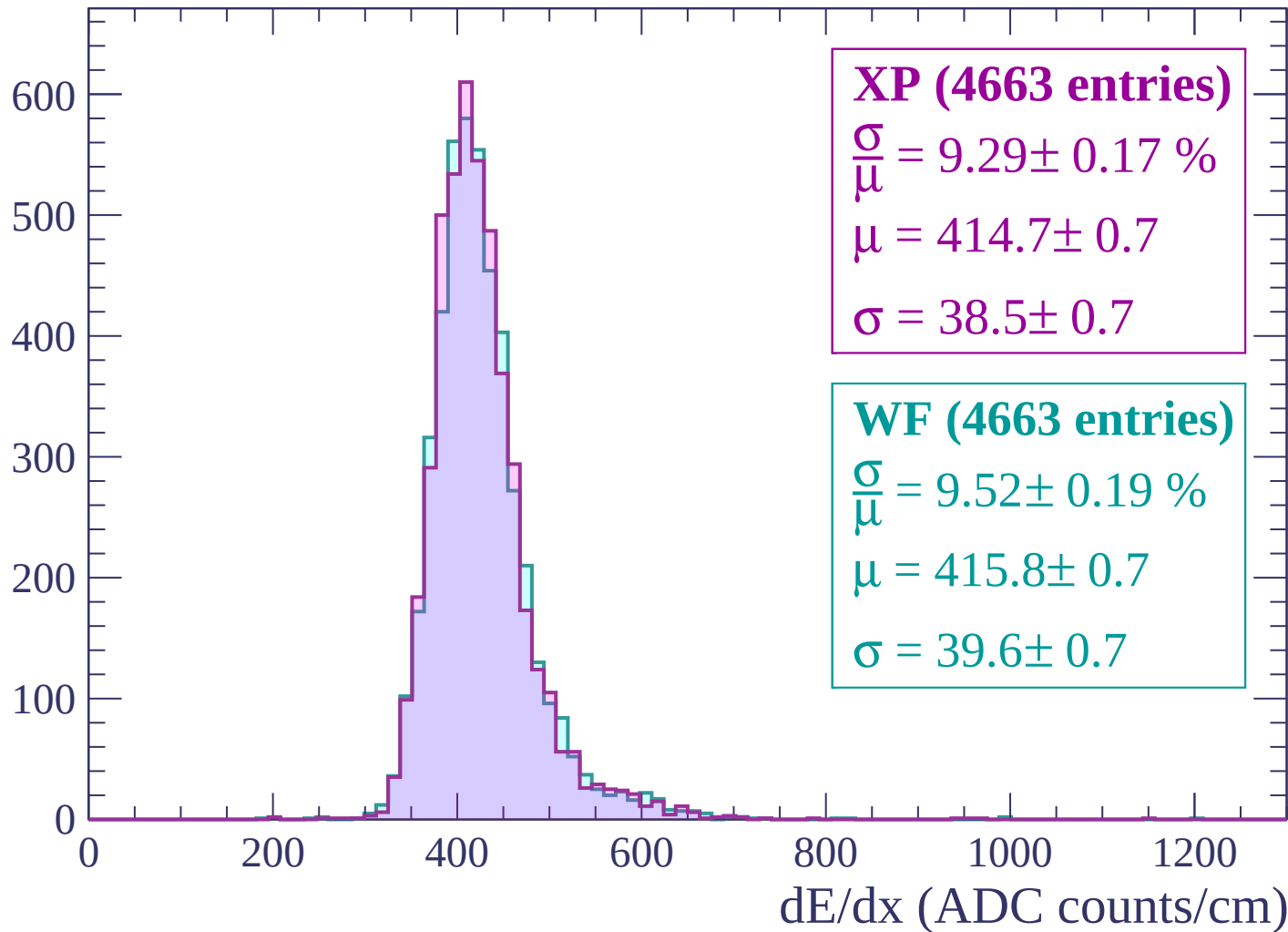
$\mu = 421.0 \pm 0.7$

$\sigma = 41.8 \pm 0.7$

dE/dx (ADC counts/cm)

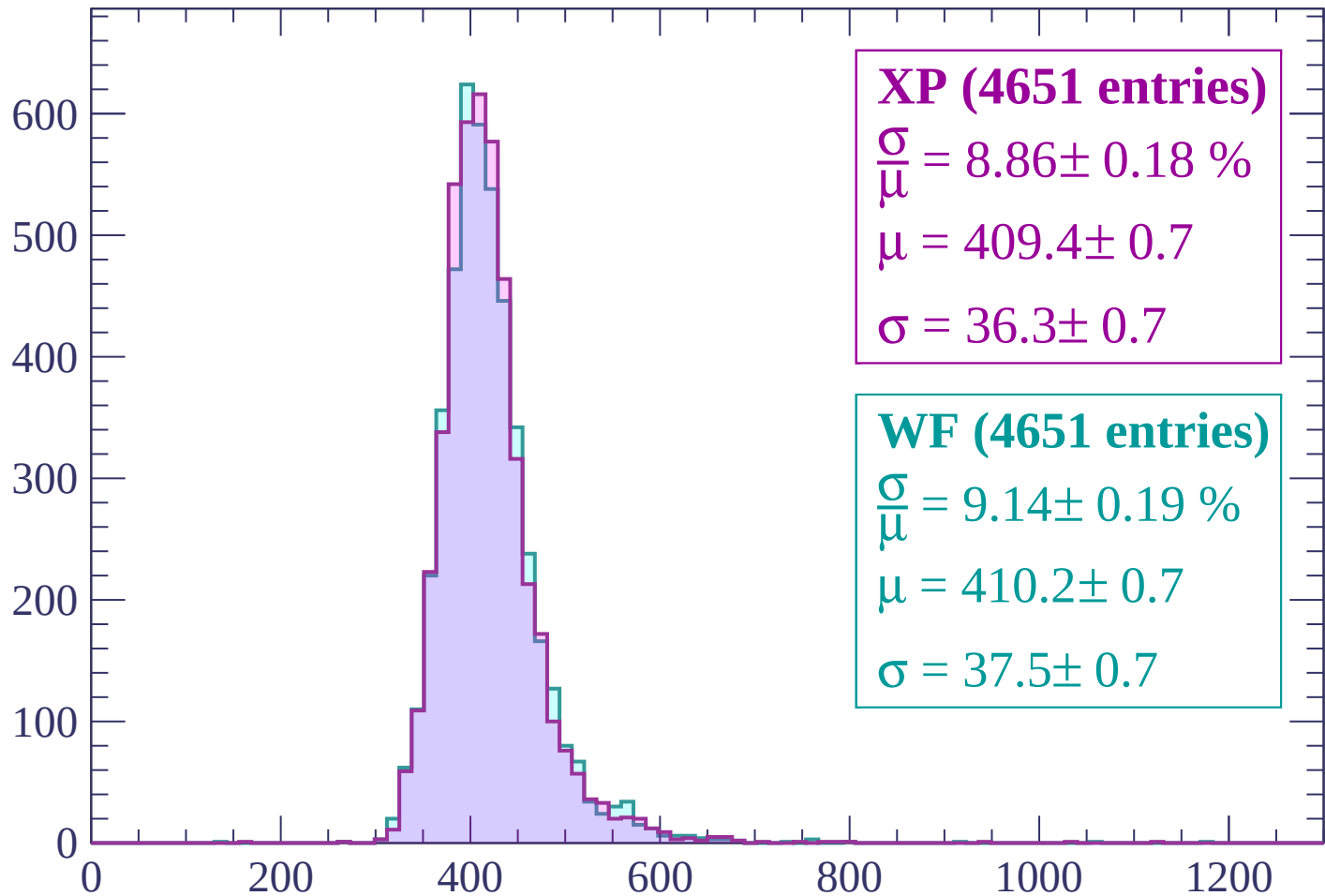
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP (4651 entries)

$$\frac{\sigma}{\mu} = 8.86 \pm 0.18 \%$$

$$\mu = 409.4 \pm 0.7$$

$$\sigma = 36.3 \pm 0.7$$

WF (4651 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.19 \%$$

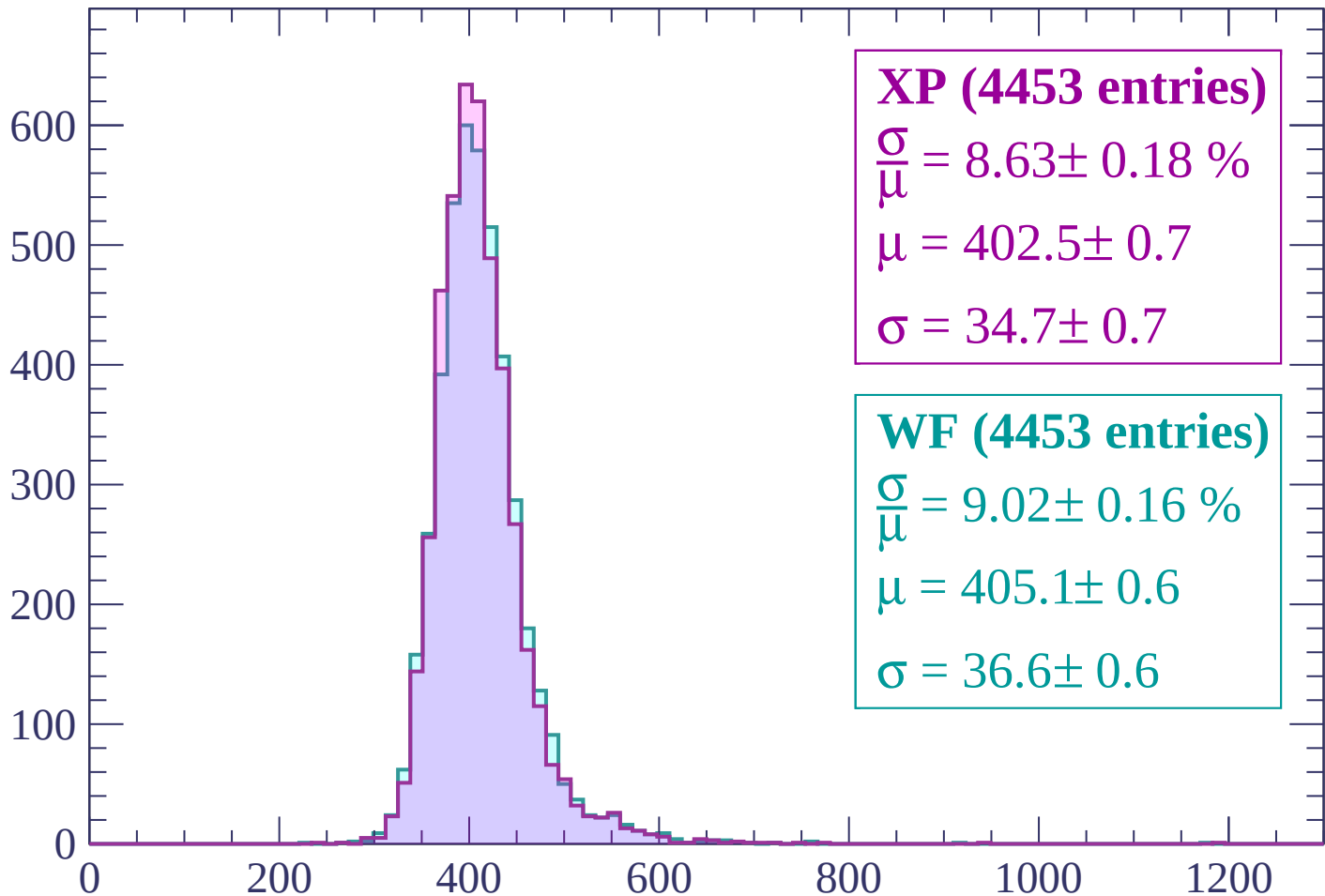
$$\mu = 410.2 \pm 0.7$$

$$\sigma = 37.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (4453 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.18 \%$$

$$\mu = 402.5 \pm 0.7$$

$$\sigma = 34.7 \pm 0.7$$

WF (4453 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$$

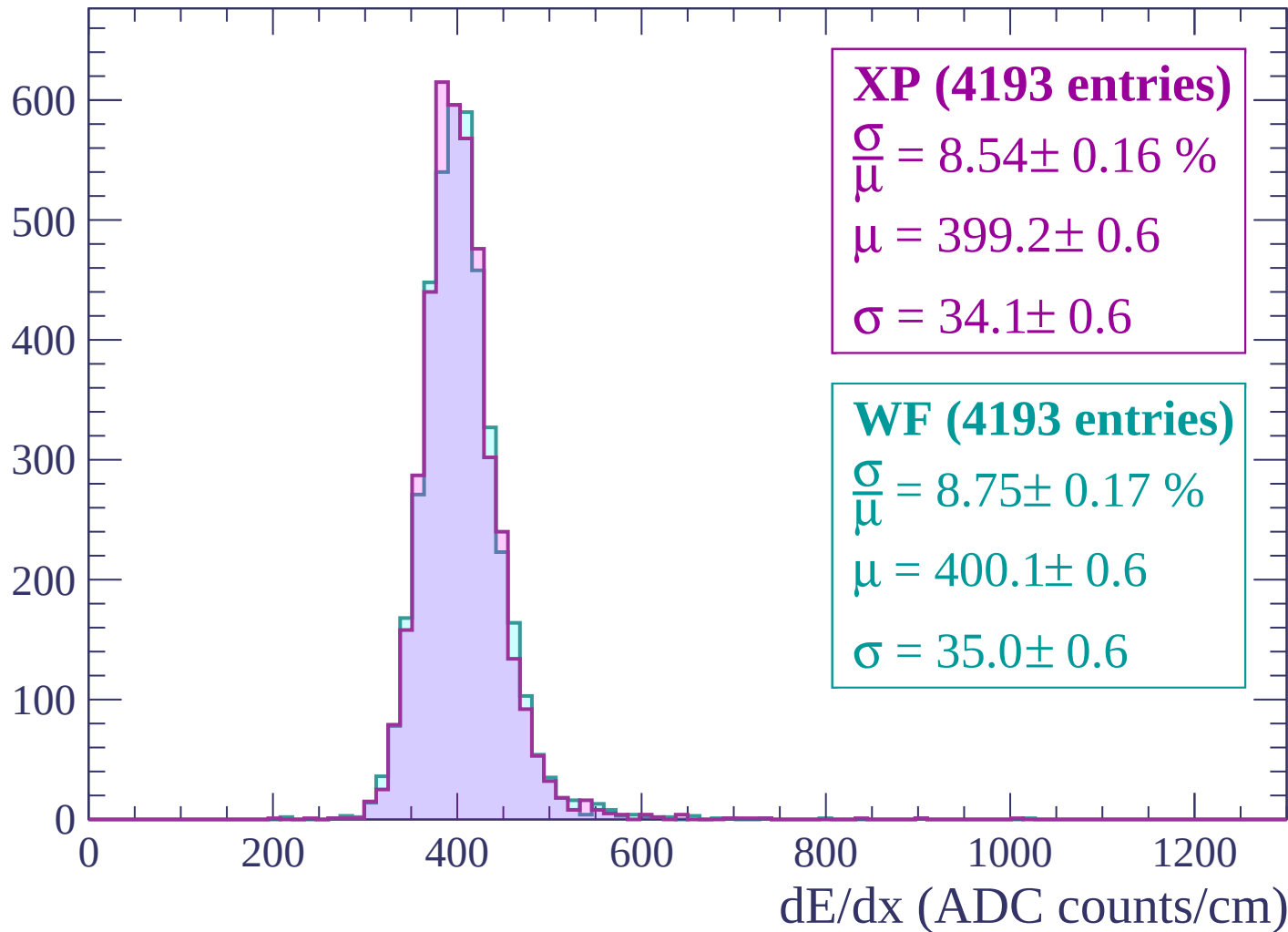
$$\mu = 405.1 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



Energy loss | $-420 < p < -360$

Count

600

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (3945 entries)

$\frac{\sigma}{\mu} = 8.84 \pm 0.17 \%$

$\mu = 395.7 \pm 0.6$

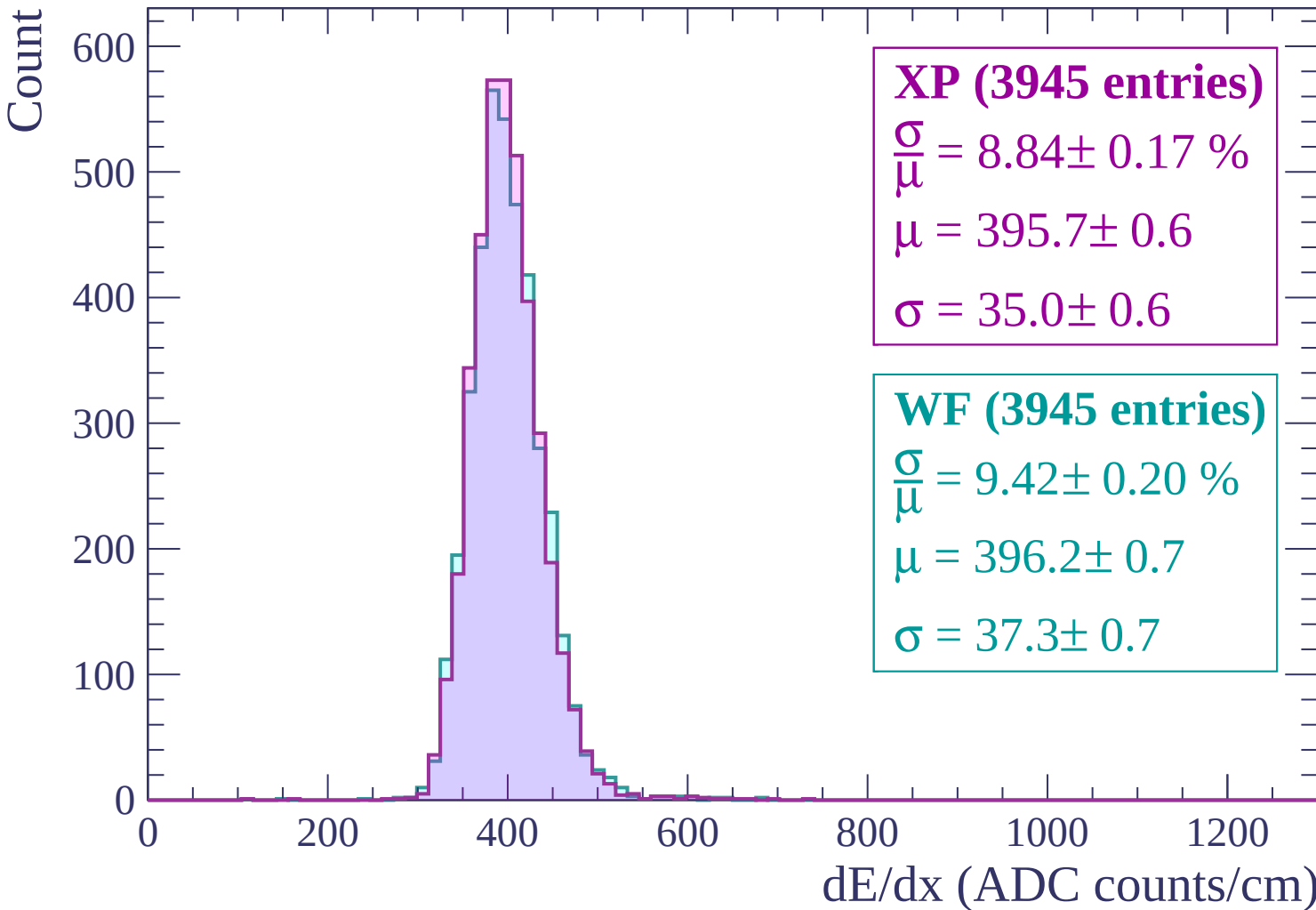
$\sigma = 35.0 \pm 0.6$

WF (3945 entries)

$\frac{\sigma}{\mu} = 9.42 \pm 0.20 \%$

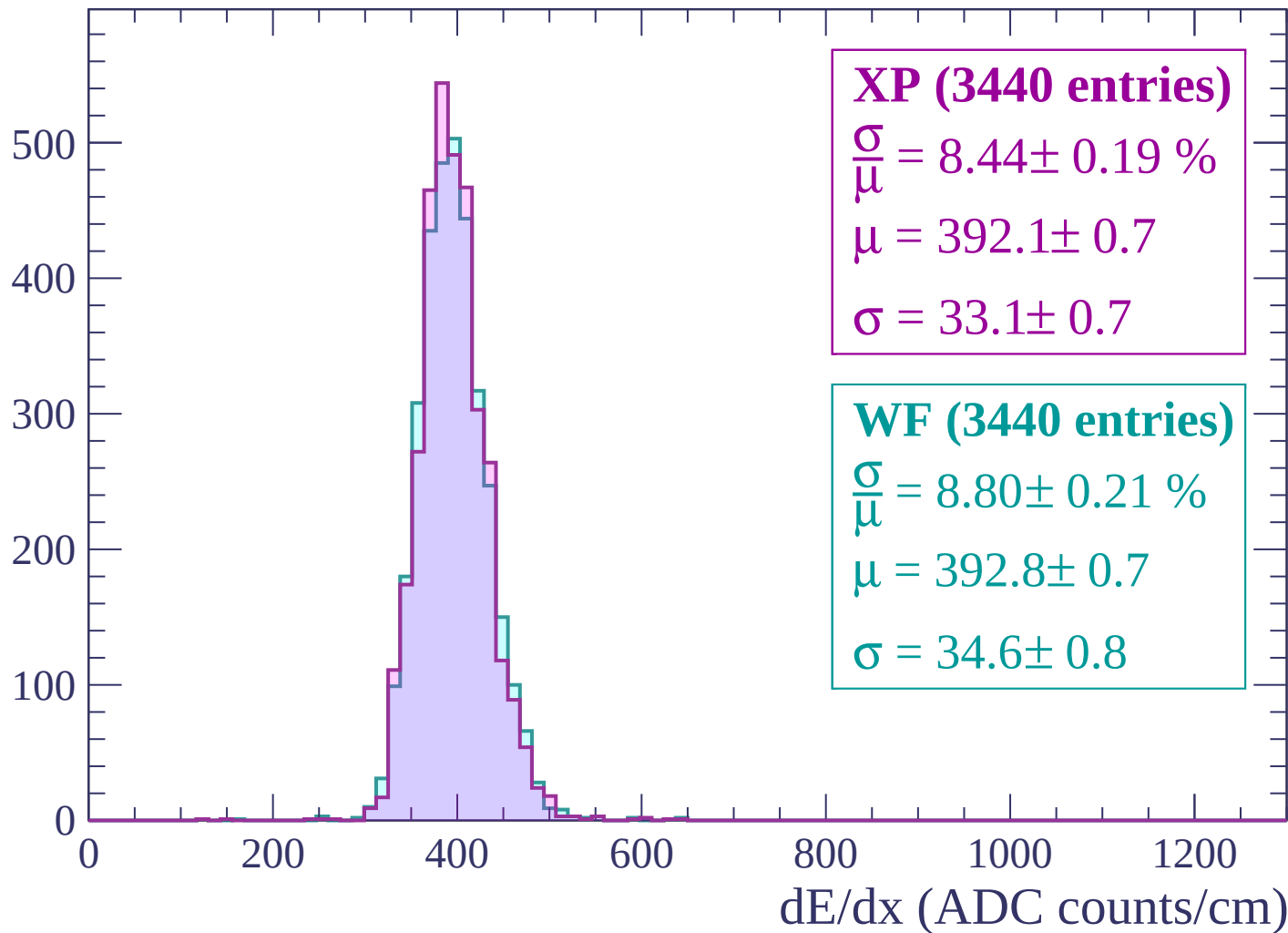
$\mu = 396.2 \pm 0.7$

$\sigma = 37.3 \pm 0.7$



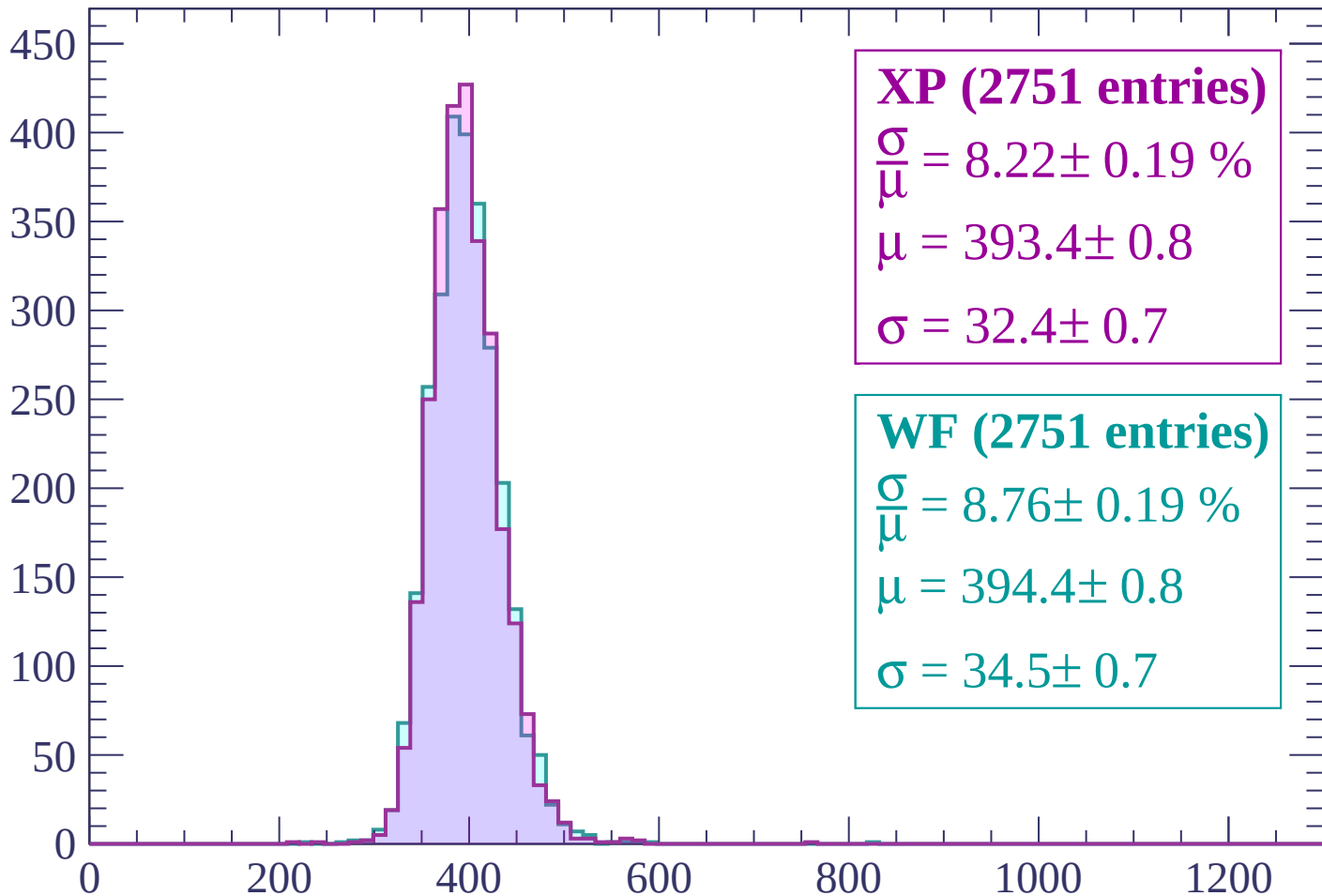
Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count



XP (2751 entries)

$$\frac{\sigma}{\mu} = 8.22 \pm 0.19 \%$$

$$\mu = 393.4 \pm 0.8$$

$$\sigma = 32.4 \pm 0.7$$

WF (2751 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.19 \%$$

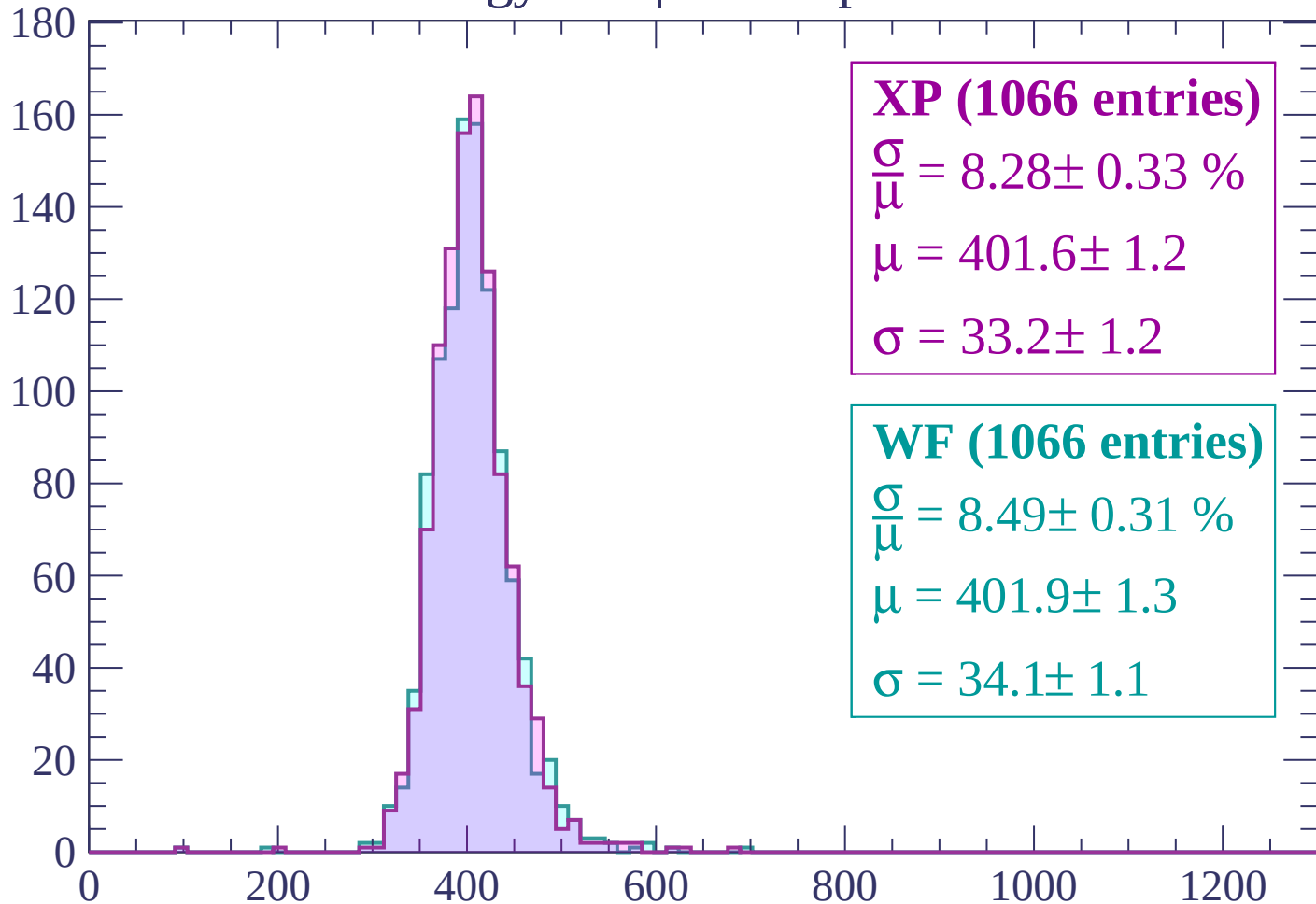
$$\mu = 394.4 \pm 0.8$$

$$\sigma = 34.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (1066 entries)

$$\frac{\sigma}{\mu} = 8.28 \pm 0.33 \%$$

$$\mu = 401.6 \pm 1.2$$

$$\sigma = 33.2 \pm 1.2$$

WF (1066 entries)

$$\frac{\sigma}{\mu} = 8.49 \pm 0.31 \%$$

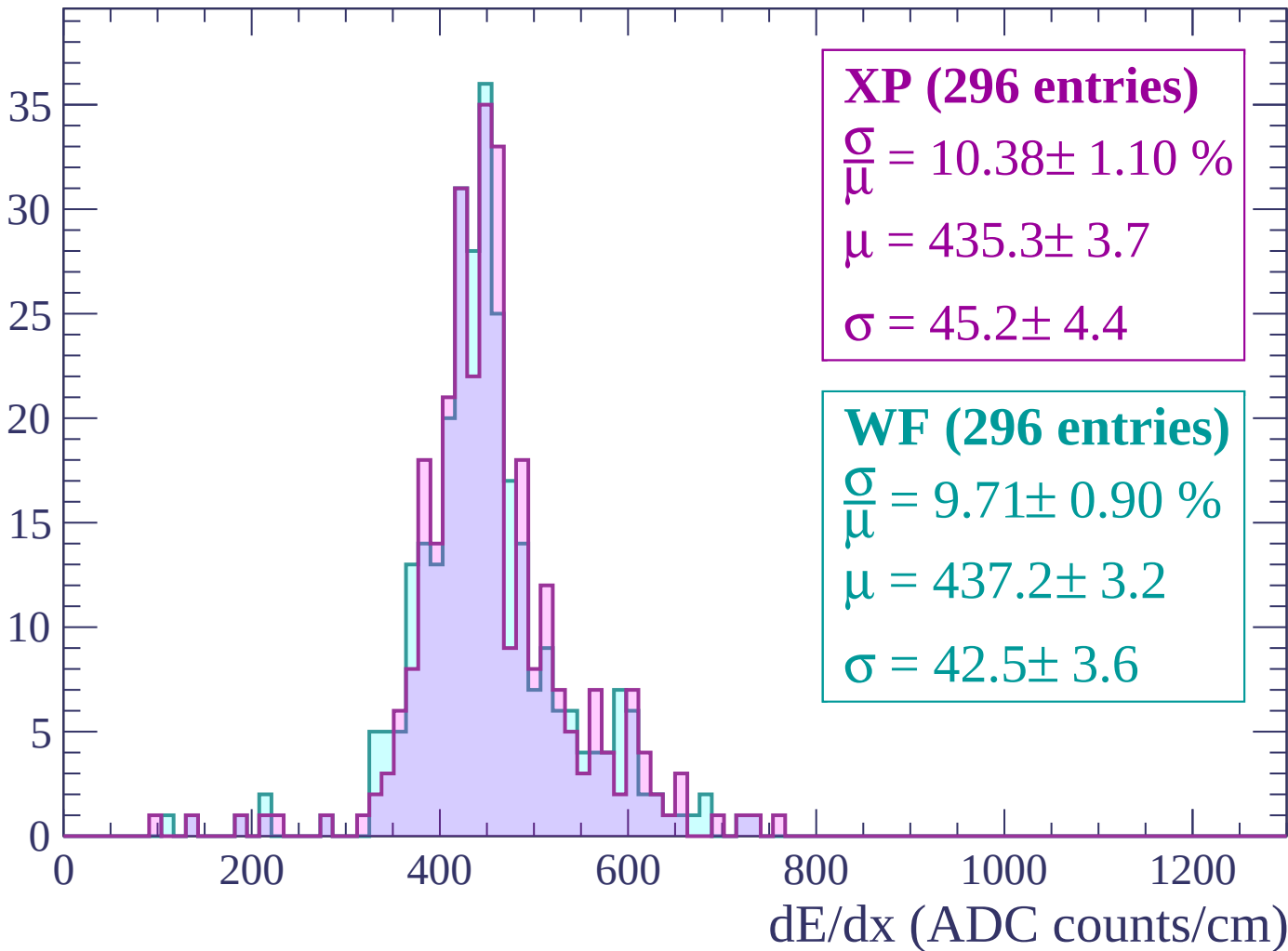
$$\mu = 401.9 \pm 1.3$$

$$\sigma = 34.1 \pm 1.1$$

dE/dx (ADC counts/cm)

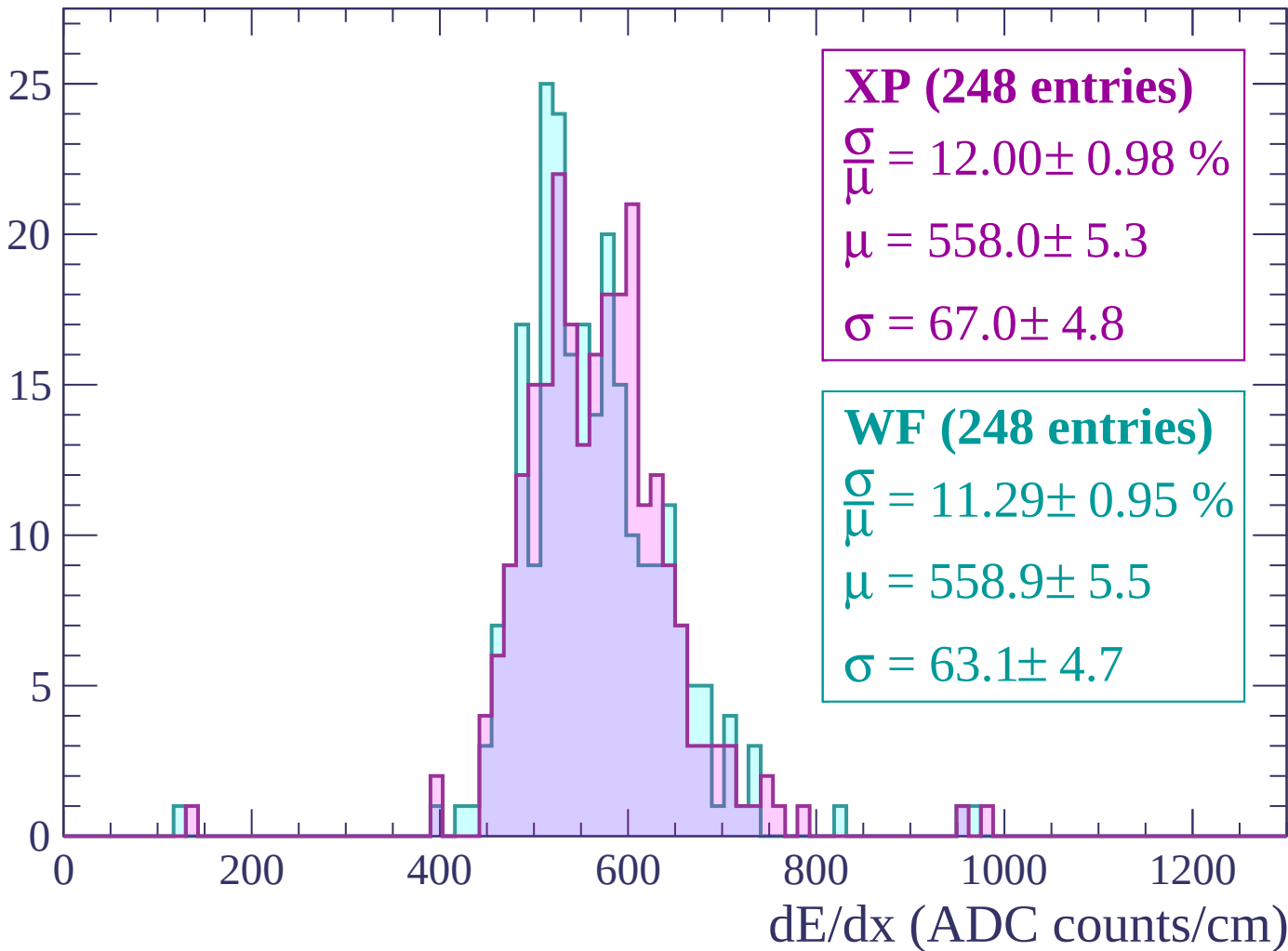
Energy loss | $-180 < p < -120$

Count



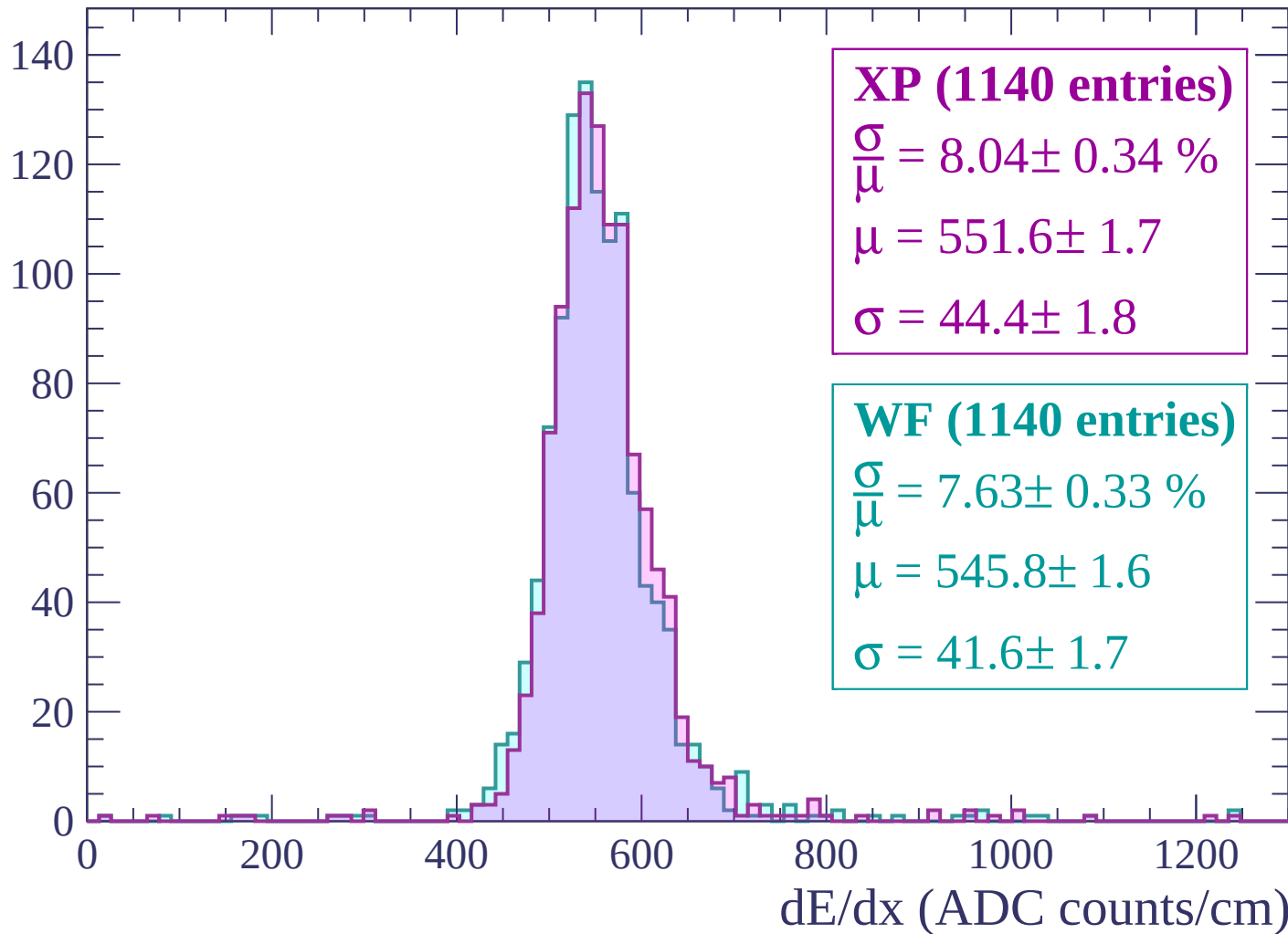
Energy loss | $-120 < p < -60$

Count



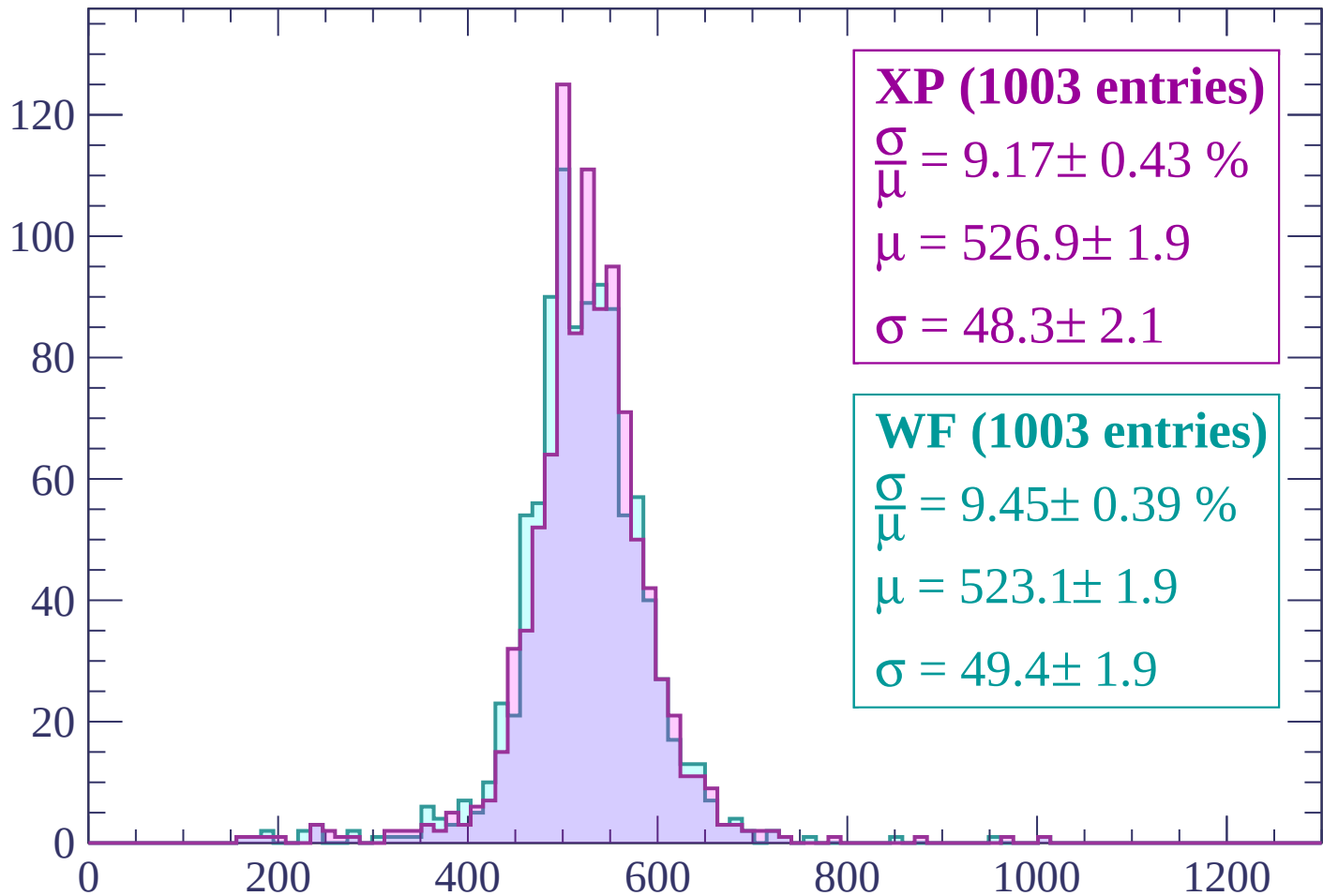
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP (1003 entries)

$$\frac{\sigma}{\mu} = 9.17 \pm 0.43 \%$$

$$\mu = 526.9 \pm 1.9$$

$$\sigma = 48.3 \pm 2.1$$

WF (1003 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.39 \%$$

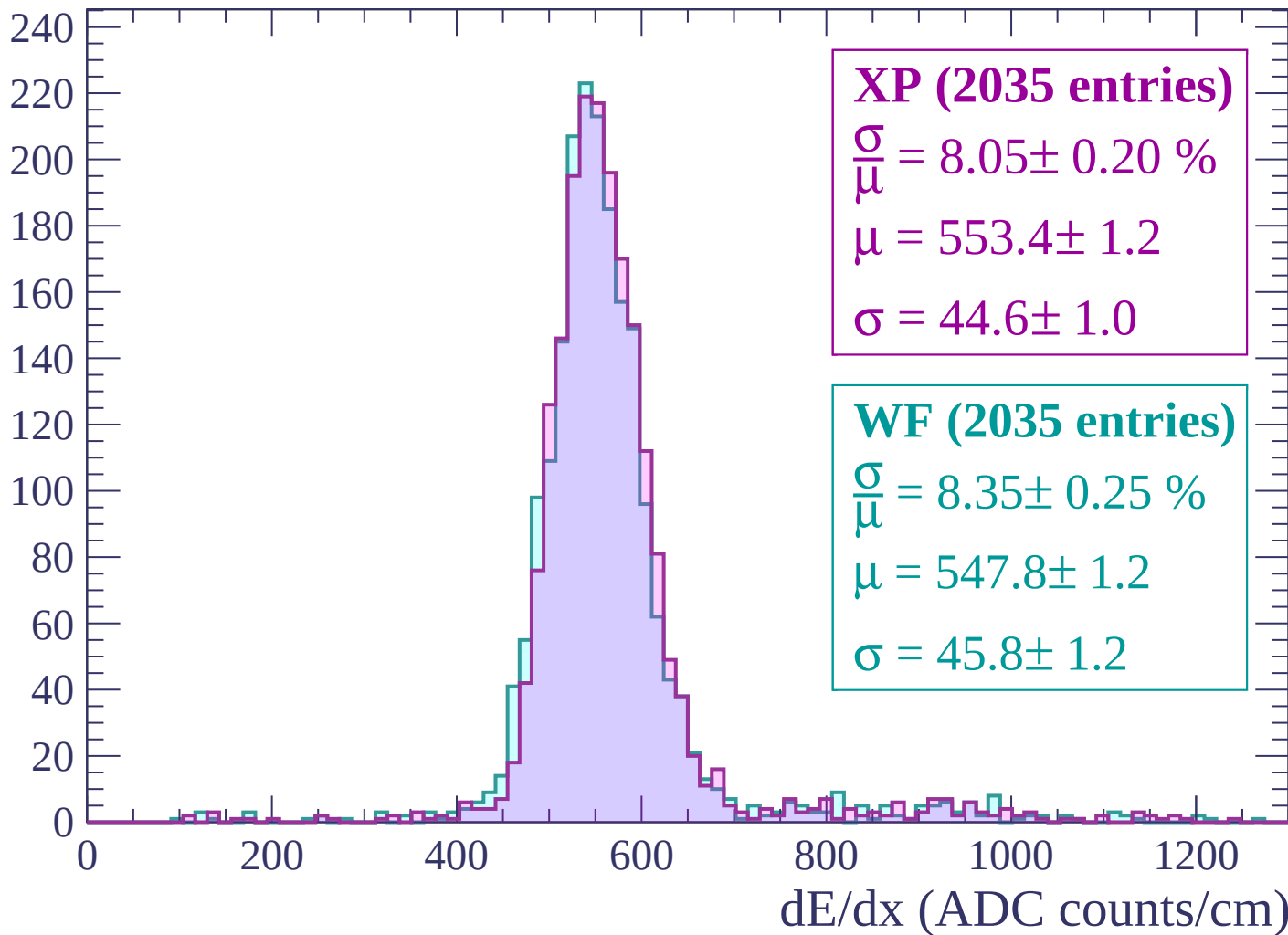
$$\mu = 523.1 \pm 1.9$$

$$\sigma = 49.4 \pm 1.9$$

dE/dx (ADC counts/cm)

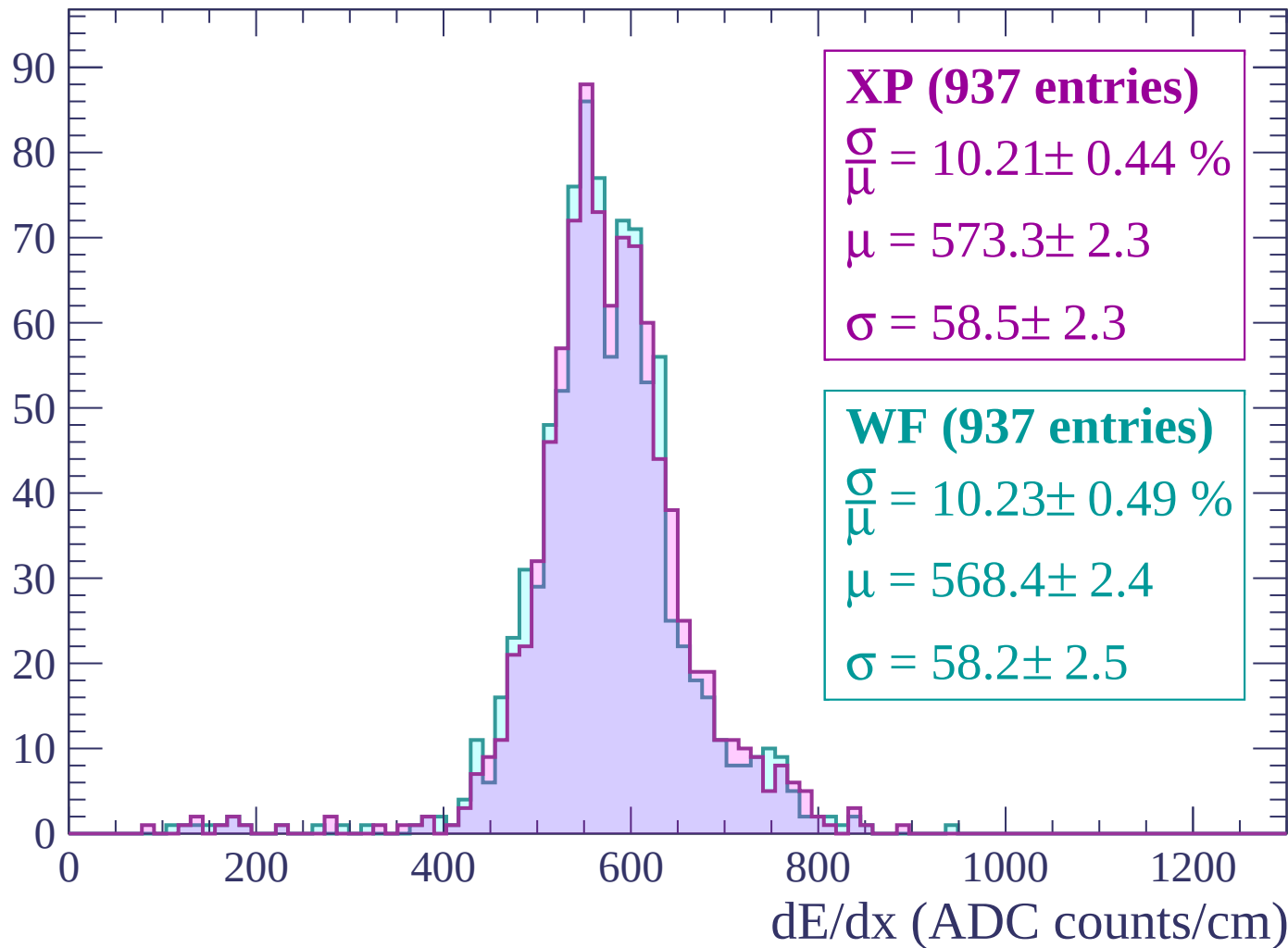
Energy loss | $60 < p < 120$

Count



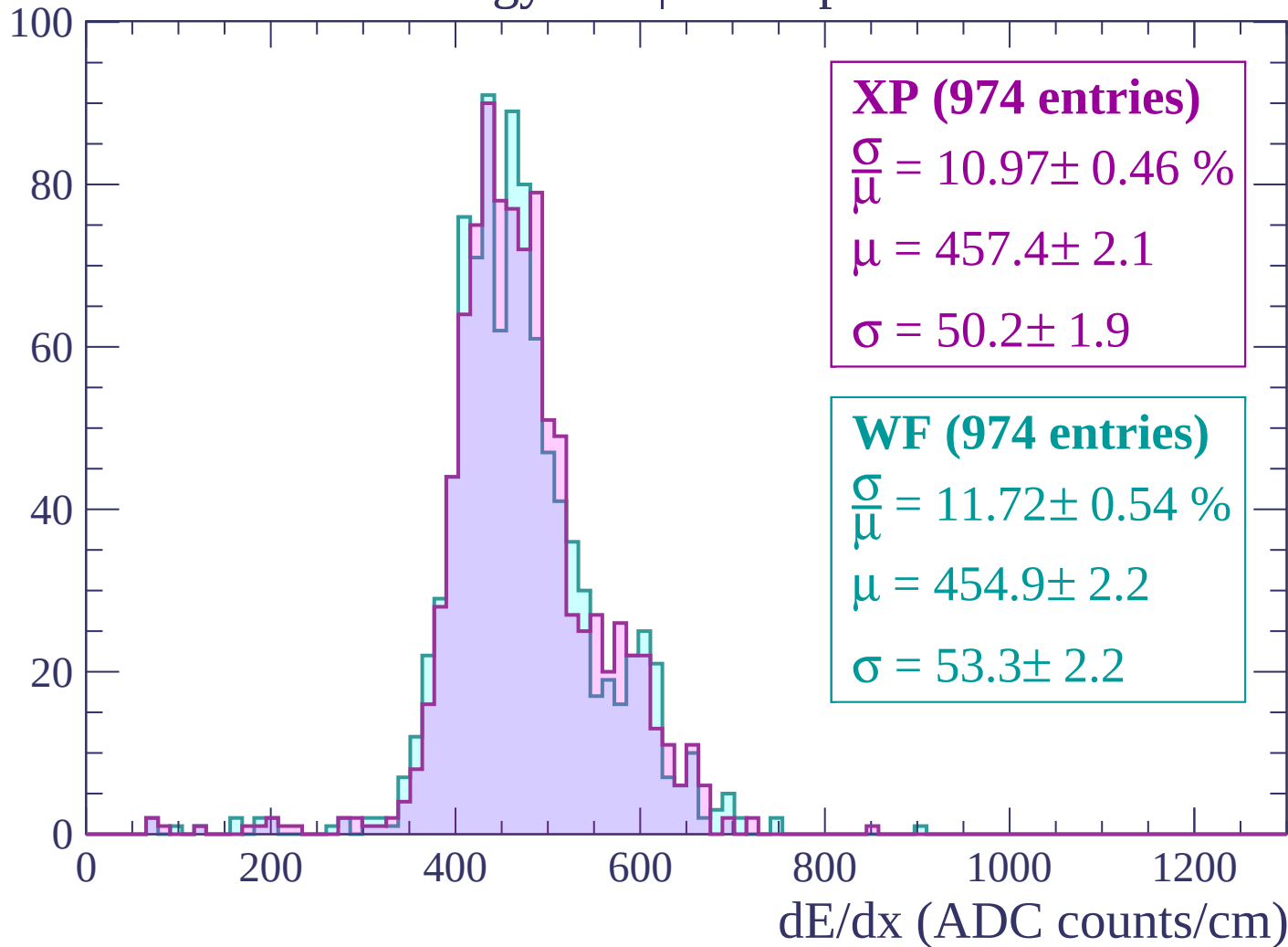
Energy loss | $120 < p < 180$

Count



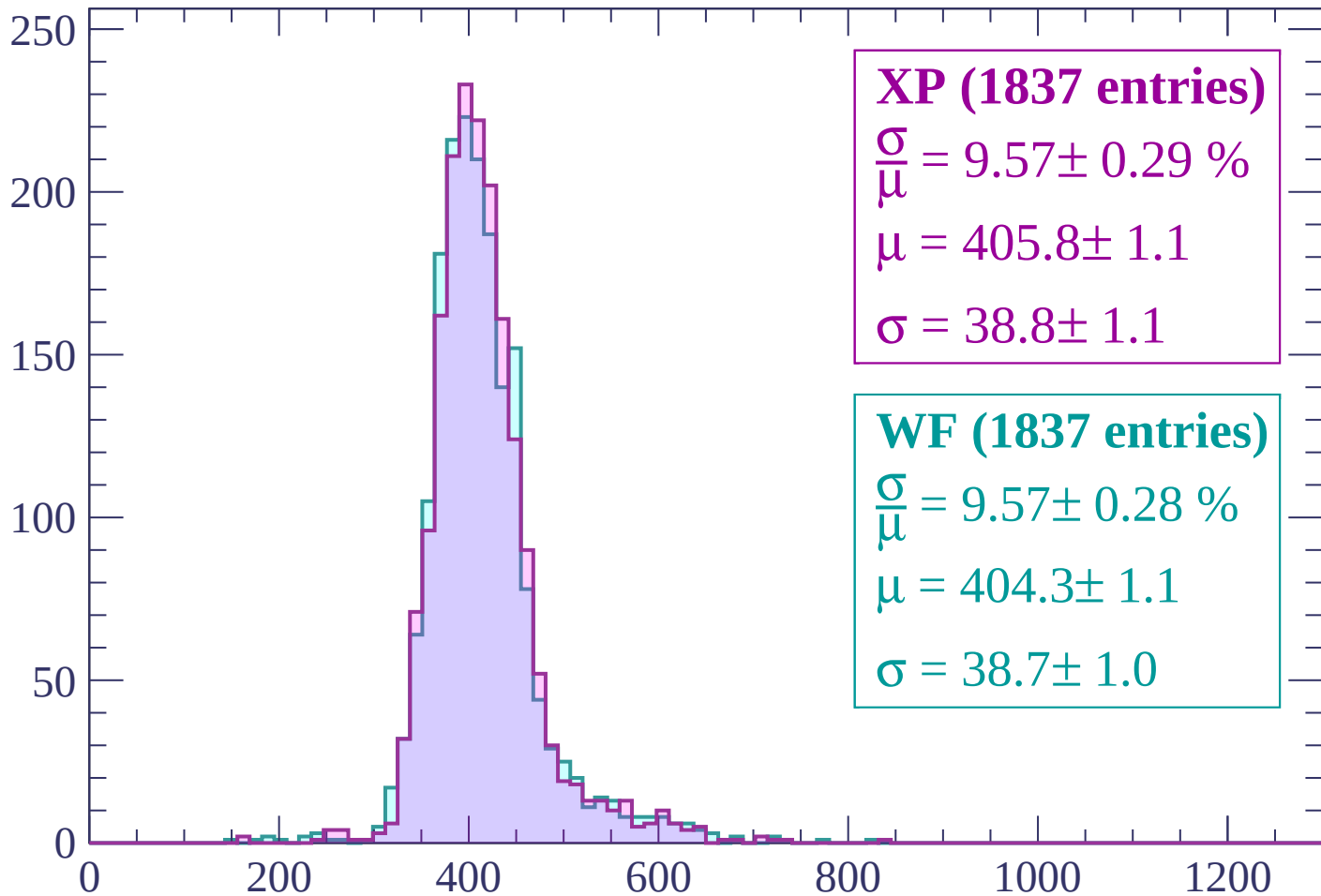
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP (1837 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.29 \%$$

$$\mu = 405.8 \pm 1.1$$

$$\sigma = 38.8 \pm 1.1$$

WF (1837 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.28 \%$$

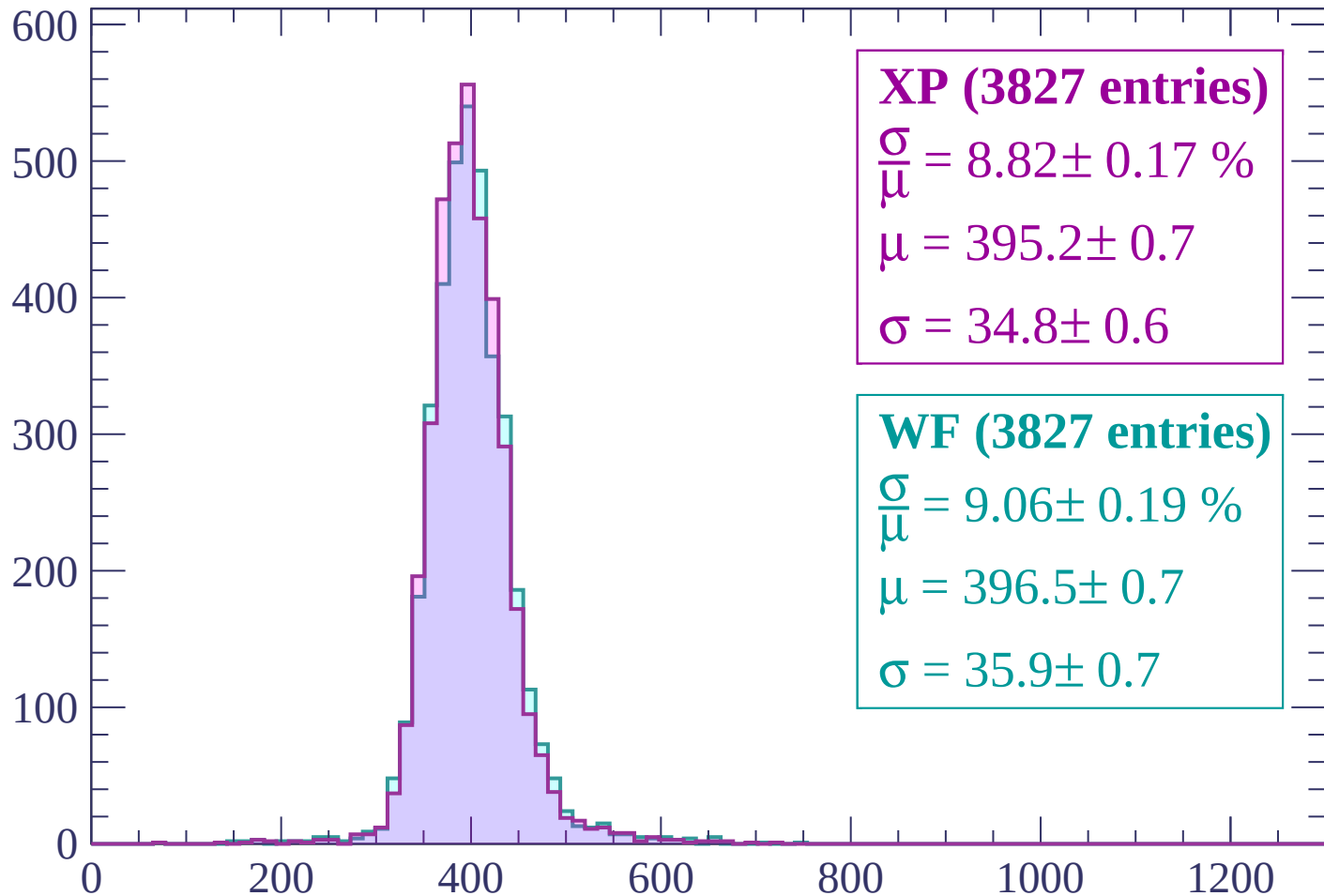
$$\mu = 404.3 \pm 1.1$$

$$\sigma = 38.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (3827 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.17 \%$$

$$\mu = 395.2 \pm 0.7$$

$$\sigma = 34.8 \pm 0.6$$

WF (3827 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.19 \%$$

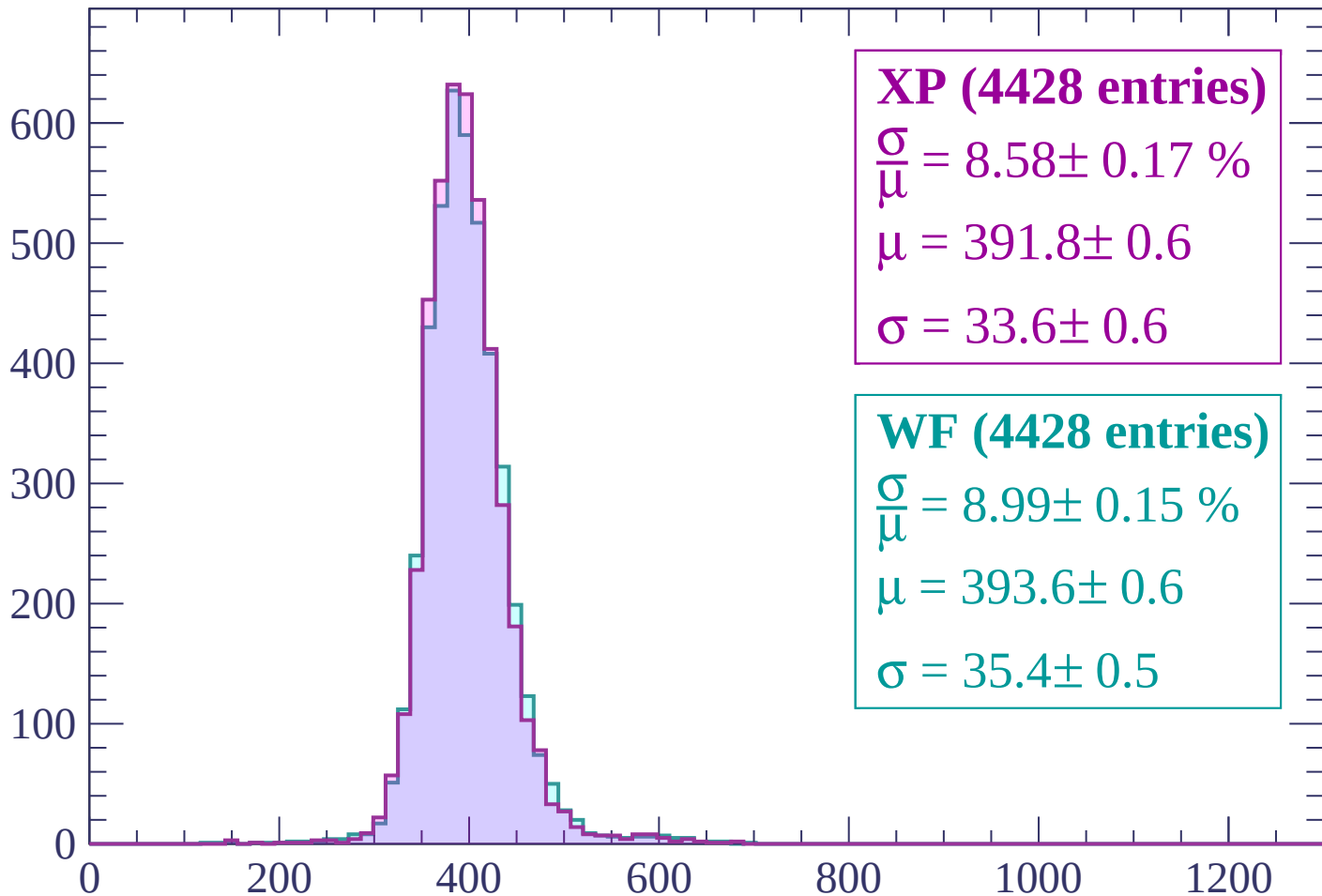
$$\mu = 396.5 \pm 0.7$$

$$\sigma = 35.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (4428 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.17 \%$$

$$\mu = 391.8 \pm 0.6$$

$$\sigma = 33.6 \pm 0.6$$

WF (4428 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.15 \%$$

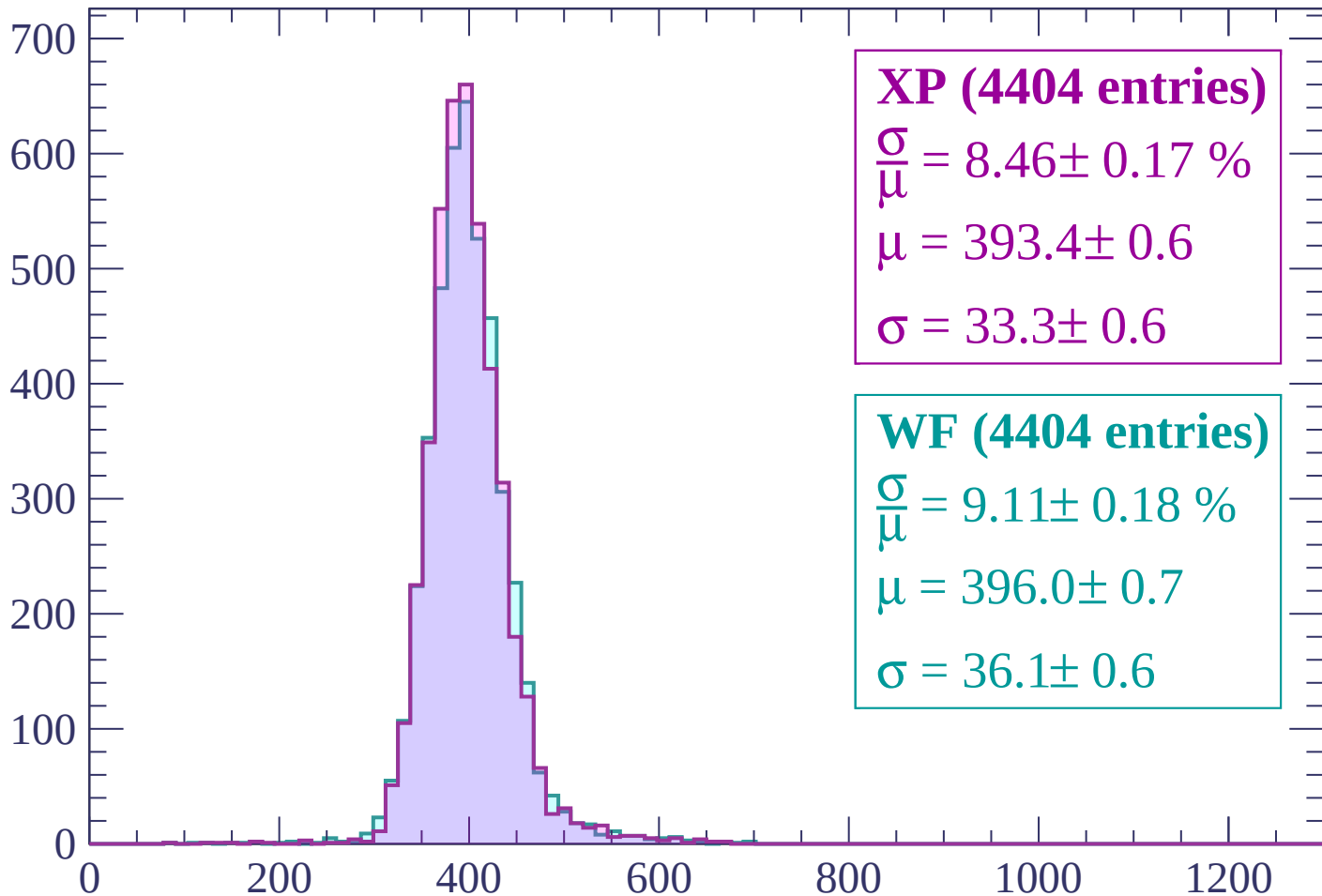
$$\mu = 393.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (4404 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

$$\mu = 393.4 \pm 0.6$$

$$\sigma = 33.3 \pm 0.6$$

WF (4404 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.18 \%$$

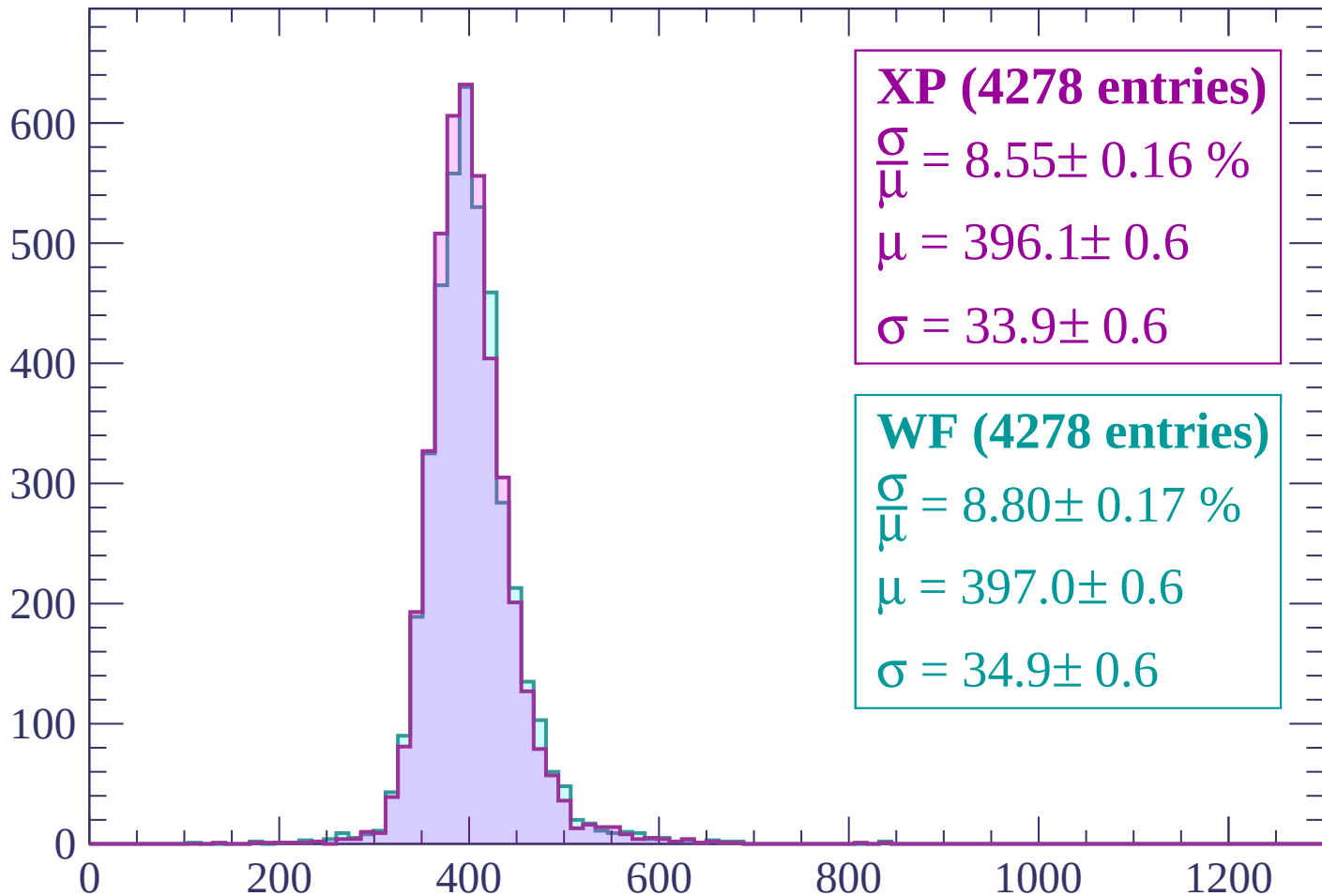
$$\mu = 396.0 \pm 0.7$$

$$\sigma = 36.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (4278 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.16 \%$$

$$\mu = 396.1 \pm 0.6$$

$$\sigma = 33.9 \pm 0.6$$

WF (4278 entries)

$$\frac{\sigma}{\mu} = 8.80 \pm 0.17 \%$$

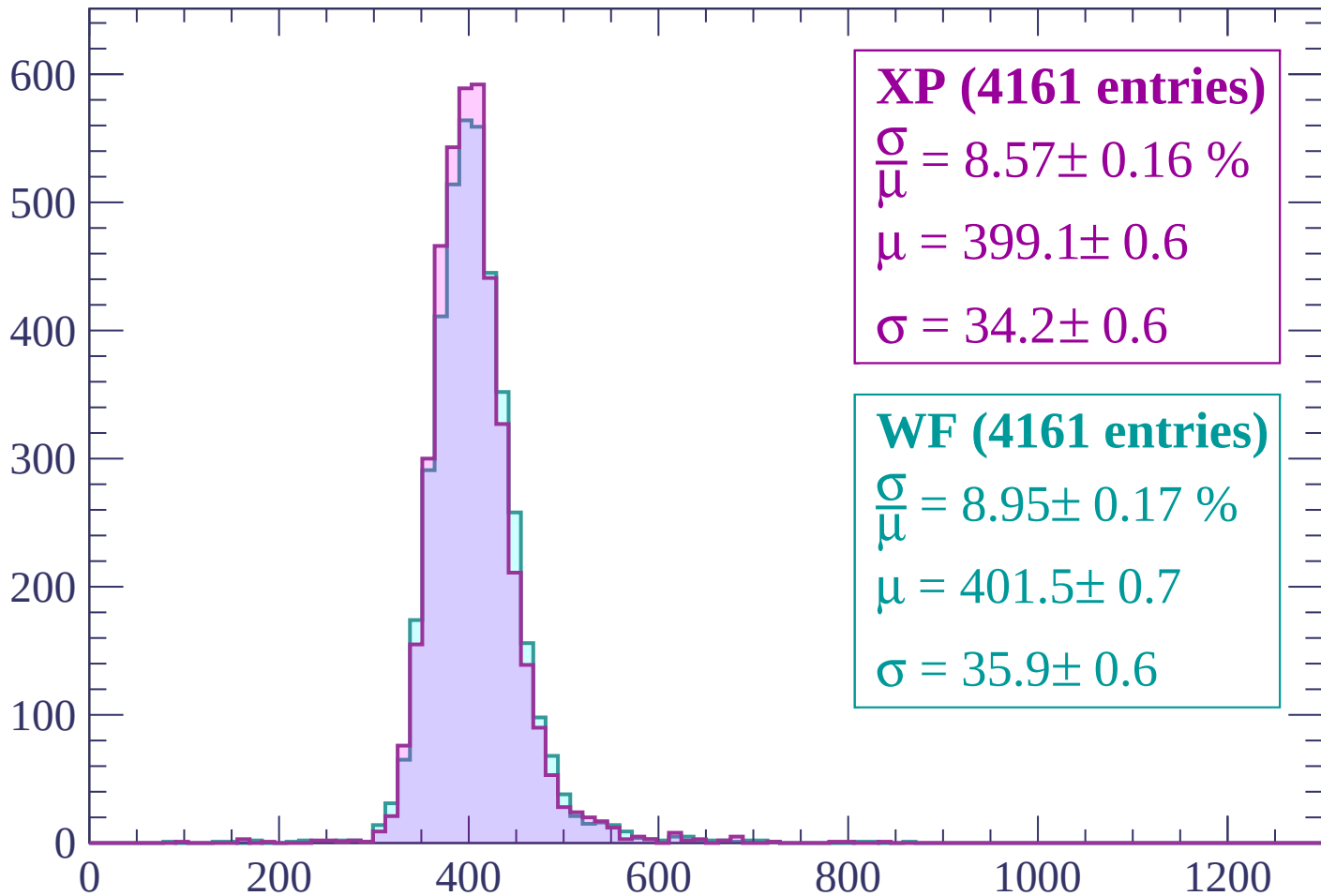
$$\mu = 397.0 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (4161 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.16 \%$$

$$\mu = 399.1 \pm 0.6$$

$$\sigma = 34.2 \pm 0.6$$

WF (4161 entries)

$$\frac{\sigma}{\mu} = 8.95 \pm 0.17 \%$$

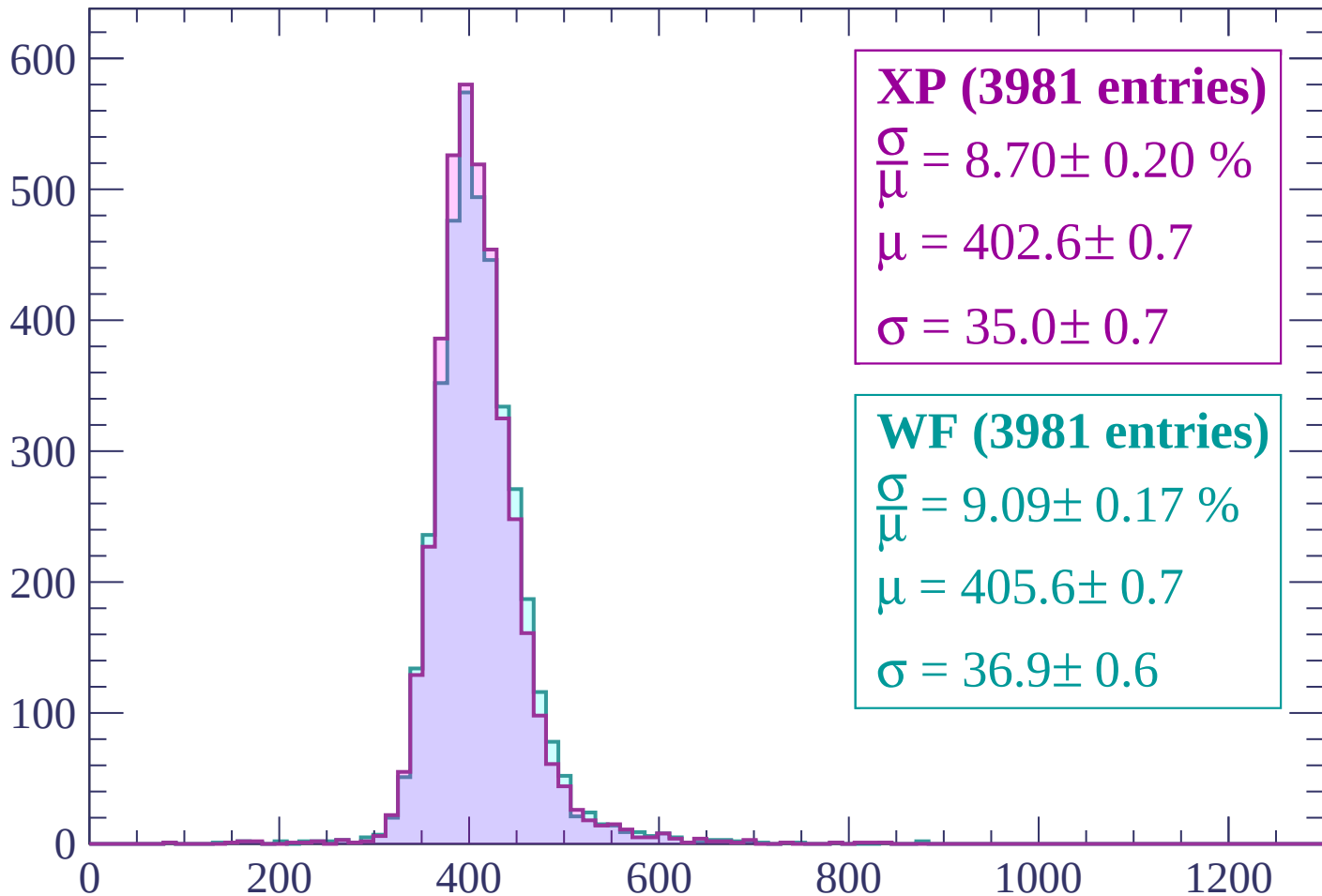
$$\mu = 401.5 \pm 0.7$$

$$\sigma = 35.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (3981 entries)

$$\frac{\sigma}{\mu} = 8.70 \pm 0.20 \%$$

$$\mu = 402.6 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

WF (3981 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.17 \%$$

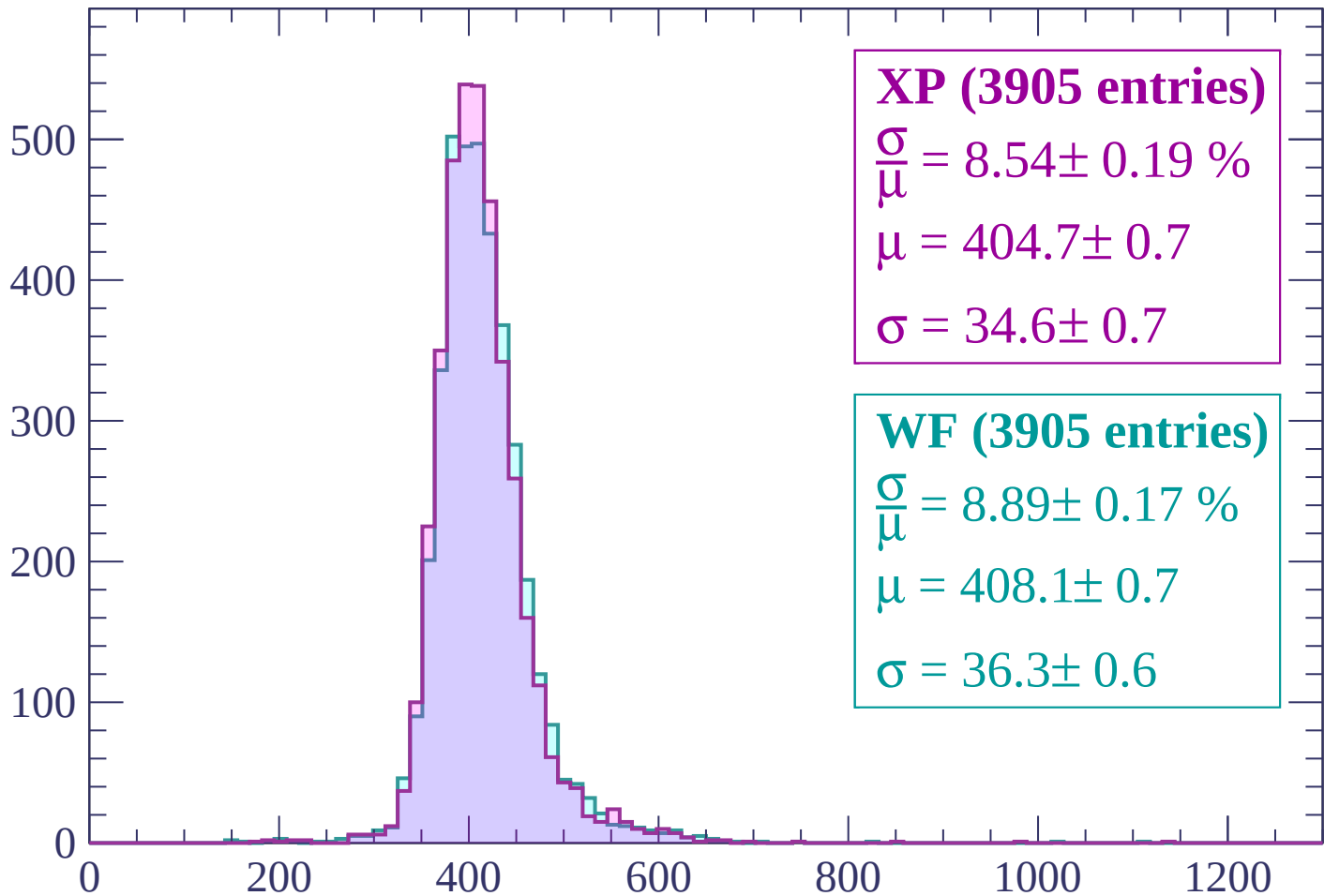
$$\mu = 405.6 \pm 0.7$$

$$\sigma = 36.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (3905 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.19 \%$$

$$\mu = 404.7 \pm 0.7$$

$$\sigma = 34.6 \pm 0.7$$

WF (3905 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

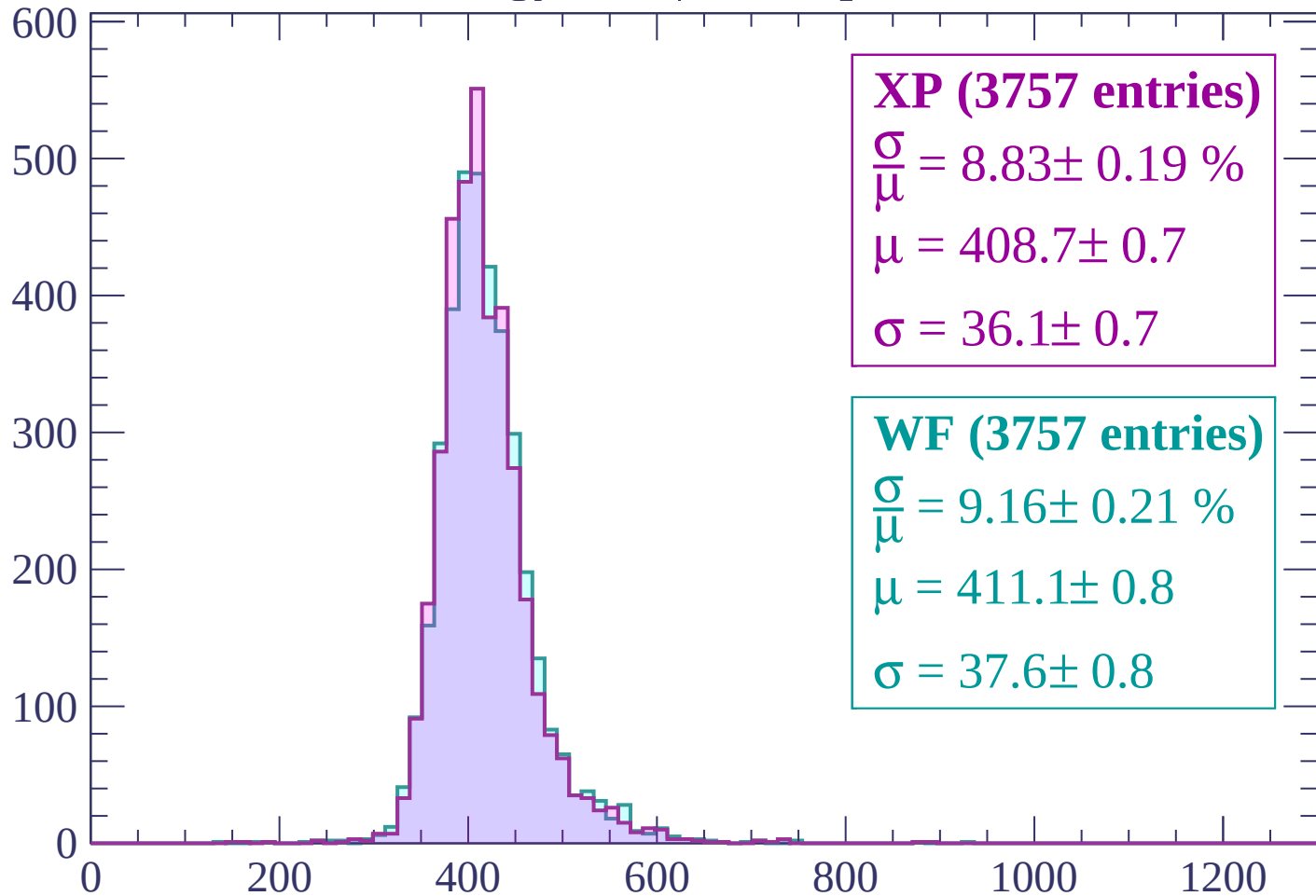
$$\mu = 408.1 \pm 0.7$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (3757 entries)

$\frac{\sigma}{\mu} = 8.83 \pm 0.19 \%$

$\mu = 408.7 \pm 0.7$

$\sigma = 36.1 \pm 0.7$

WF (3757 entries)

$\frac{\sigma}{\mu} = 9.16 \pm 0.21 \%$

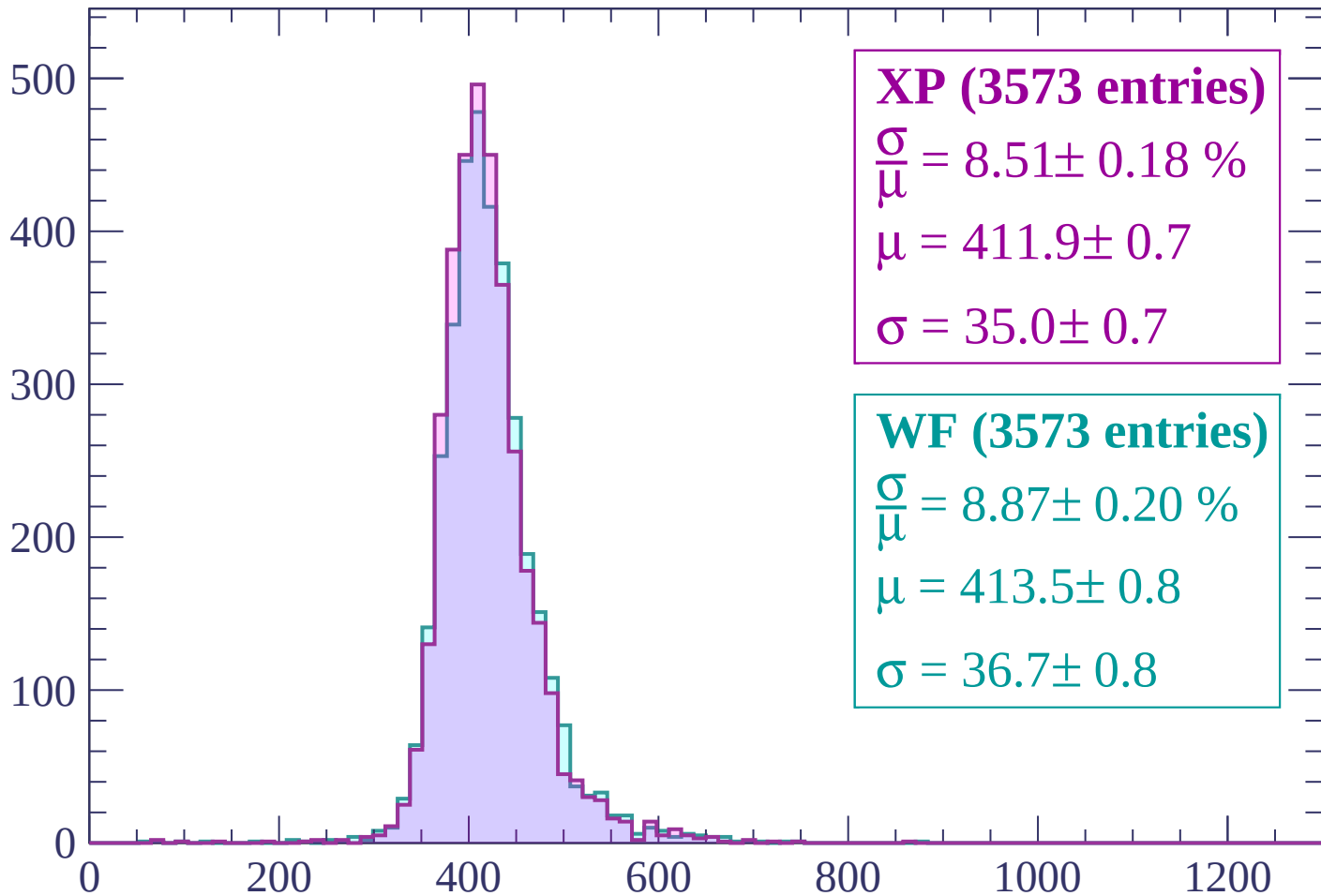
$\mu = 411.1 \pm 0.8$

$\sigma = 37.6 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (3573 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.18 \%$$

$$\mu = 411.9 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

WF (3573 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.20 \%$$

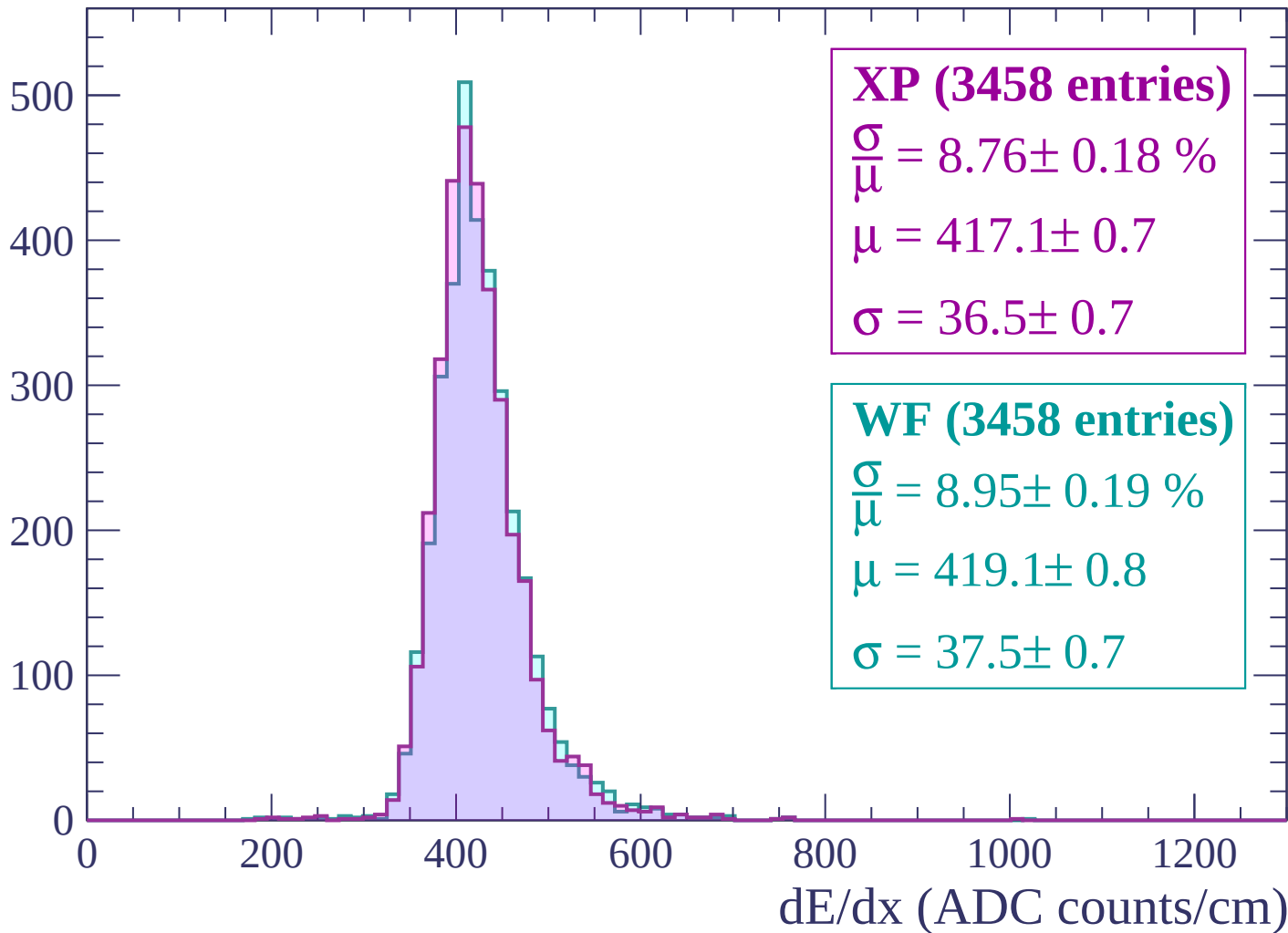
$$\mu = 413.5 \pm 0.8$$

$$\sigma = 36.7 \pm 0.8$$

dE/dx (ADC counts/cm)

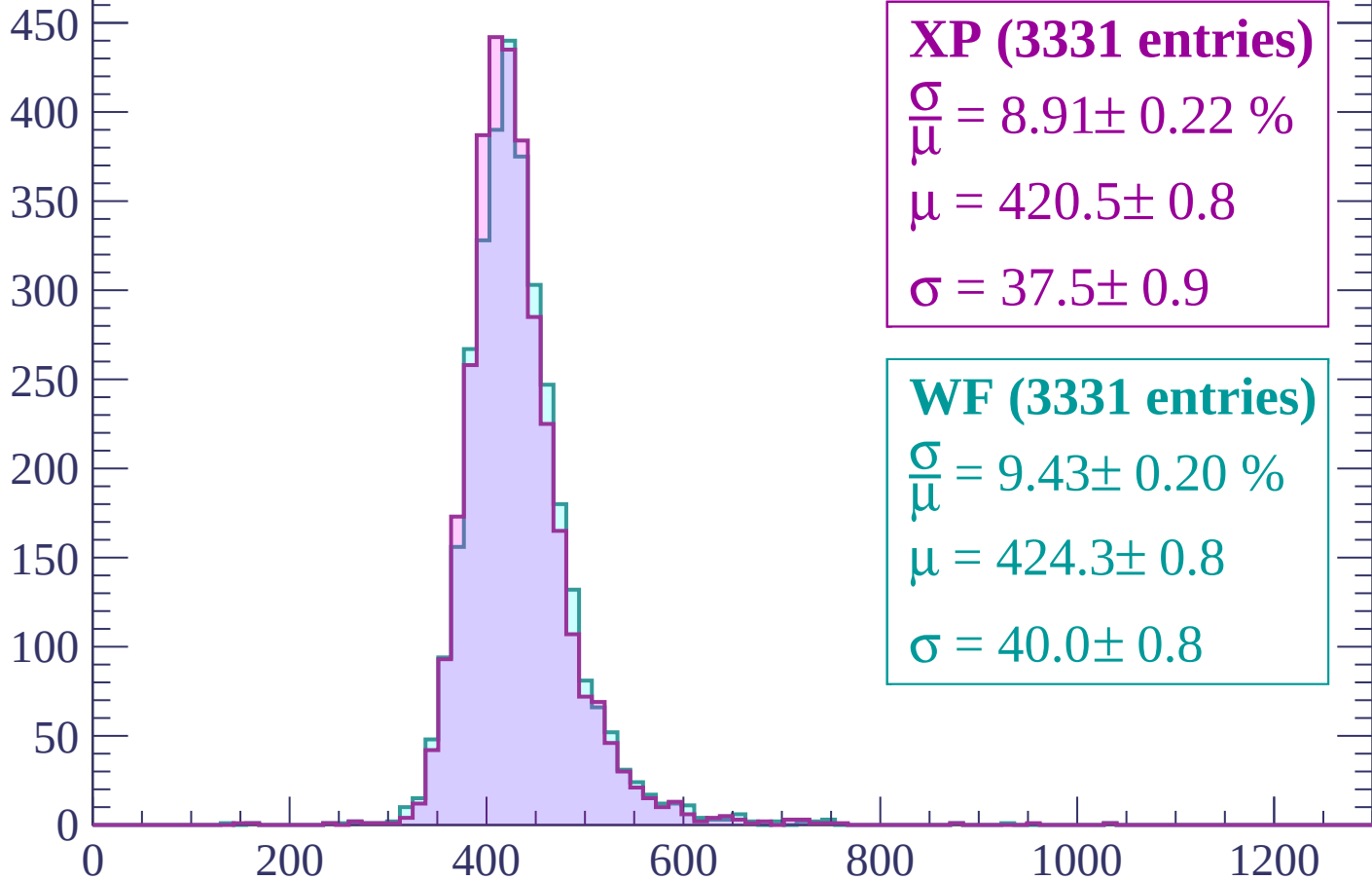
Energy loss | $840 < p < 900$

Count



Energy loss | $900 < p < 960$

Count



XP (3331 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.22 \%$$

$$\mu = 420.5 \pm 0.8$$

$$\sigma = 37.5 \pm 0.9$$

WF (3331 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.20 \%$$

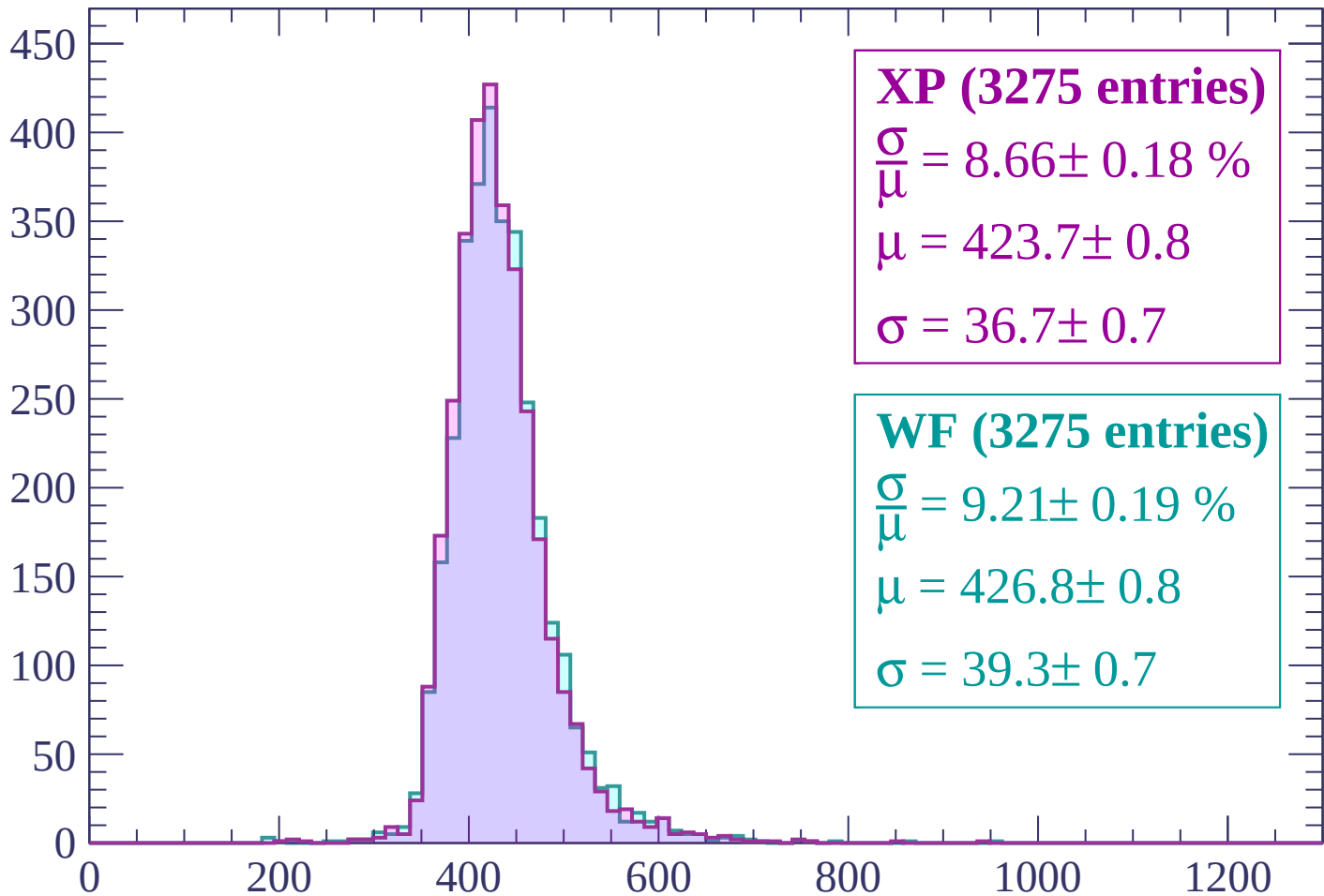
$$\mu = 424.3 \pm 0.8$$

$$\sigma = 40.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (3275 entries)

$$\frac{\sigma}{\mu} = 8.66 \pm 0.18 \%$$

$$\mu = 423.7 \pm 0.8$$

$$\sigma = 36.7 \pm 0.7$$

WF (3275 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.19 \%$$

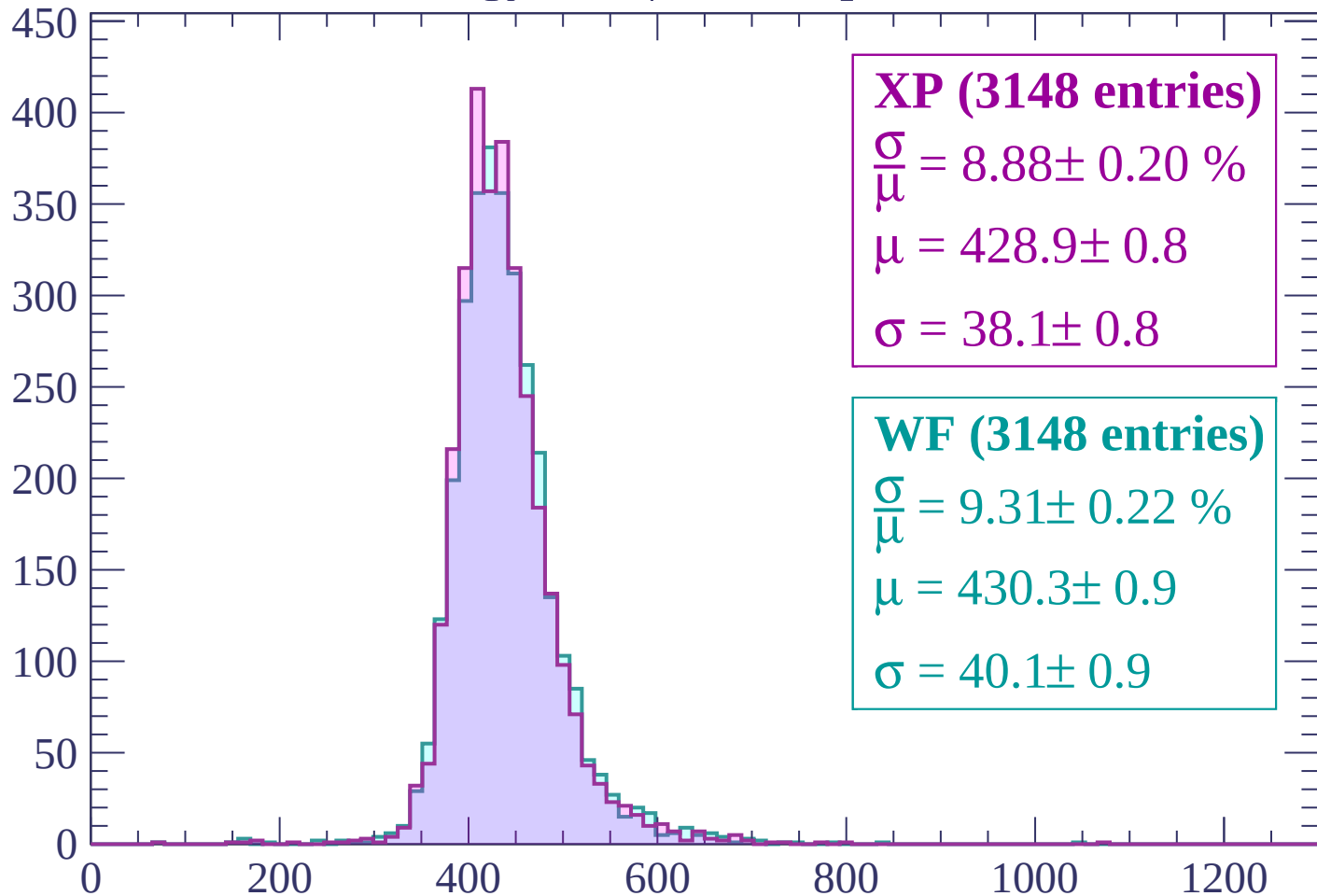
$$\mu = 426.8 \pm 0.8$$

$$\sigma = 39.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (3148 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.20 \%$$

$$\mu = 428.9 \pm 0.8$$

$$\sigma = 38.1 \pm 0.8$$

WF (3148 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.22 \%$$

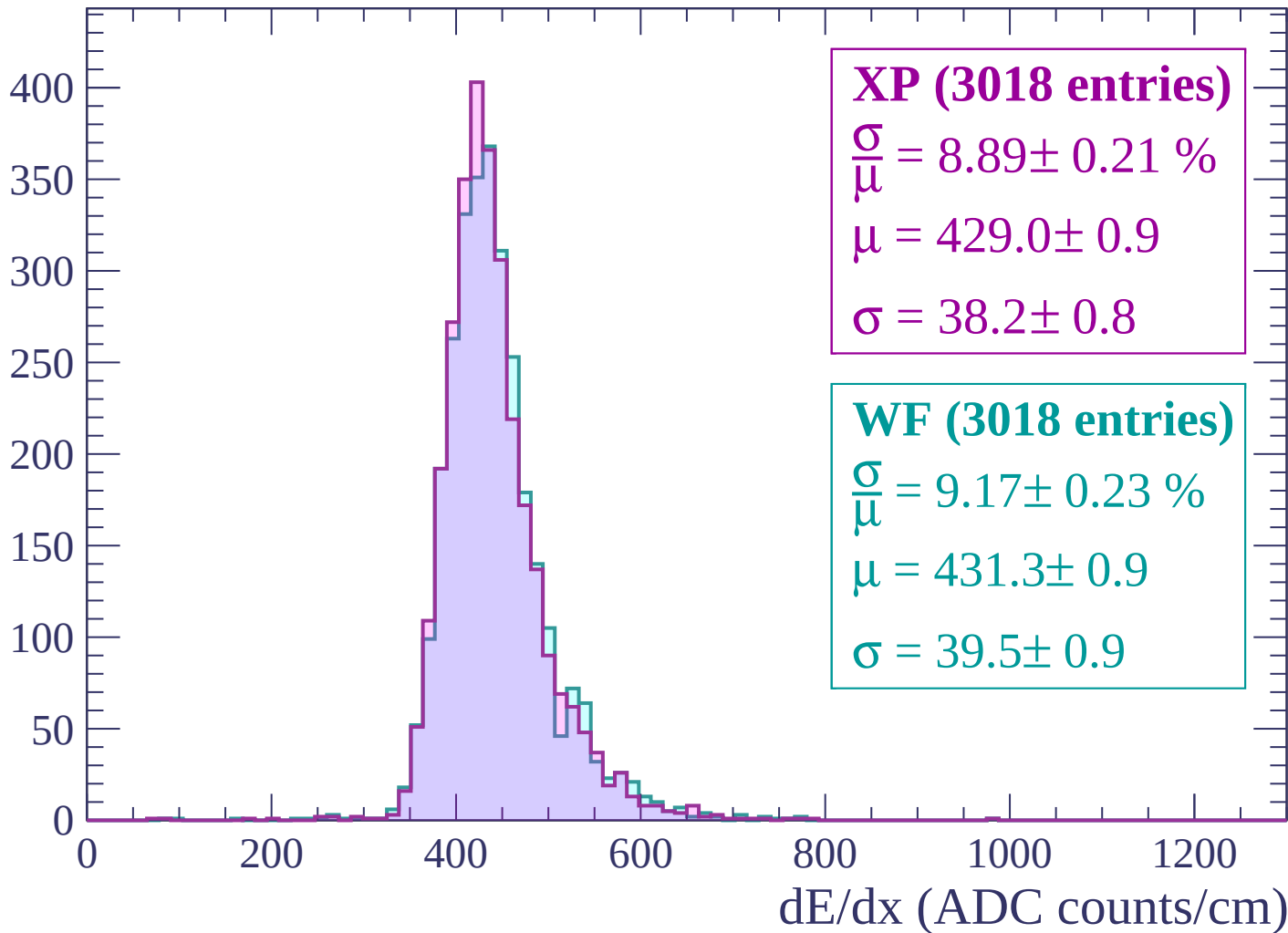
$$\mu = 430.3 \pm 0.9$$

$$\sigma = 40.1 \pm 0.9$$

dE/dx (ADC counts/cm)

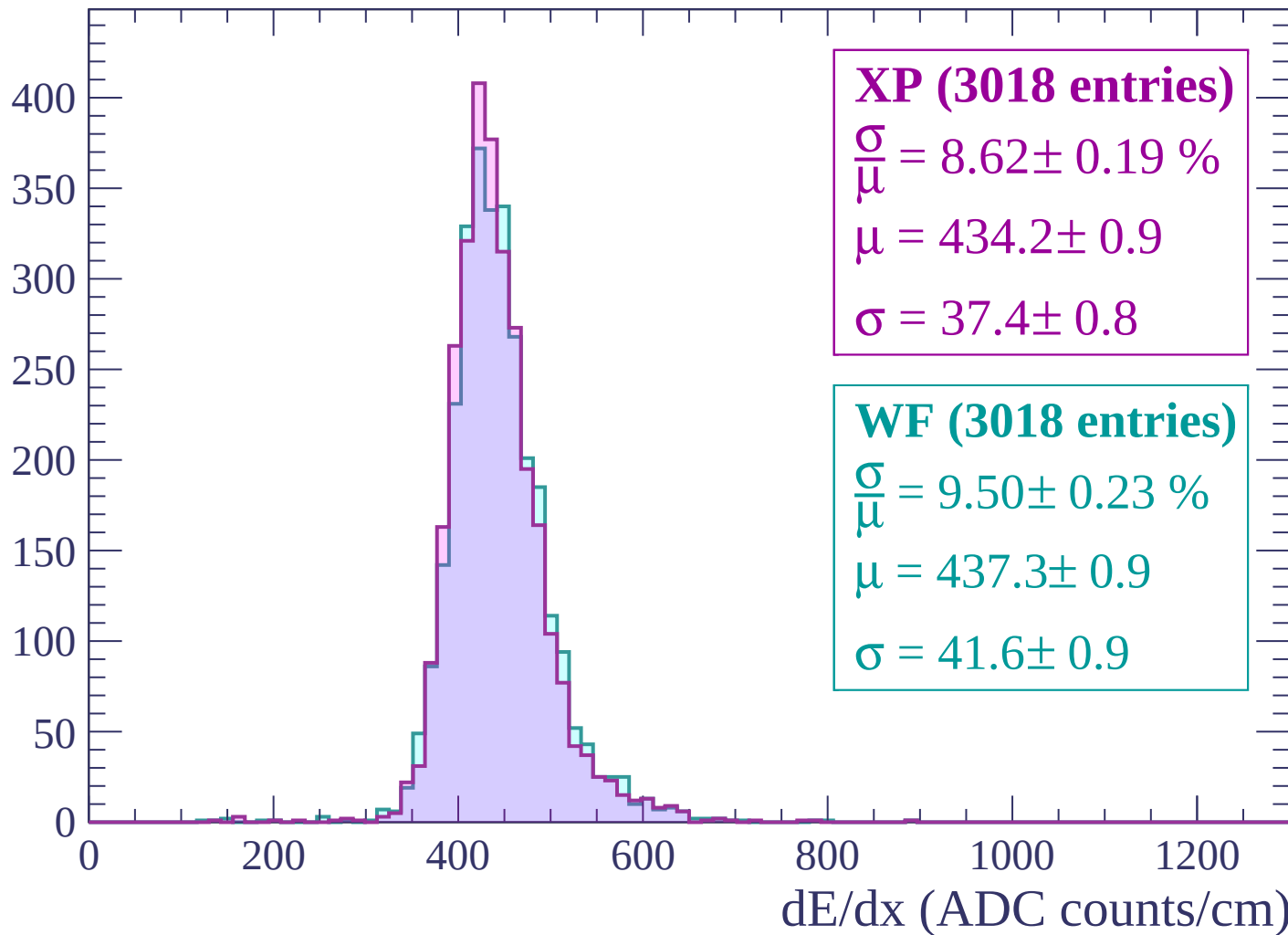
Energy loss | $1080 < p < 1140$

Count



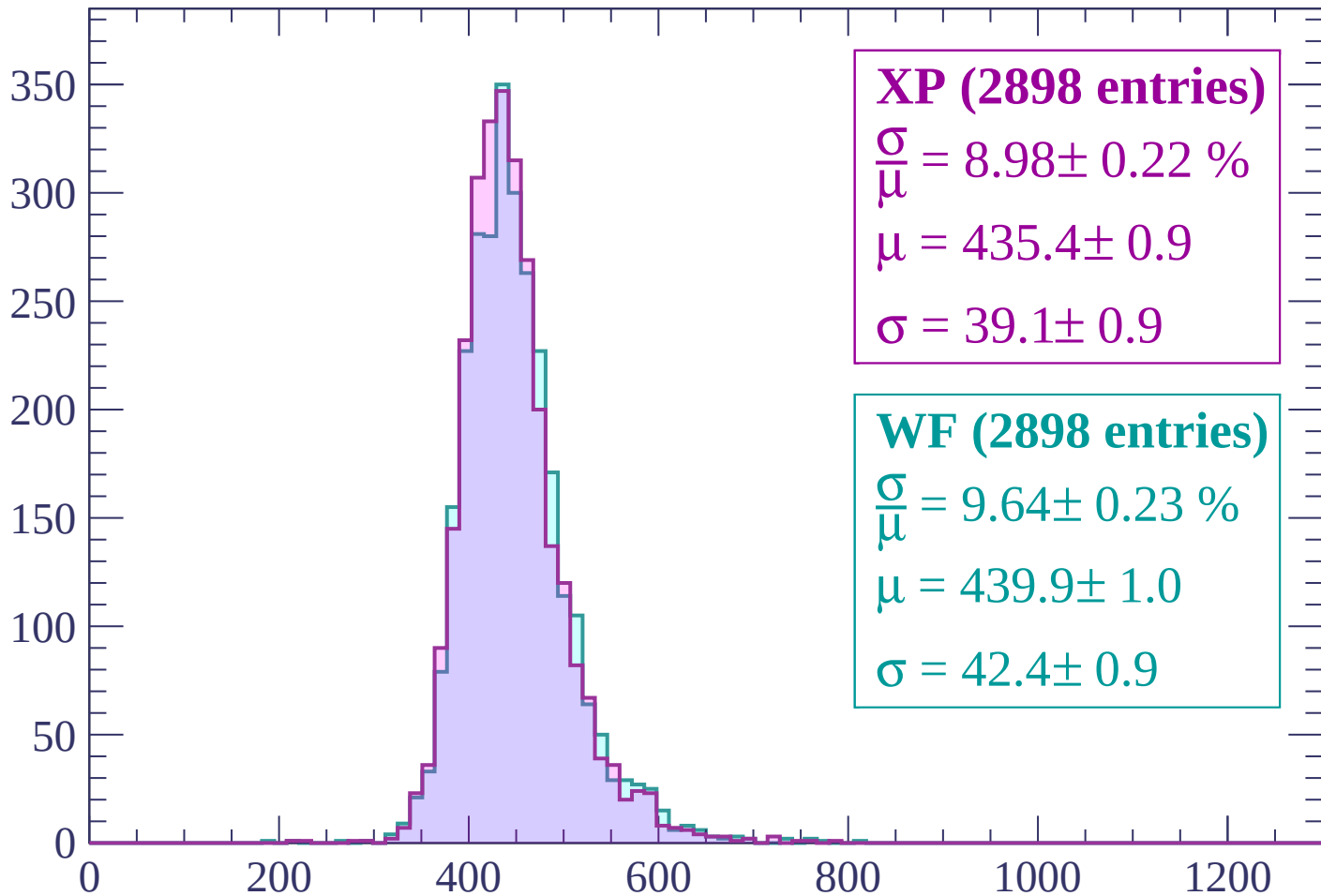
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (2898 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.22 \%$$

$$\mu = 435.4 \pm 0.9$$

$$\sigma = 39.1 \pm 0.9$$

WF (2898 entries)

$$\frac{\sigma}{\mu} = 9.64 \pm 0.23 \%$$

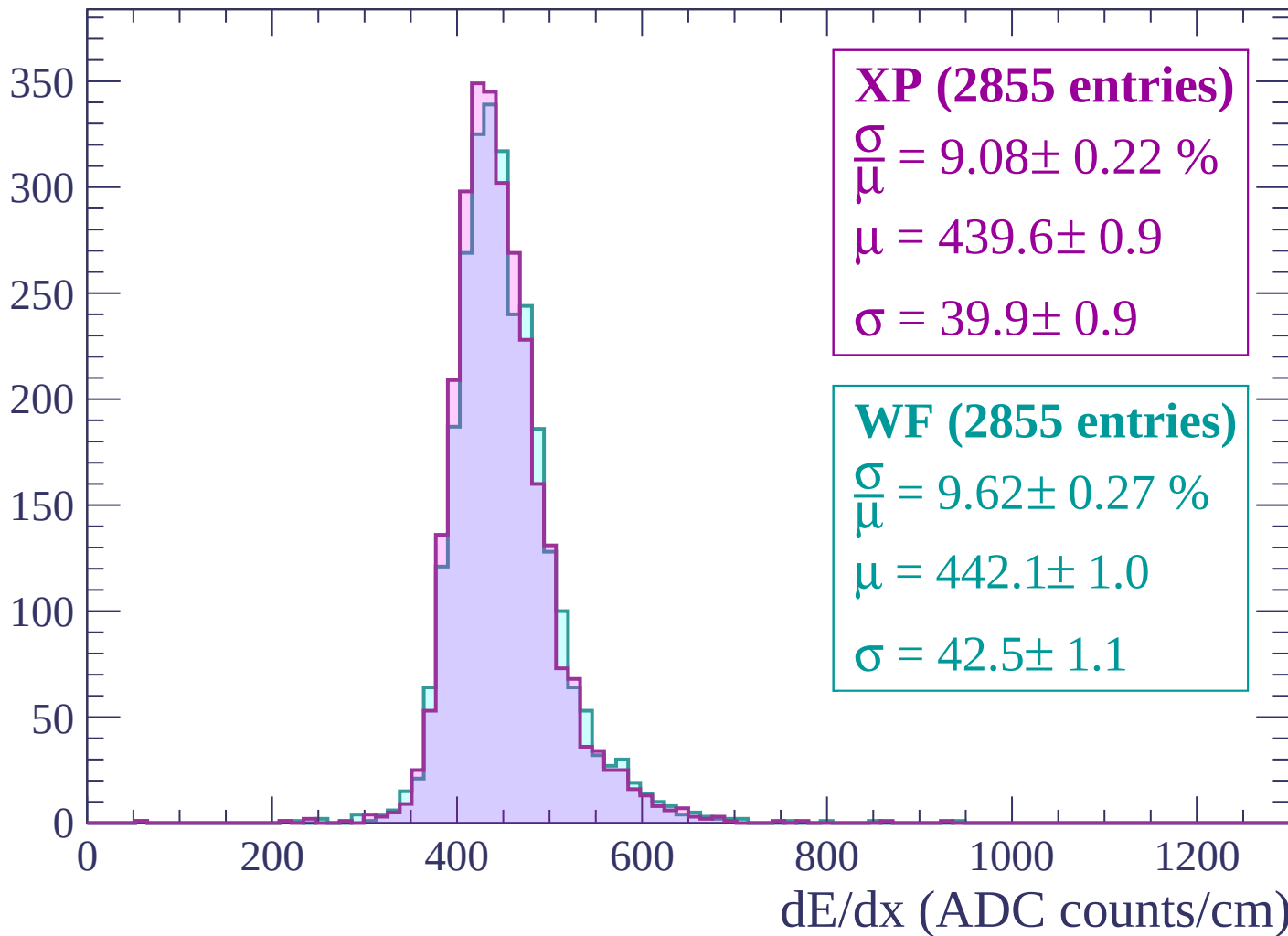
$$\mu = 439.9 \pm 1.0$$

$$\sigma = 42.4 \pm 0.9$$

dE/dx (ADC counts/cm)

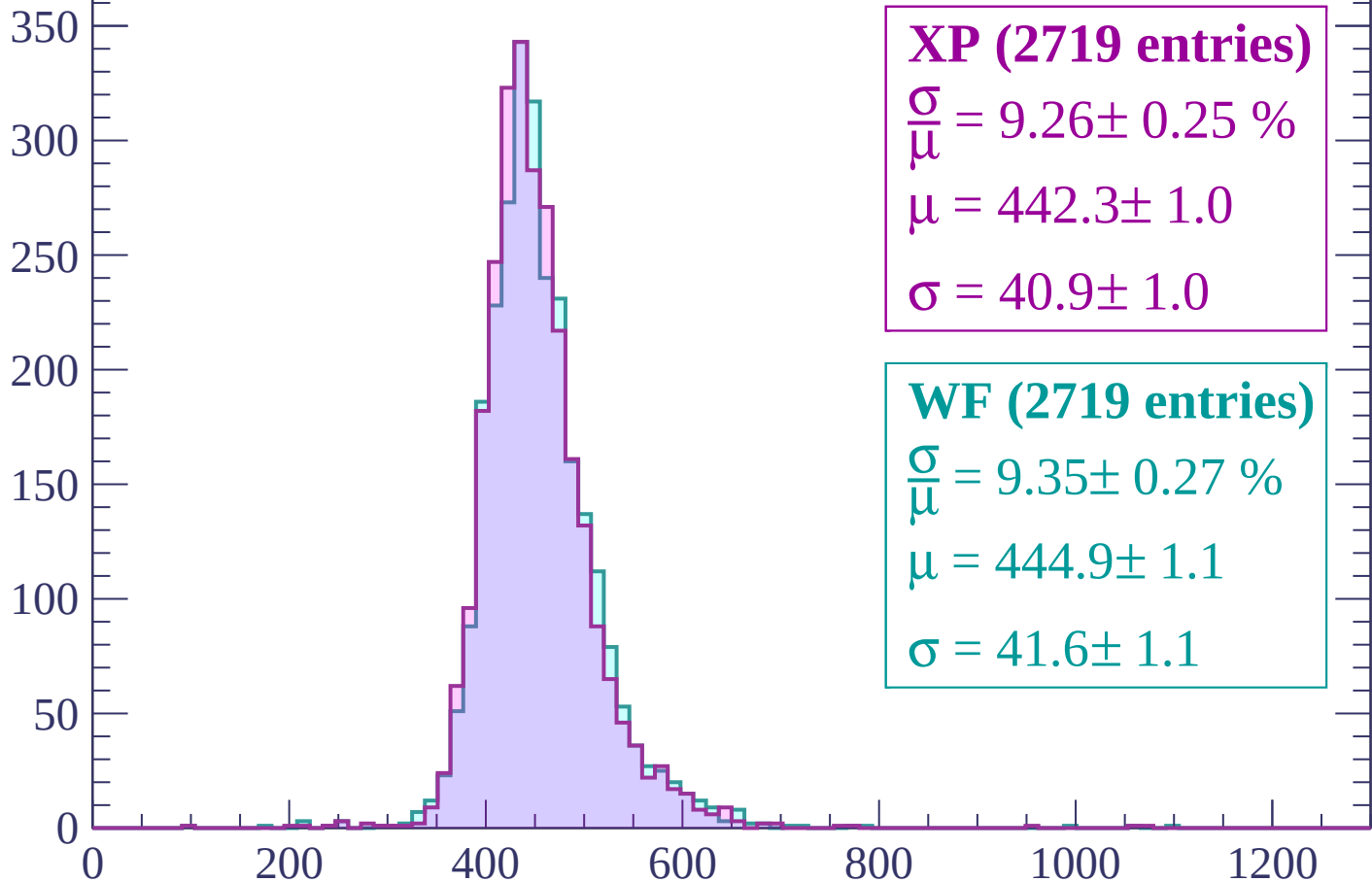
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP (2719 entries)

$$\frac{\sigma}{\mu} = 9.26 \pm 0.25 \%$$

$$\mu = 442.3 \pm 1.0$$

$$\sigma = 40.9 \pm 1.0$$

WF (2719 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.27 \%$$

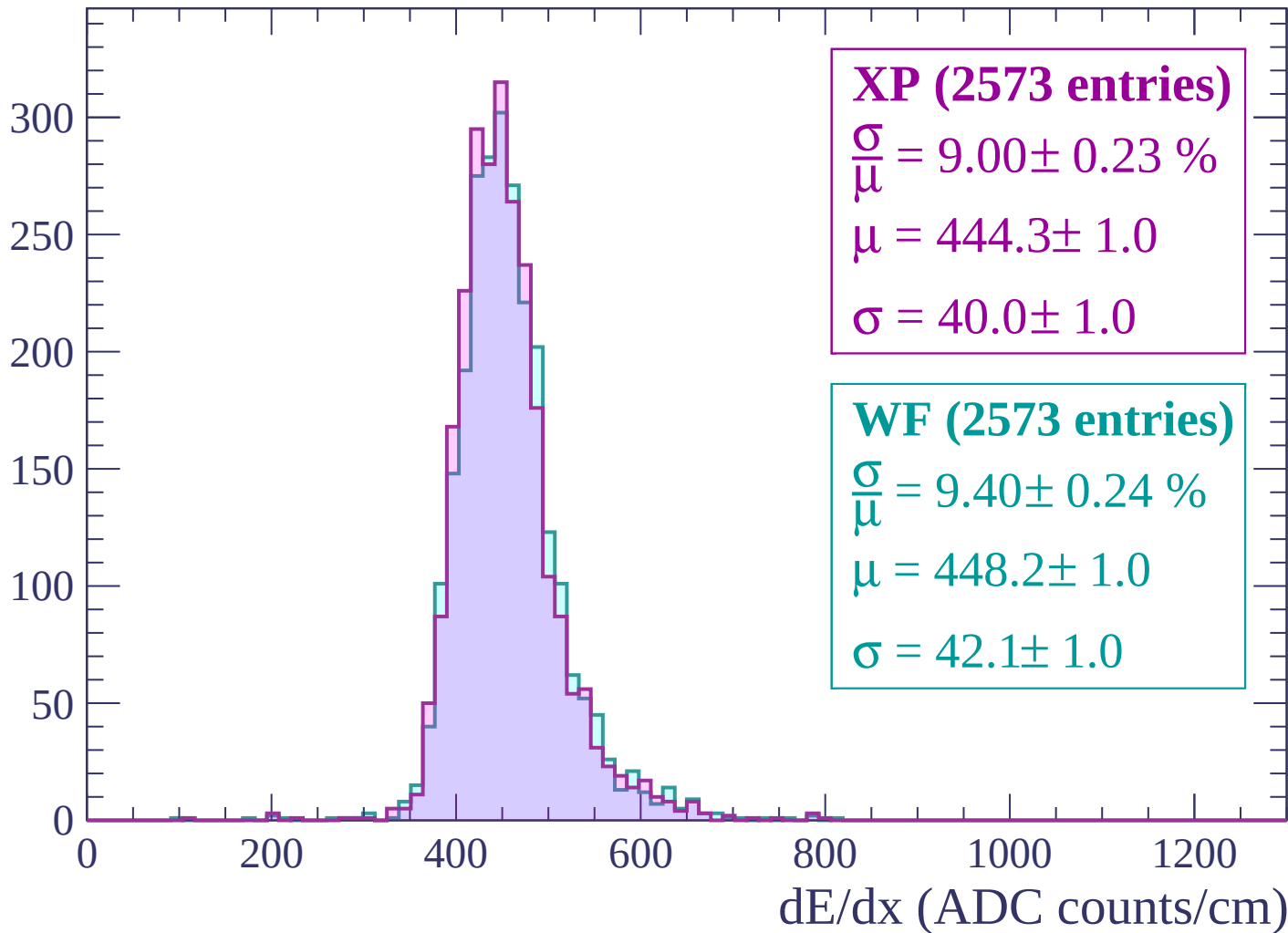
$$\mu = 444.9 \pm 1.1$$

$$\sigma = 41.6 \pm 1.1$$

dE/dx (ADC counts/cm)

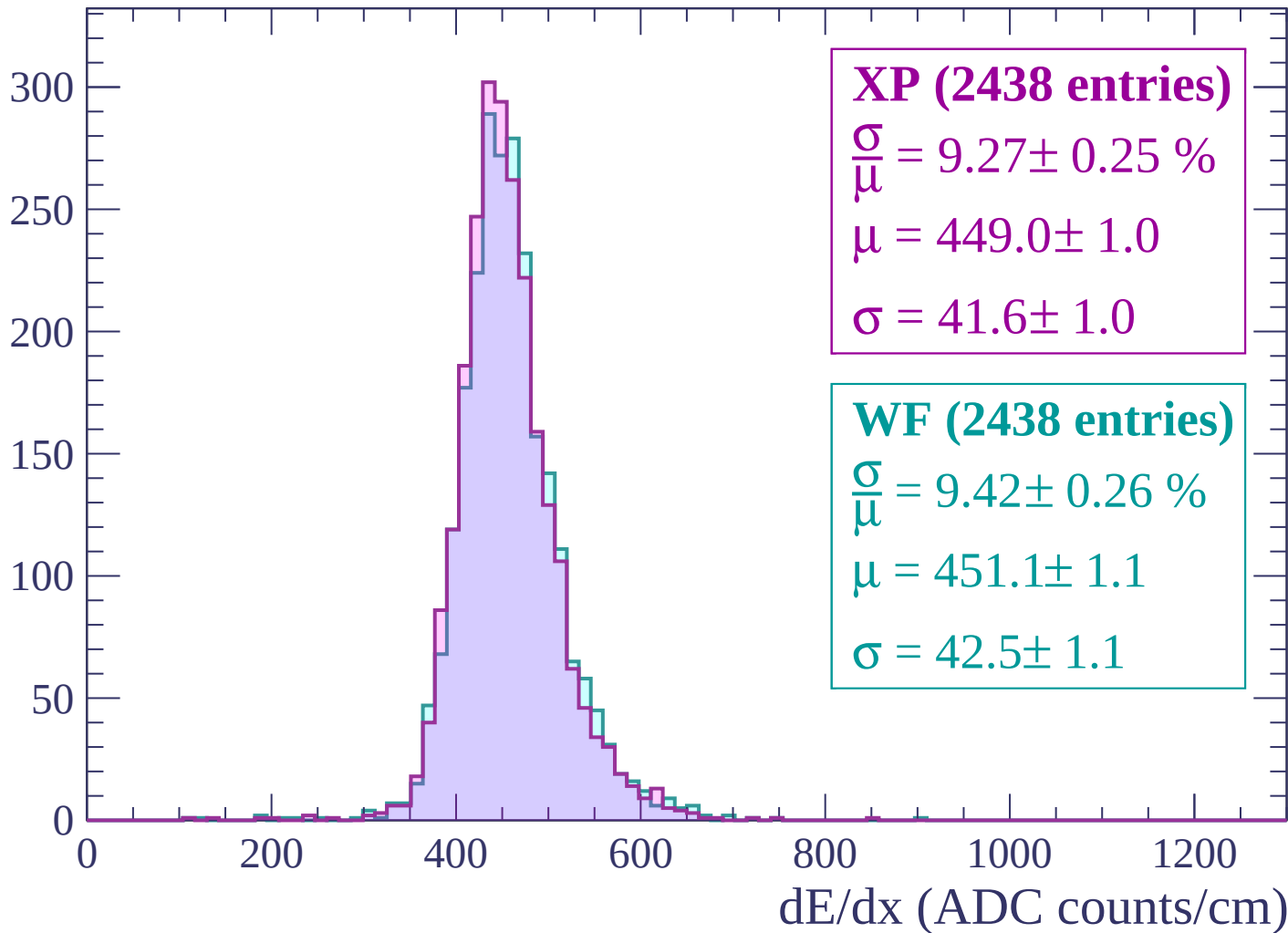
Energy loss | $1380 < p < 1440$

Count



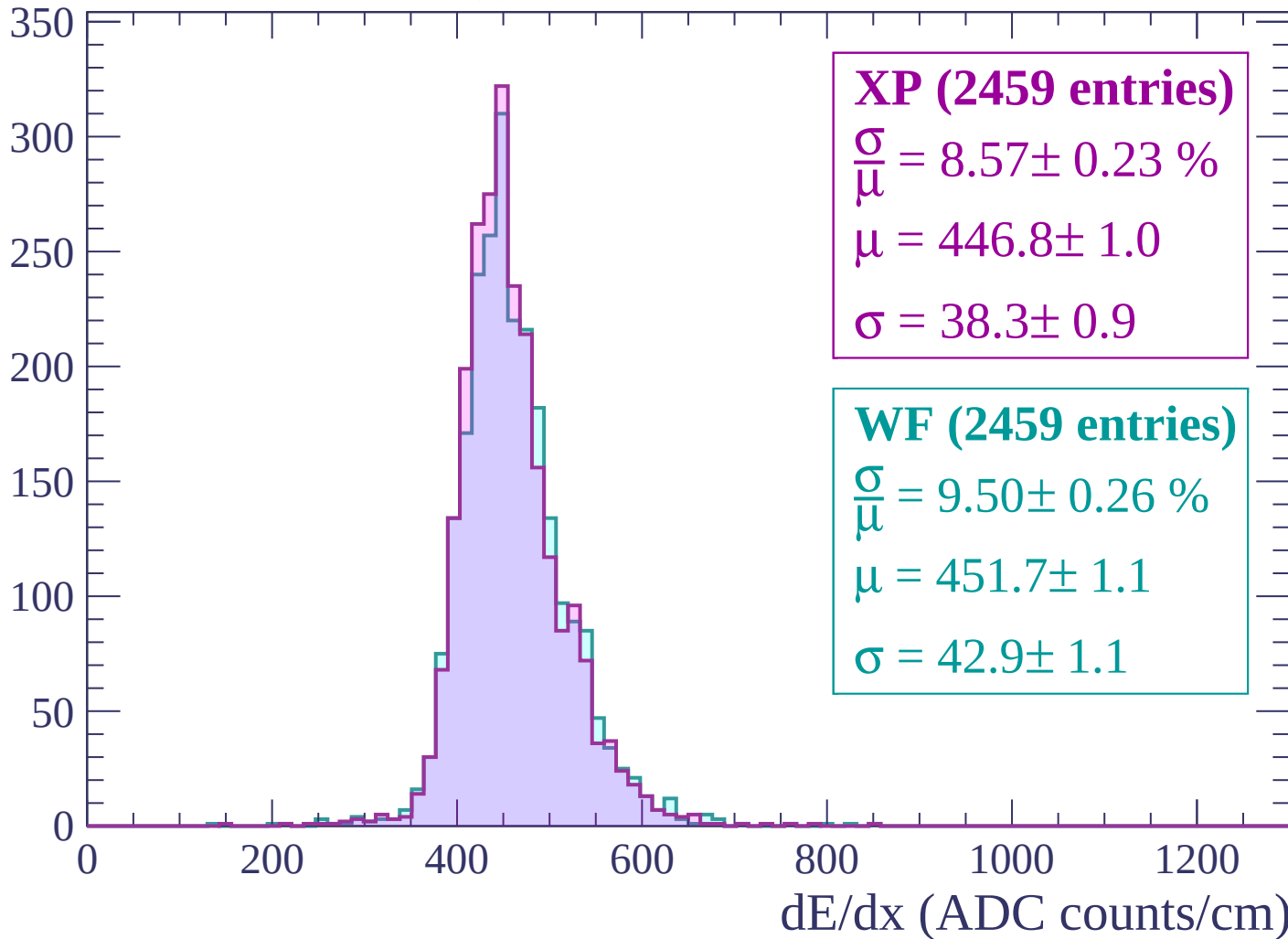
Energy loss | $1440 < p < 1500$

Count



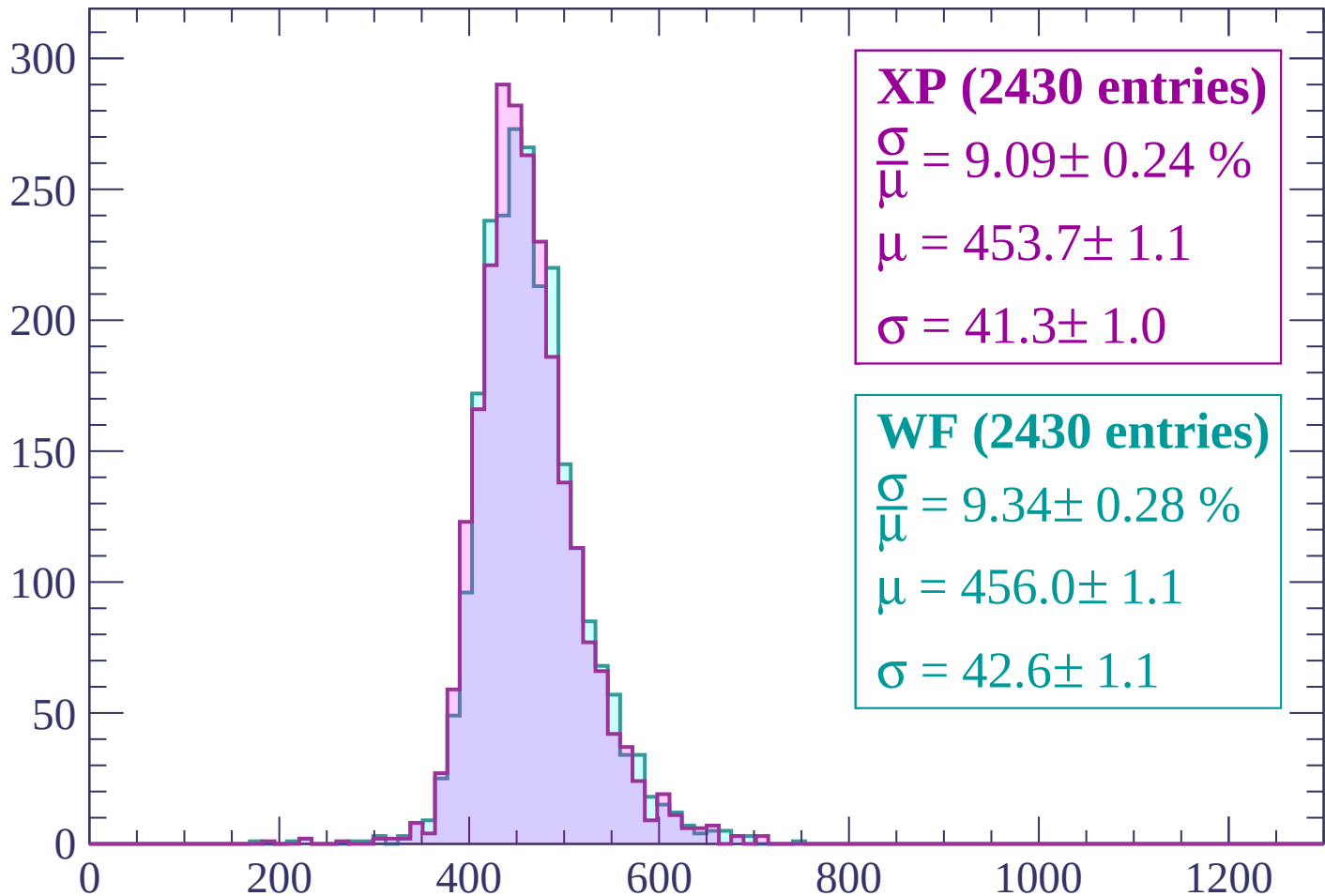
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (2430 entries)

$\frac{\sigma}{\mu} = 9.09 \pm 0.24 \%$

$\mu = 453.7 \pm 1.1$

$\sigma = 41.3 \pm 1.0$

WF (2430 entries)

$\frac{\sigma}{\mu} = 9.34 \pm 0.28 \%$

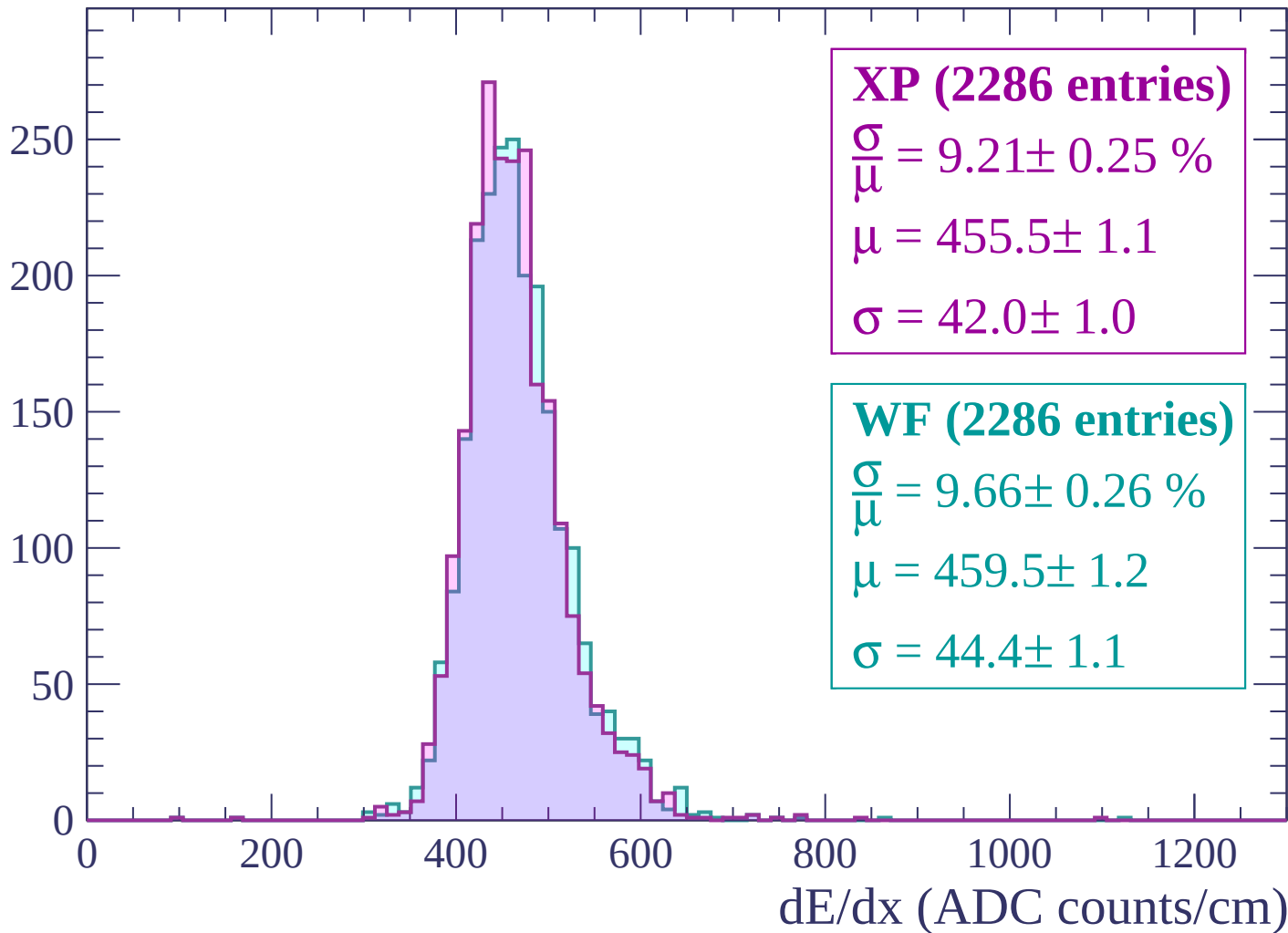
$\mu = 456.0 \pm 1.1$

$\sigma = 42.6 \pm 1.1$

dE/dx (ADC counts/cm)

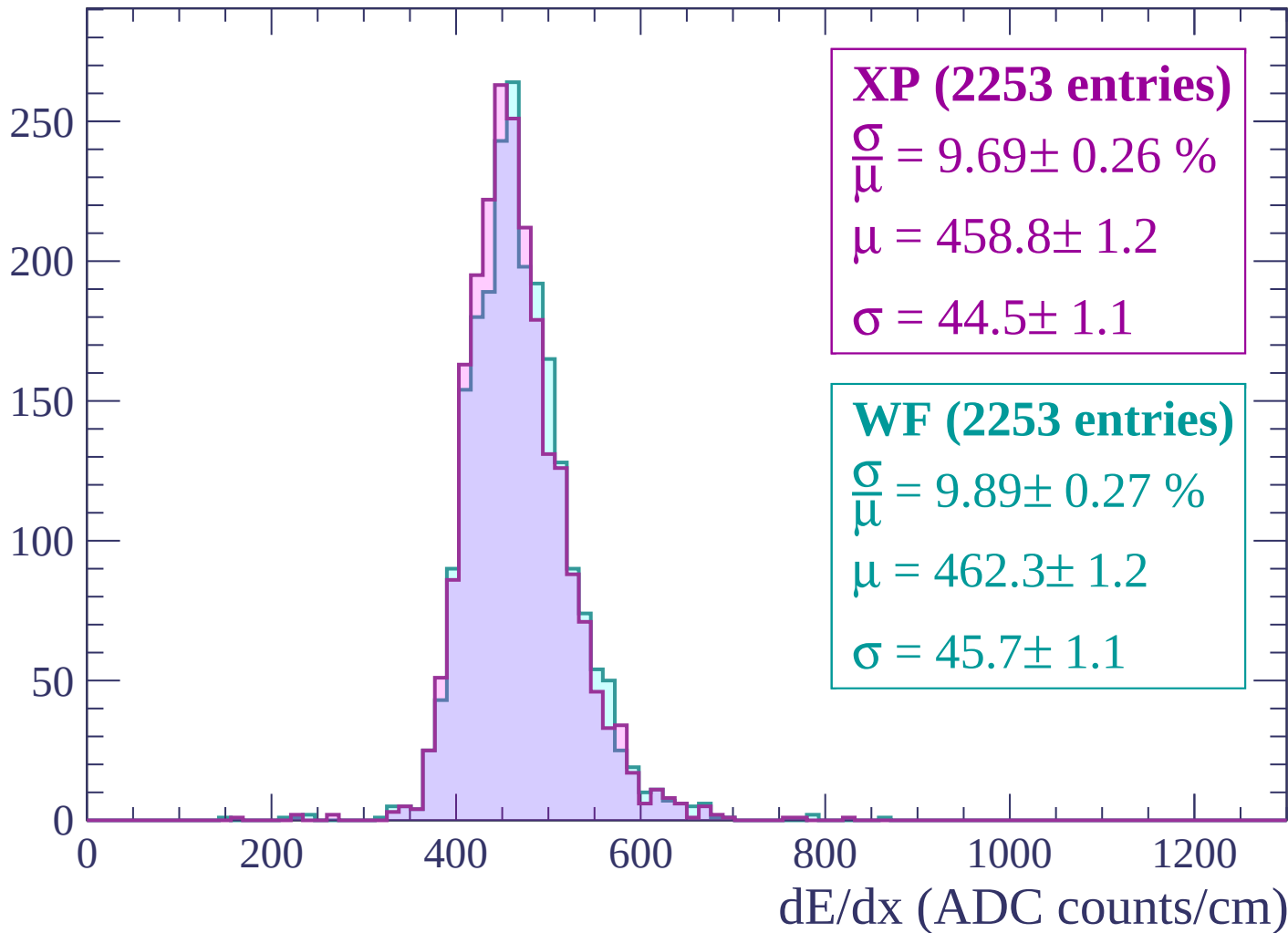
Energy loss | $1620 < p < 1680$

Count



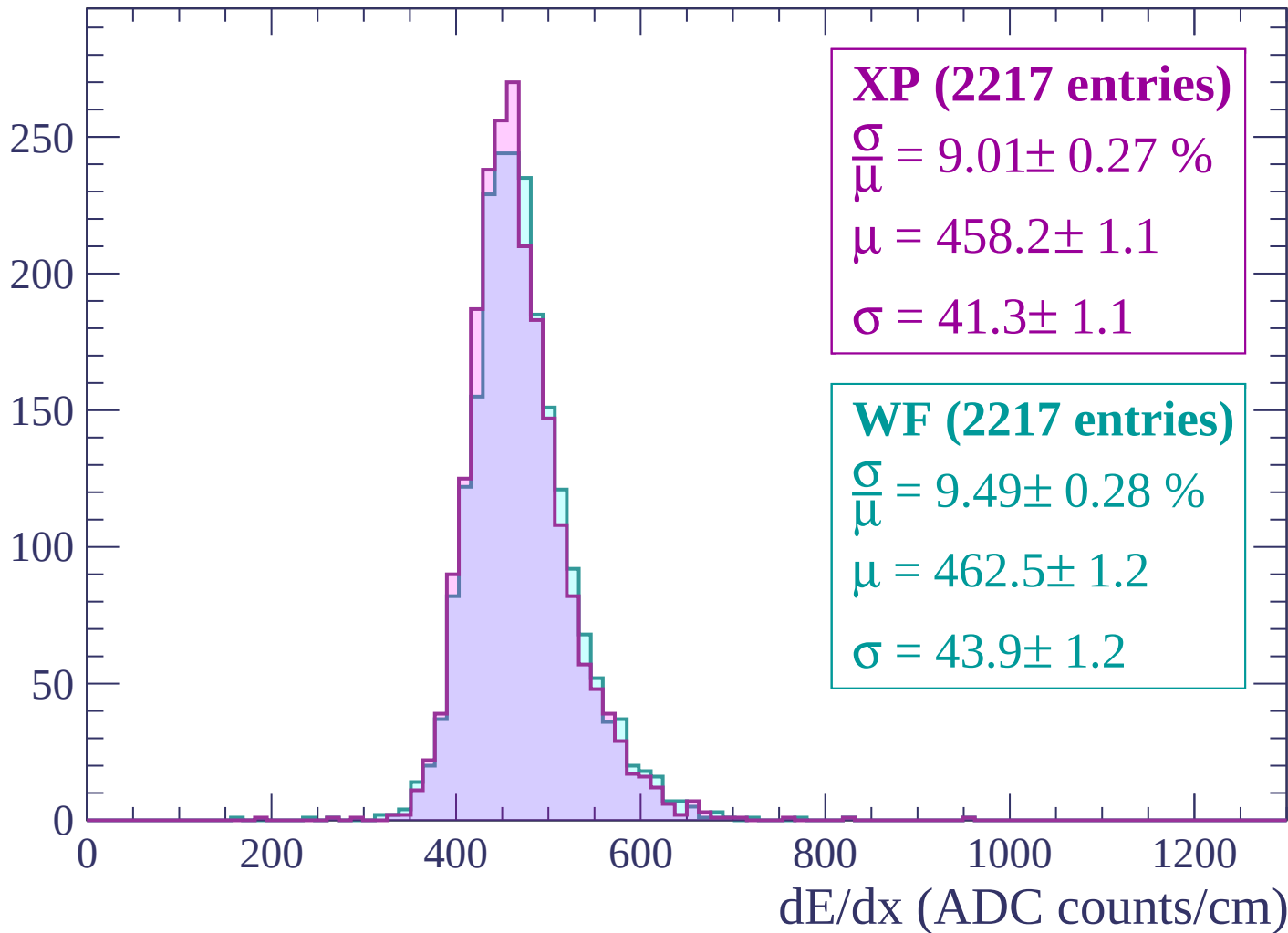
Energy loss | $1680 < p < 1740$

Count



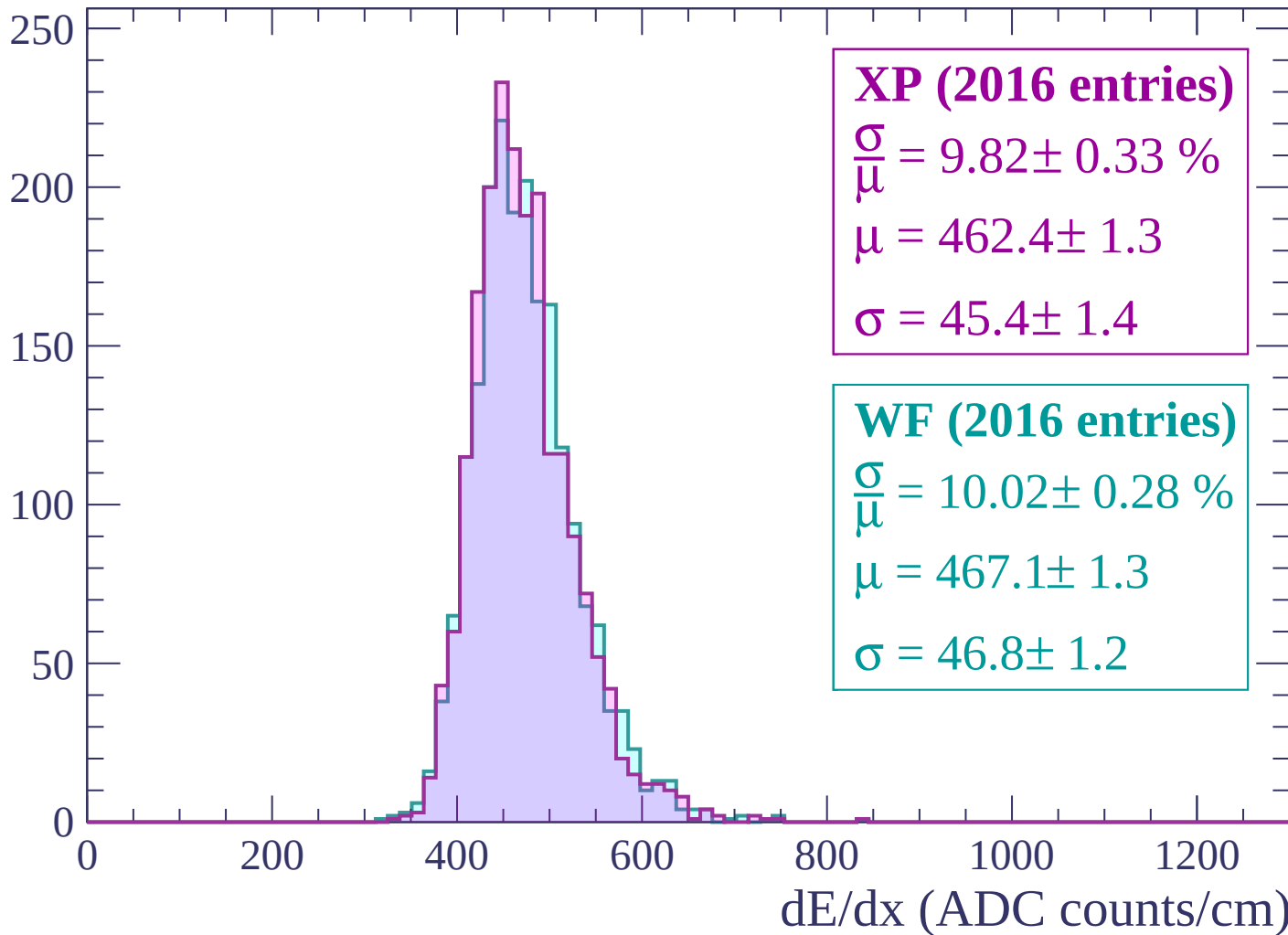
Energy loss | $1740 < p < 1800$

Count



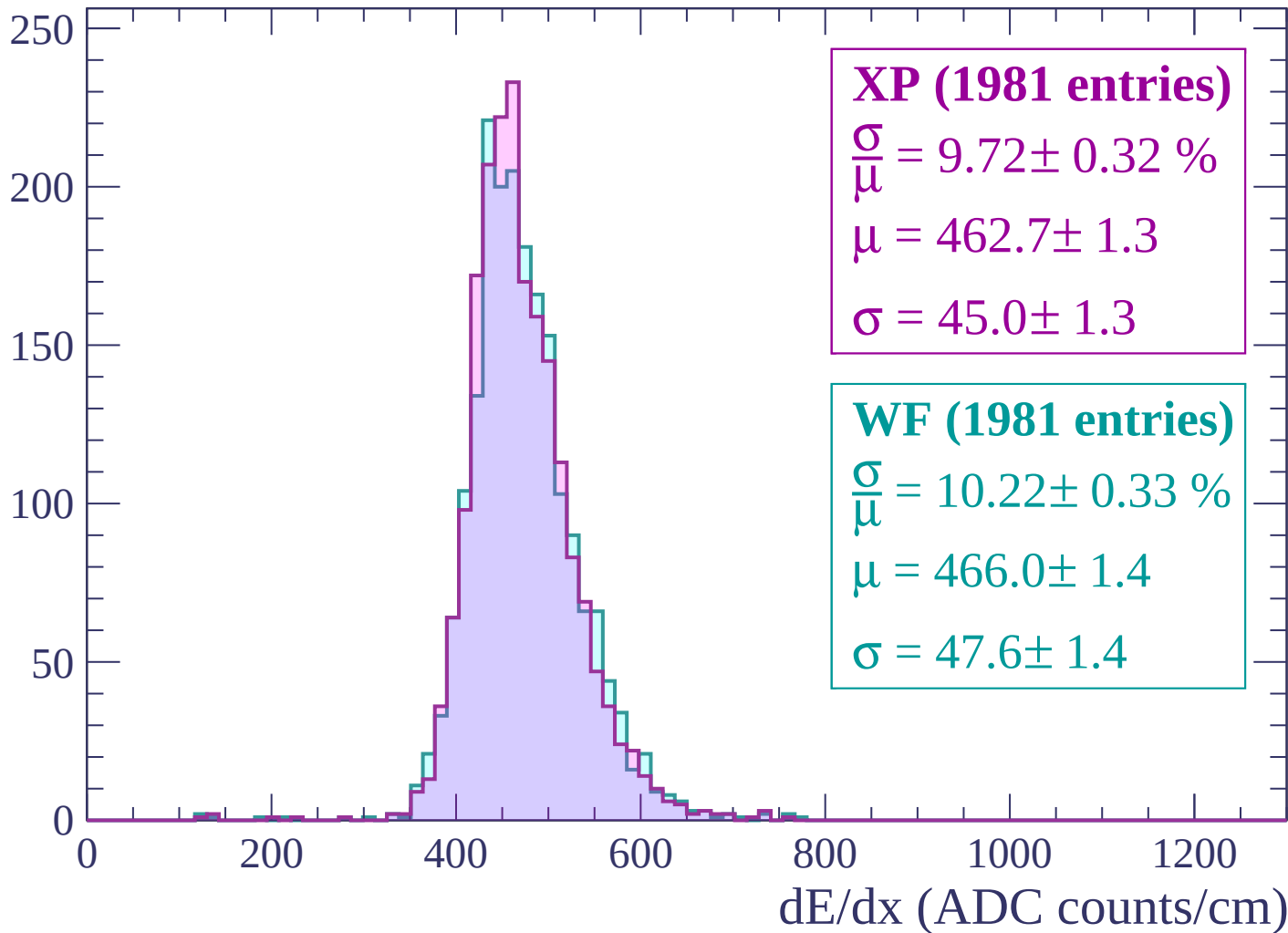
Energy loss | $1800 < p < 1860$

Count



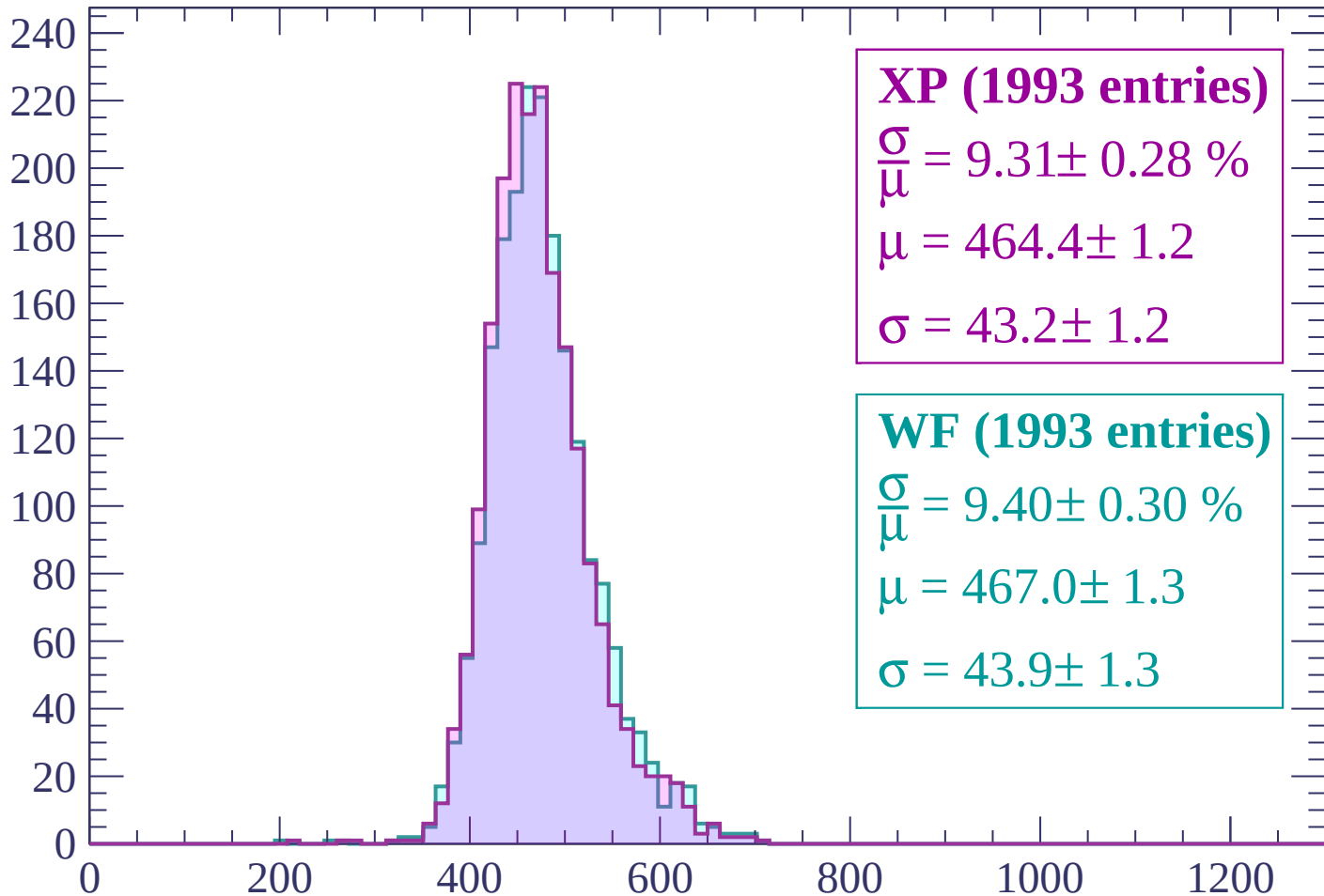
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (1993 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.28 \%$$

$$\mu = 464.4 \pm 1.2$$

$$\sigma = 43.2 \pm 1.2$$

WF (1993 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.30 \%$$

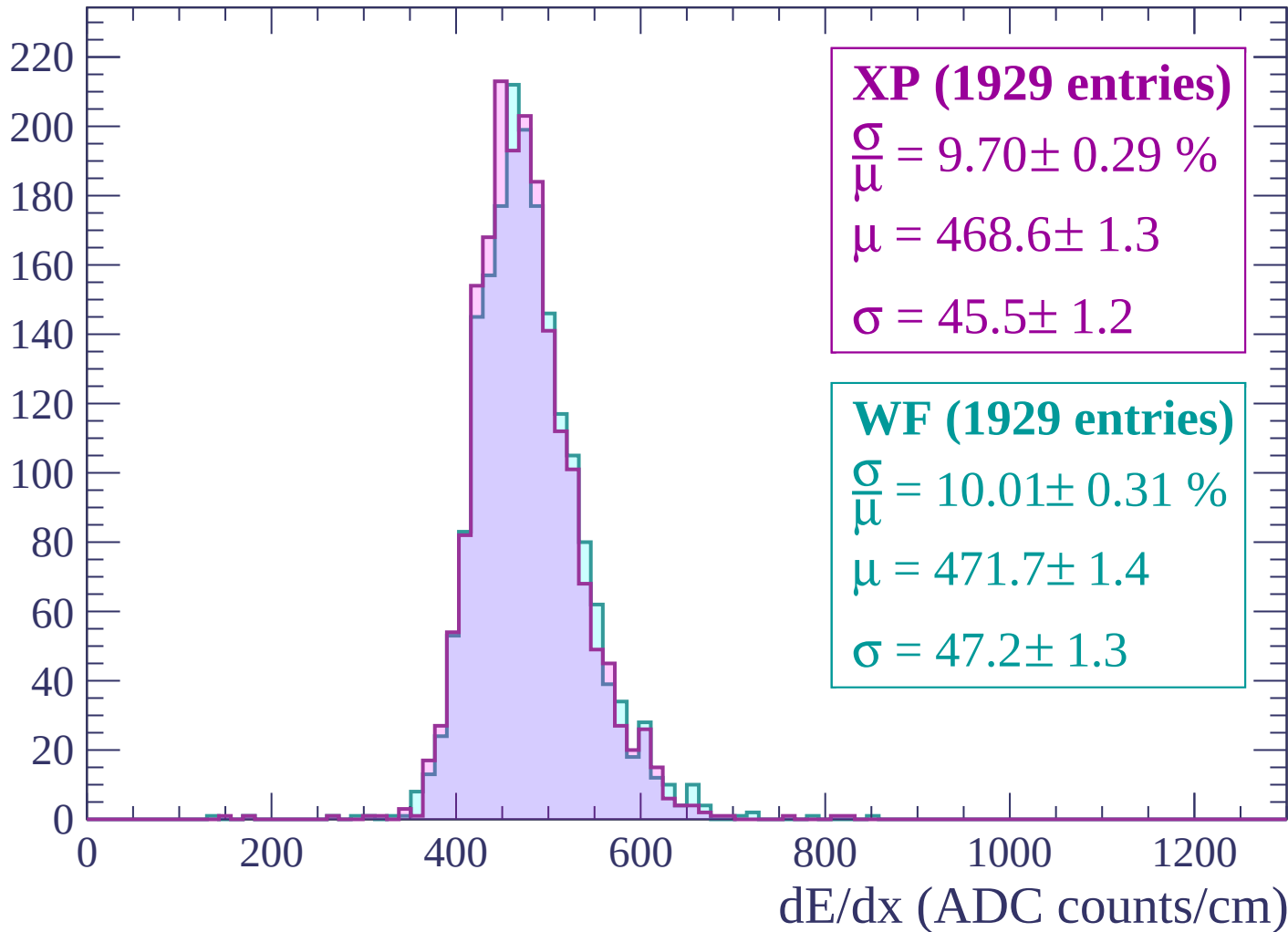
$$\mu = 467.0 \pm 1.3$$

$$\sigma = 43.9 \pm 1.3$$

dE/dx (ADC counts/cm)

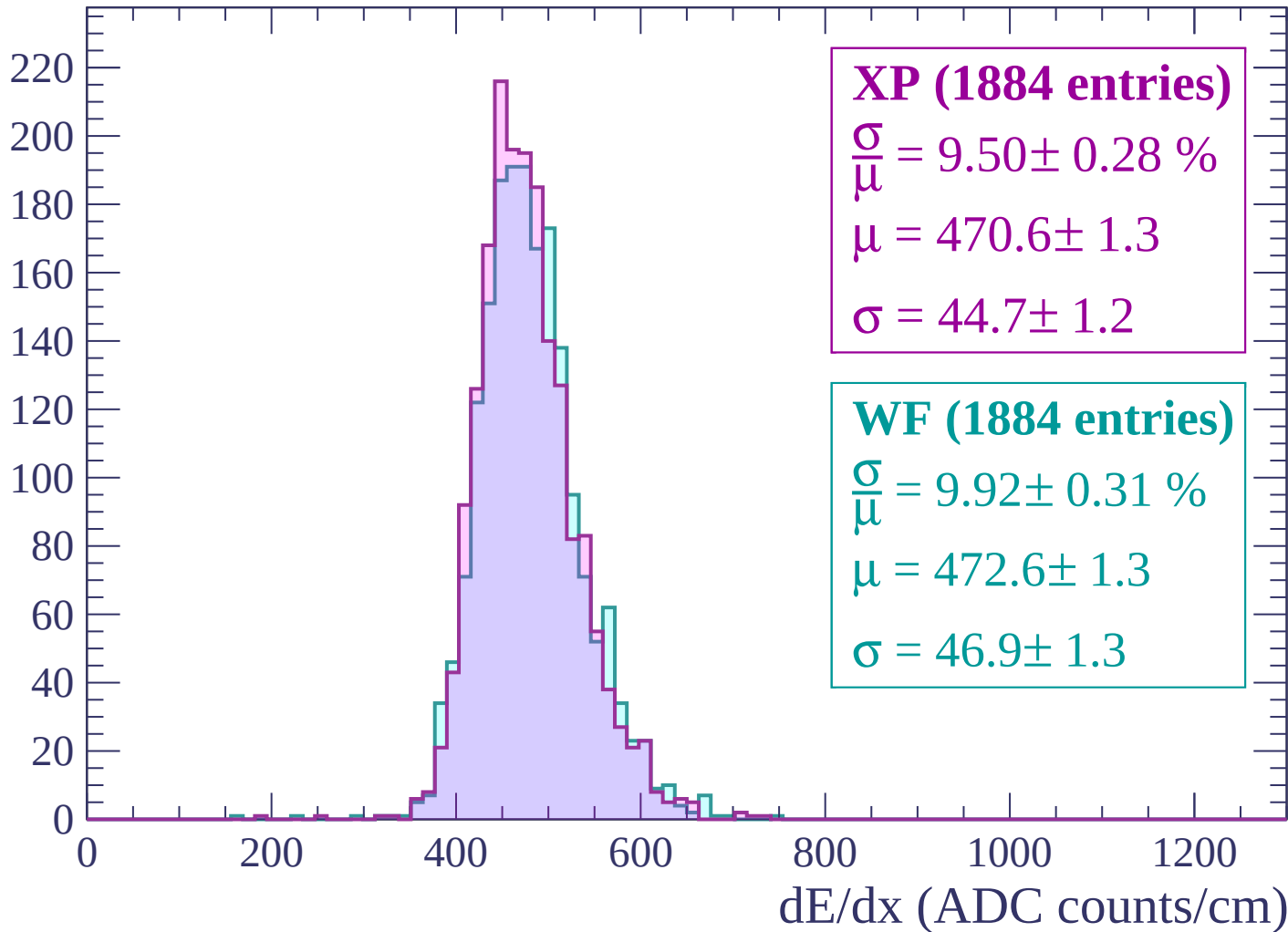
Energy loss | $1980 < p < 2040$

Count



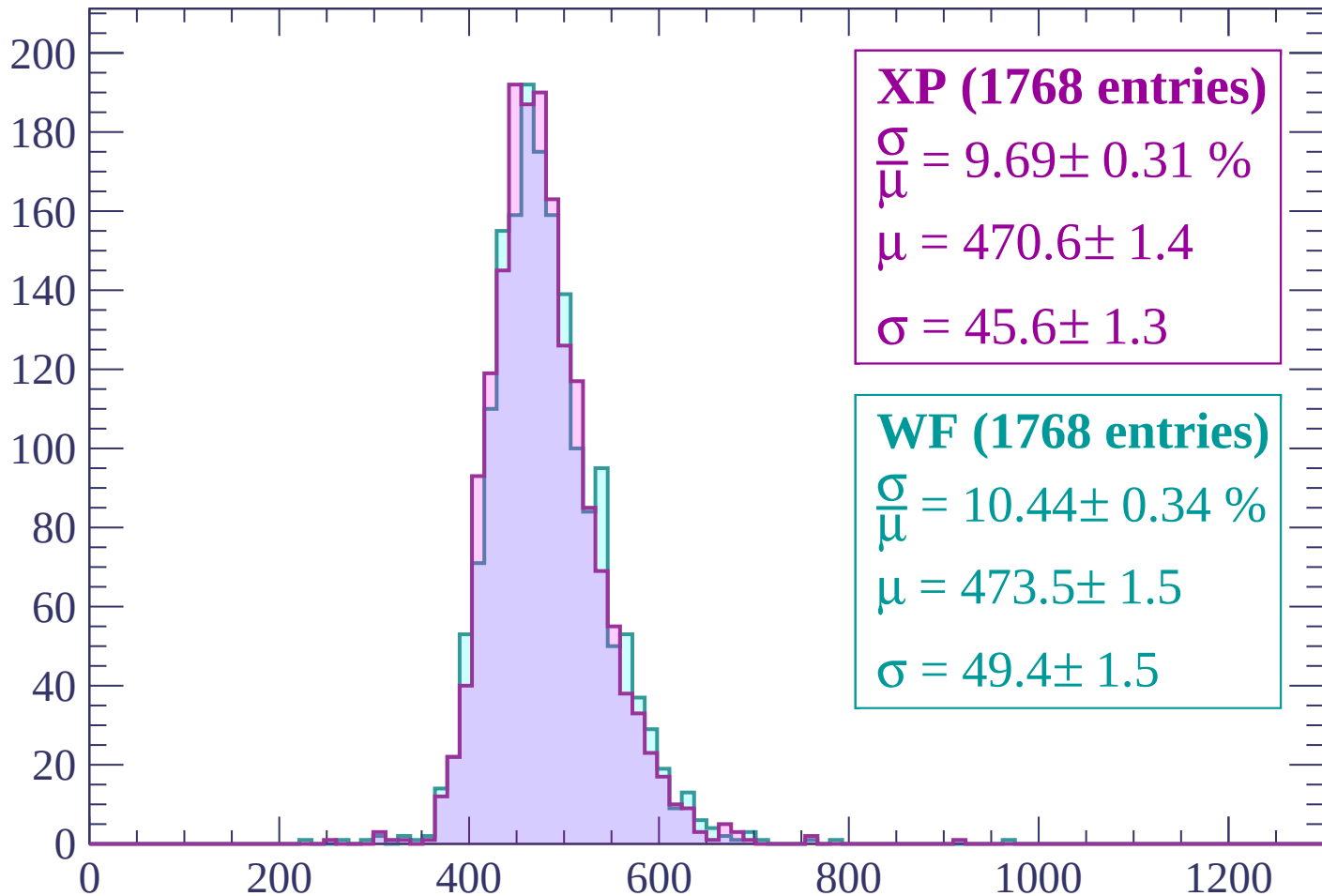
Energy loss | $2040 < p < 2100$

Count



Energy loss | $2100 < p < 2160$

Count



XP (1768 entries)

$$\frac{\sigma}{\mu} = 9.69 \pm 0.31 \%$$

$$\mu = 470.6 \pm 1.4$$

$$\sigma = 45.6 \pm 1.3$$

WF (1768 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 0.34 \%$$

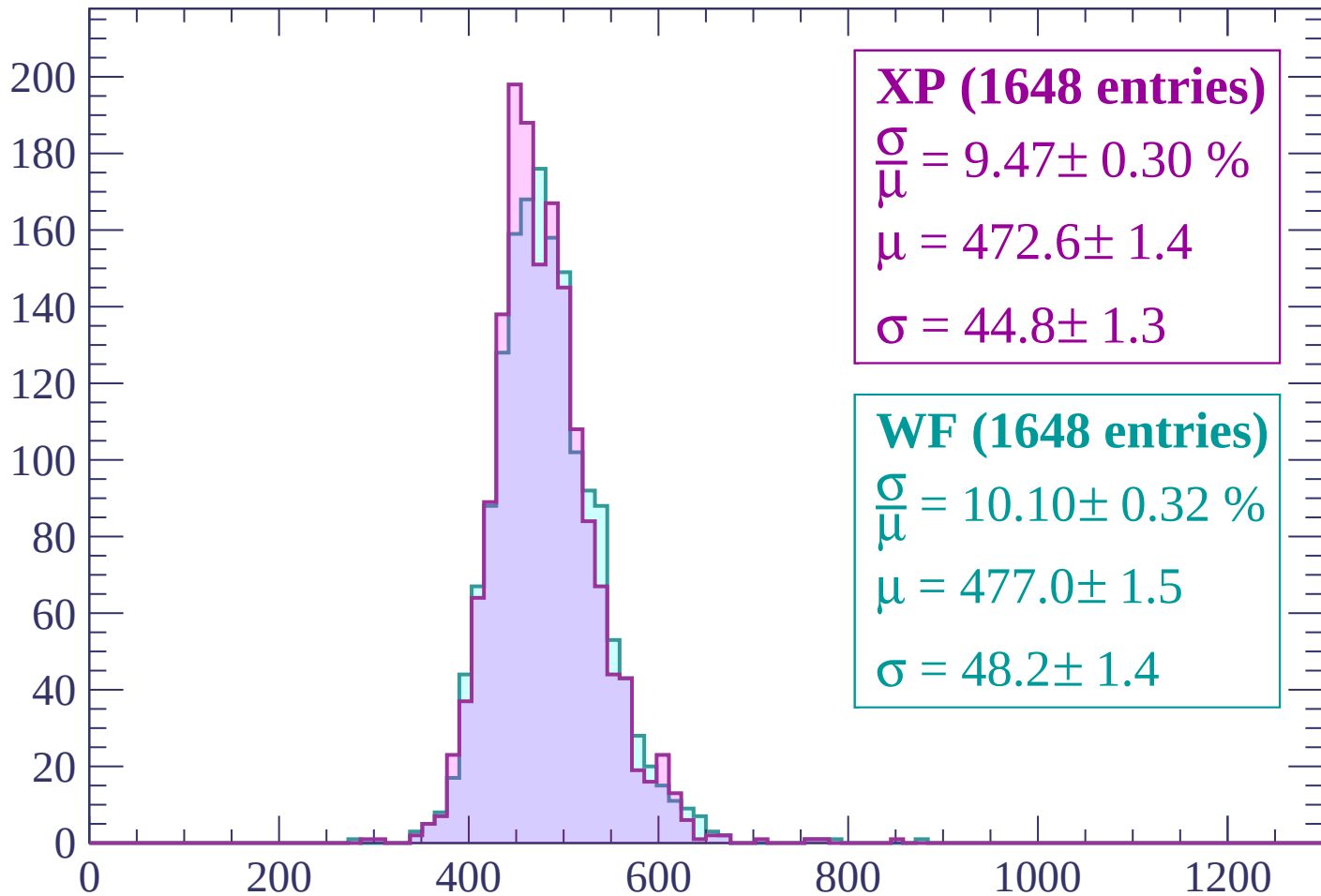
$$\mu = 473.5 \pm 1.5$$

$$\sigma = 49.4 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (1648 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.30 \%$$

$$\mu = 472.6 \pm 1.4$$

$$\sigma = 44.8 \pm 1.3$$

WF (1648 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.32 \%$$

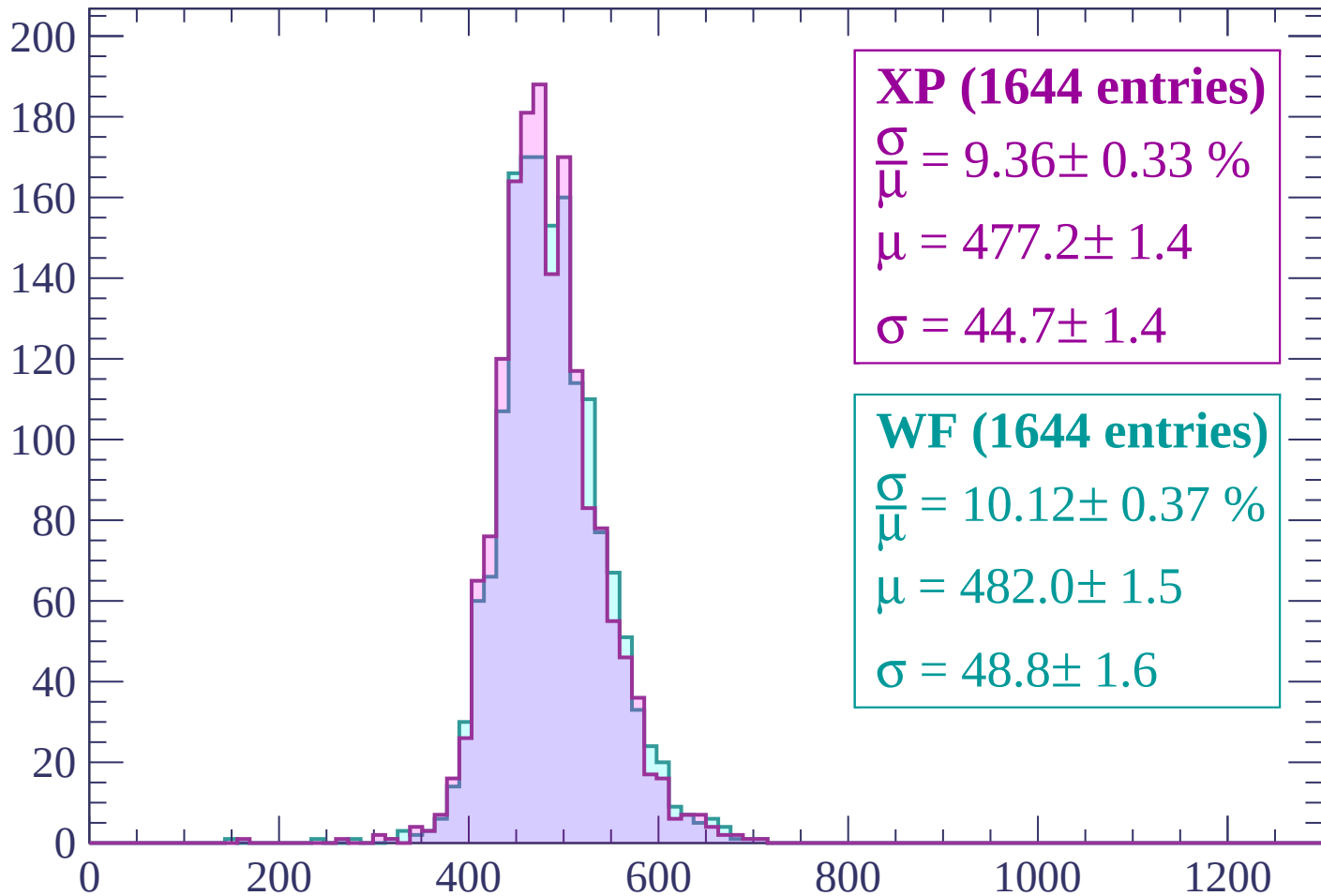
$$\mu = 477.0 \pm 1.5$$

$$\sigma = 48.2 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (1644 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.33 \%$$

$$\mu = 477.2 \pm 1.4$$

$$\sigma = 44.7 \pm 1.4$$

WF (1644 entries)

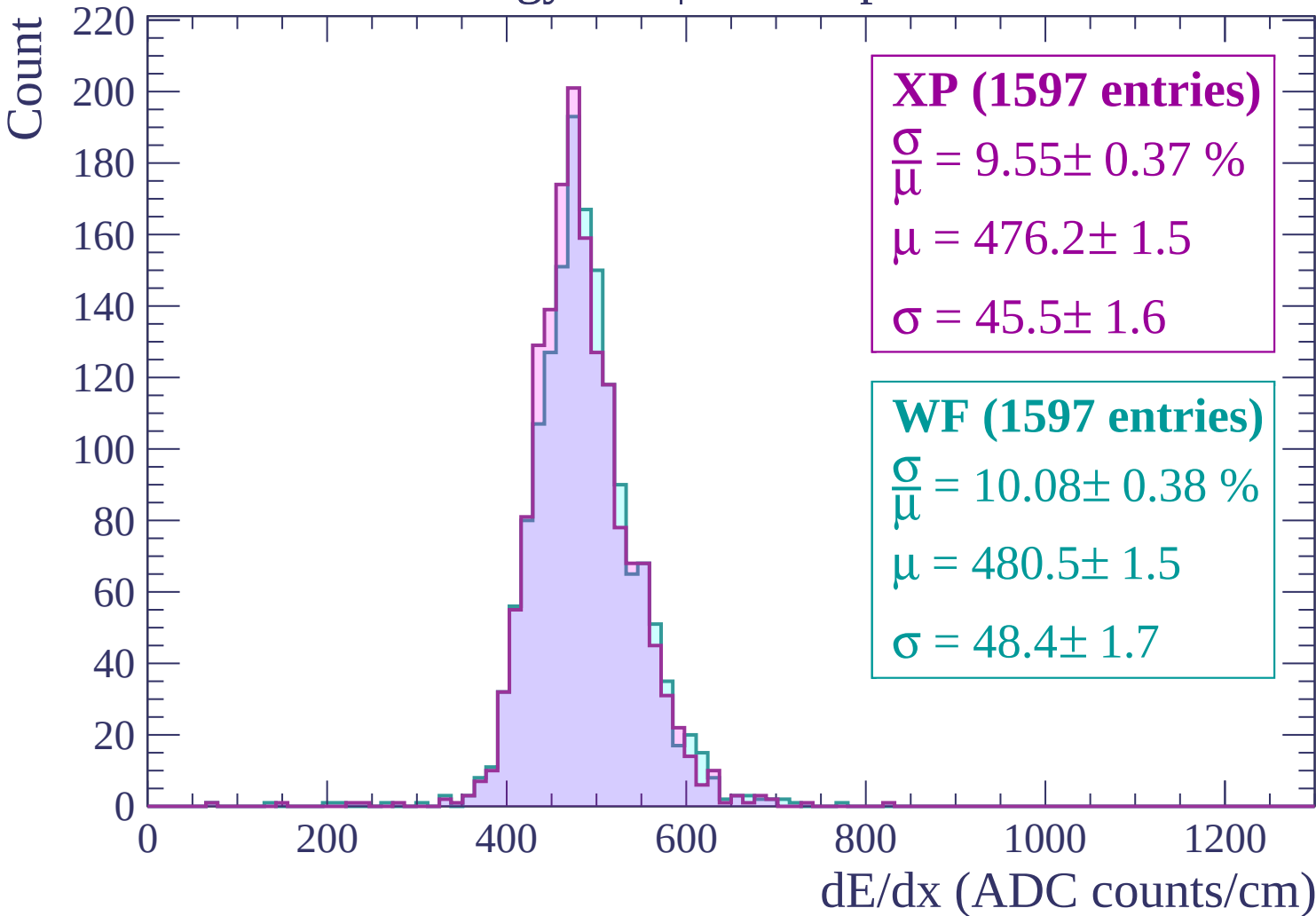
$$\frac{\sigma}{\mu} = 10.12 \pm 0.37 \%$$

$$\mu = 482.0 \pm 1.5$$

$$\sigma = 48.8 \pm 1.6$$

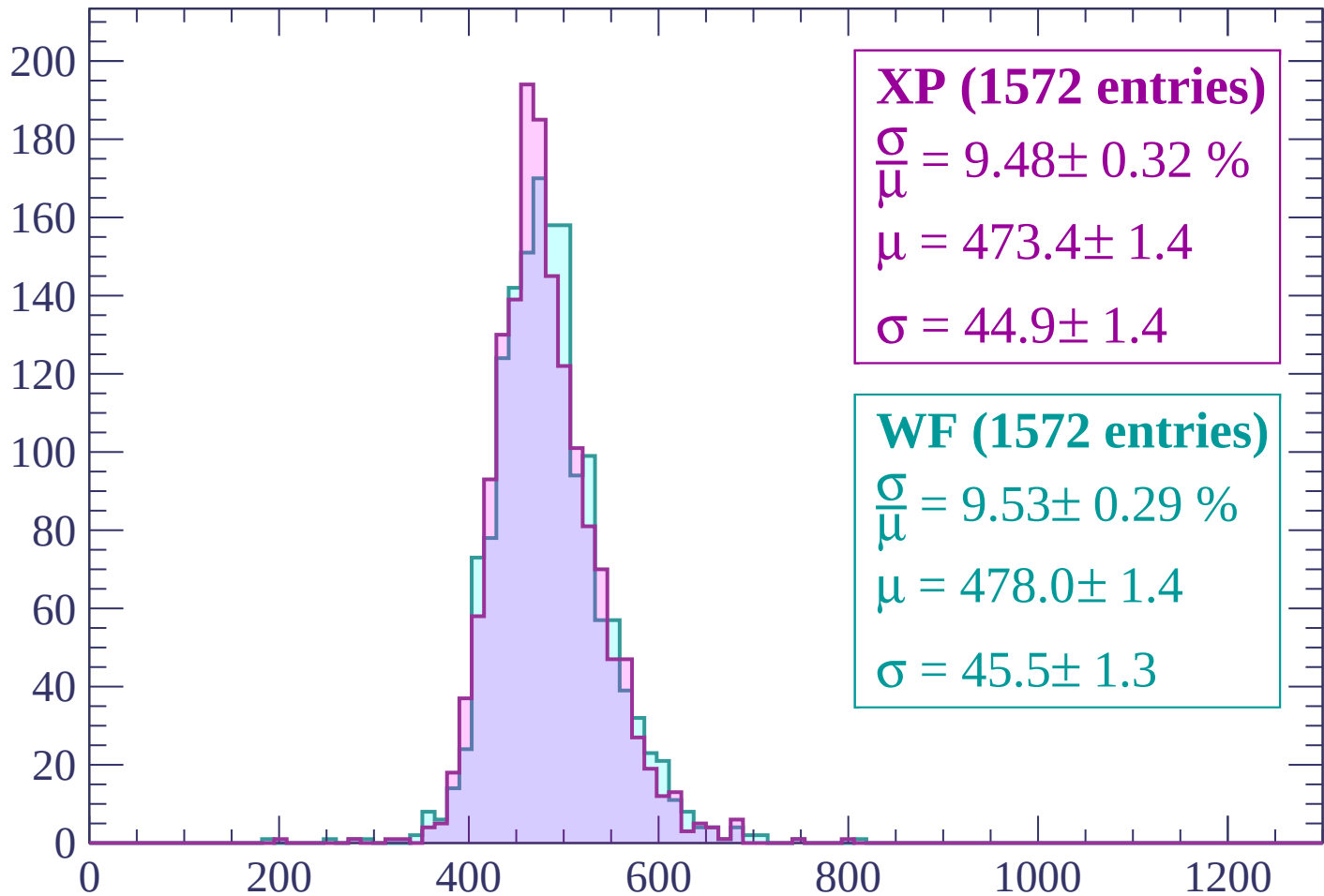
dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$



Energy loss | $2340 < p < 2400$

Count



XP (1572 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.32 \%$$

$$\mu = 473.4 \pm 1.4$$

$$\sigma = 44.9 \pm 1.4$$

WF (1572 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.29 \%$$

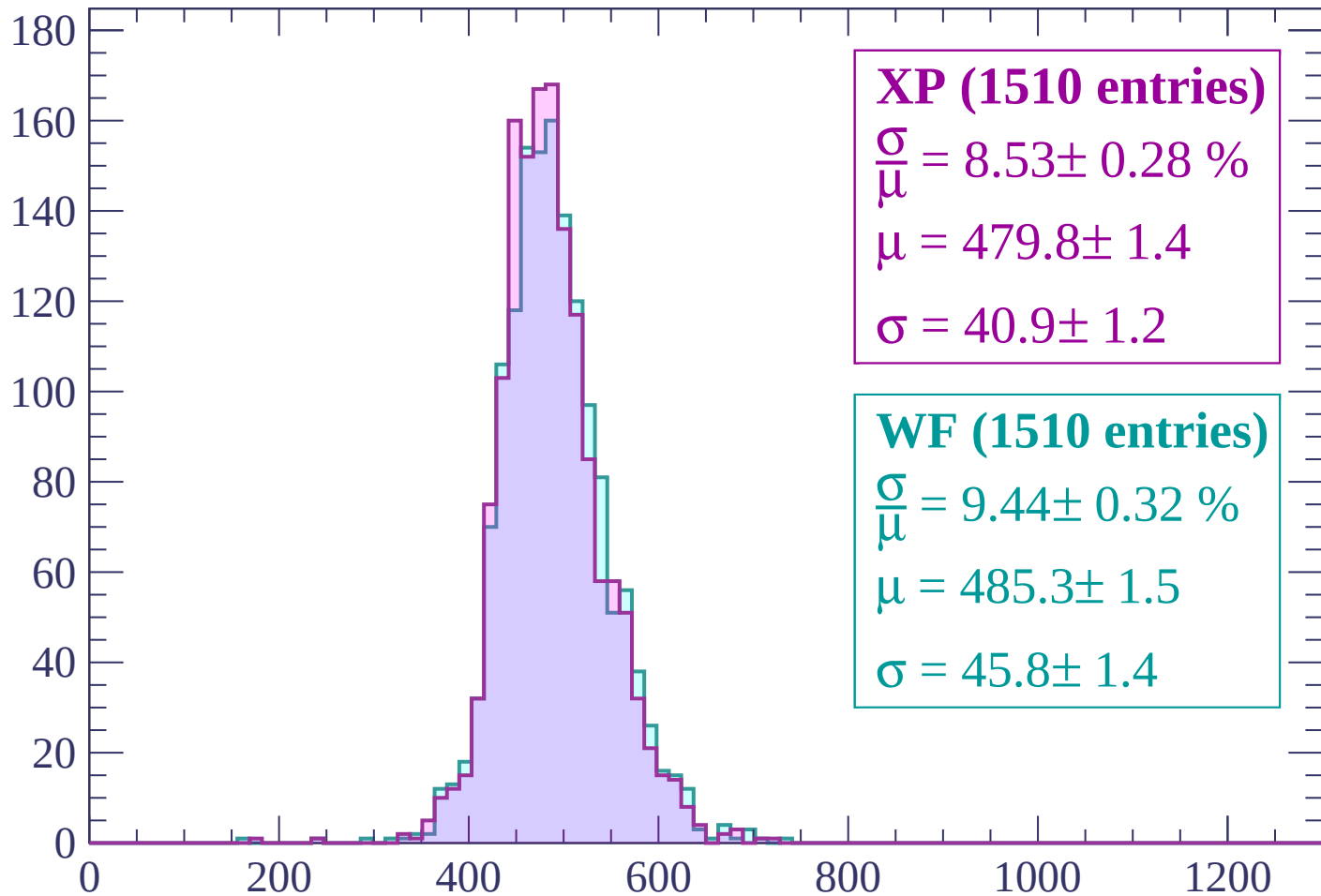
$$\mu = 478.0 \pm 1.4$$

$$\sigma = 45.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (1510 entries)

$$\frac{\sigma}{\mu} = 8.53 \pm 0.28 \%$$

$$\mu = 479.8 \pm 1.4$$

$$\sigma = 40.9 \pm 1.2$$

WF (1510 entries)

$$\frac{\sigma}{\mu} = 9.44 \pm 0.32 \%$$

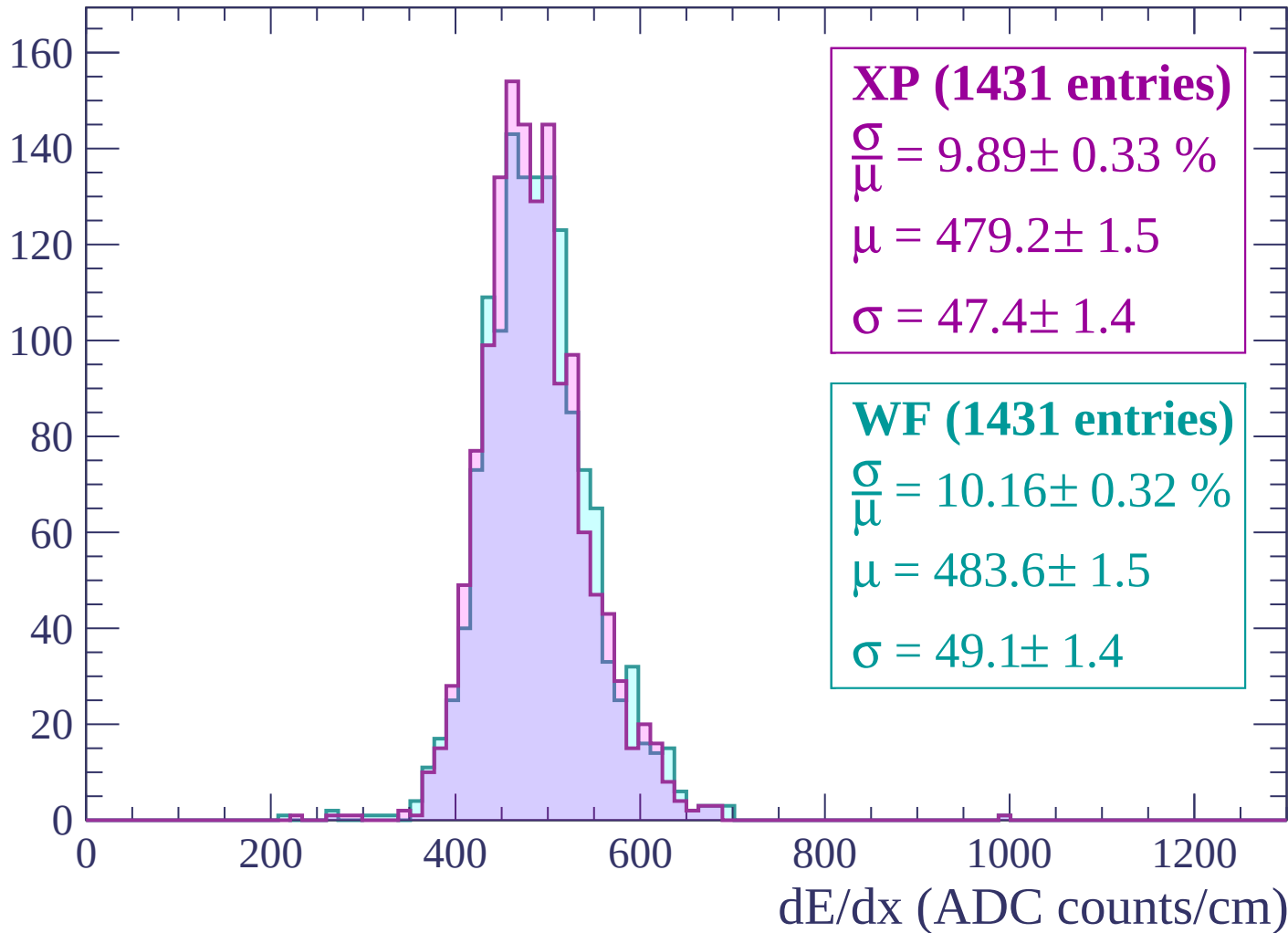
$$\mu = 485.3 \pm 1.5$$

$$\sigma = 45.8 \pm 1.4$$

dE/dx (ADC counts/cm)

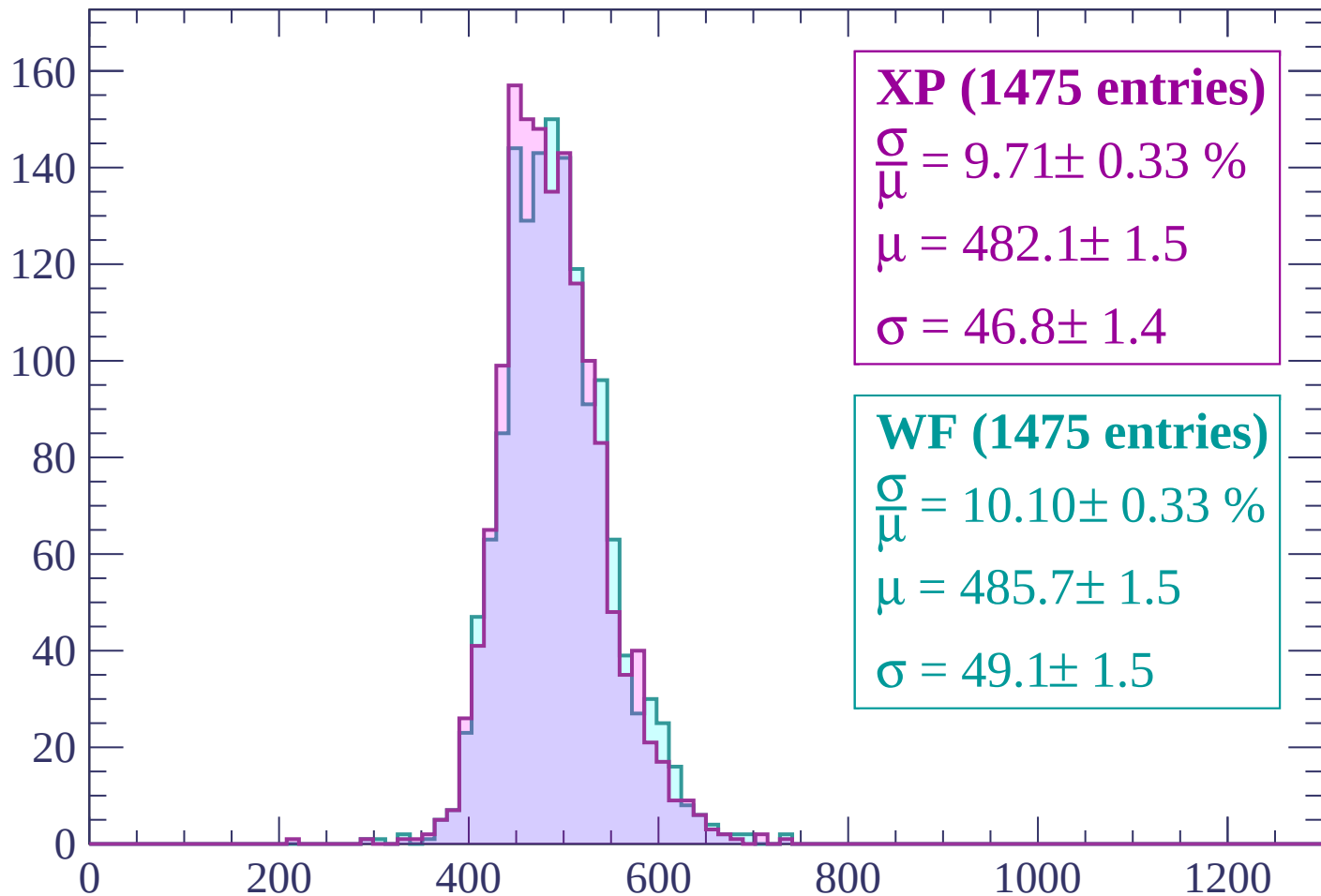
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (1475 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.33 \%$$

$$\mu = 482.1 \pm 1.5$$

$$\sigma = 46.8 \pm 1.4$$

WF (1475 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.33 \%$$

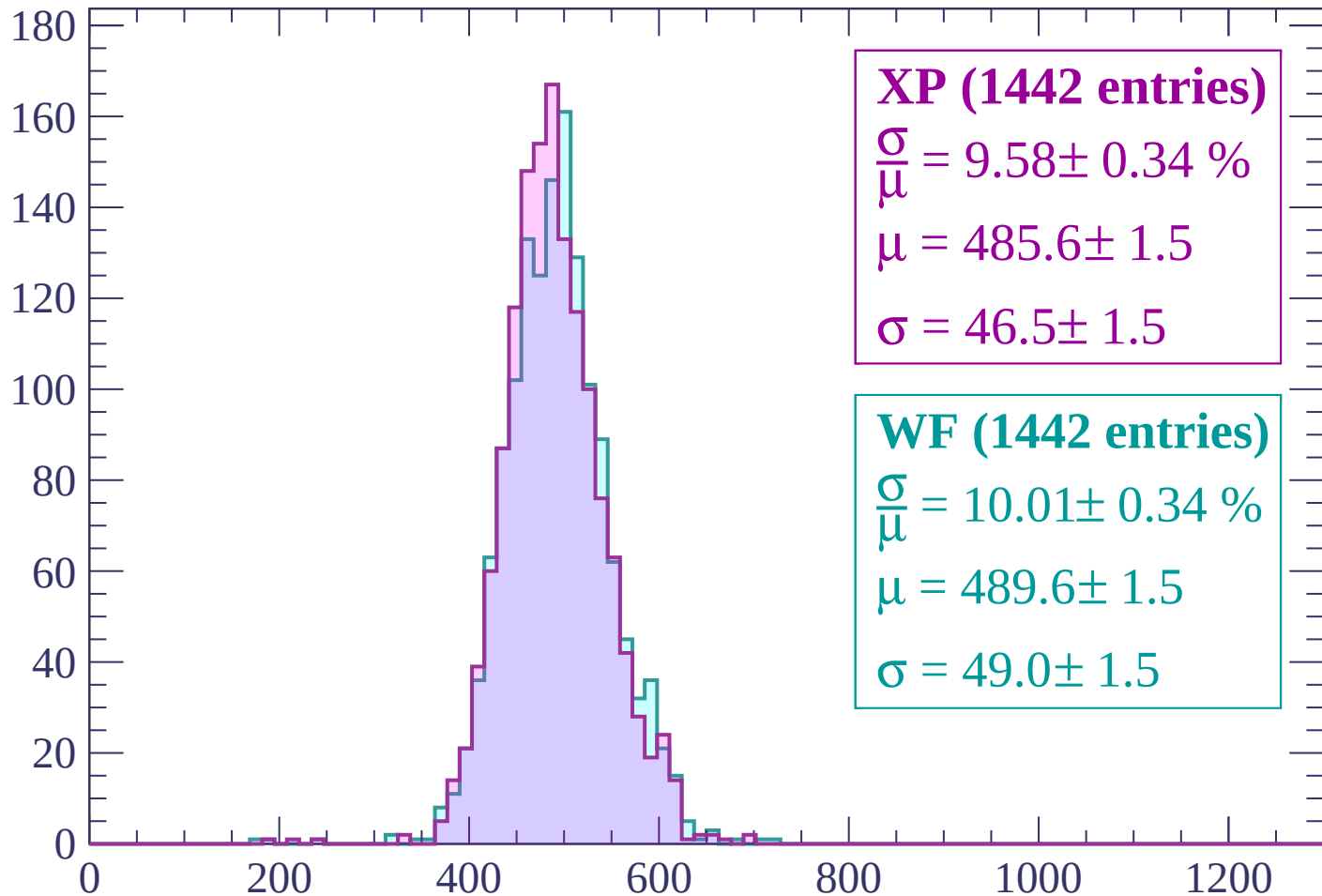
$$\mu = 485.7 \pm 1.5$$

$$\sigma = 49.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (1442 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.34 \%$$

$$\mu = 485.6 \pm 1.5$$

$$\sigma = 46.5 \pm 1.5$$

WF (1442 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.34 \%$$

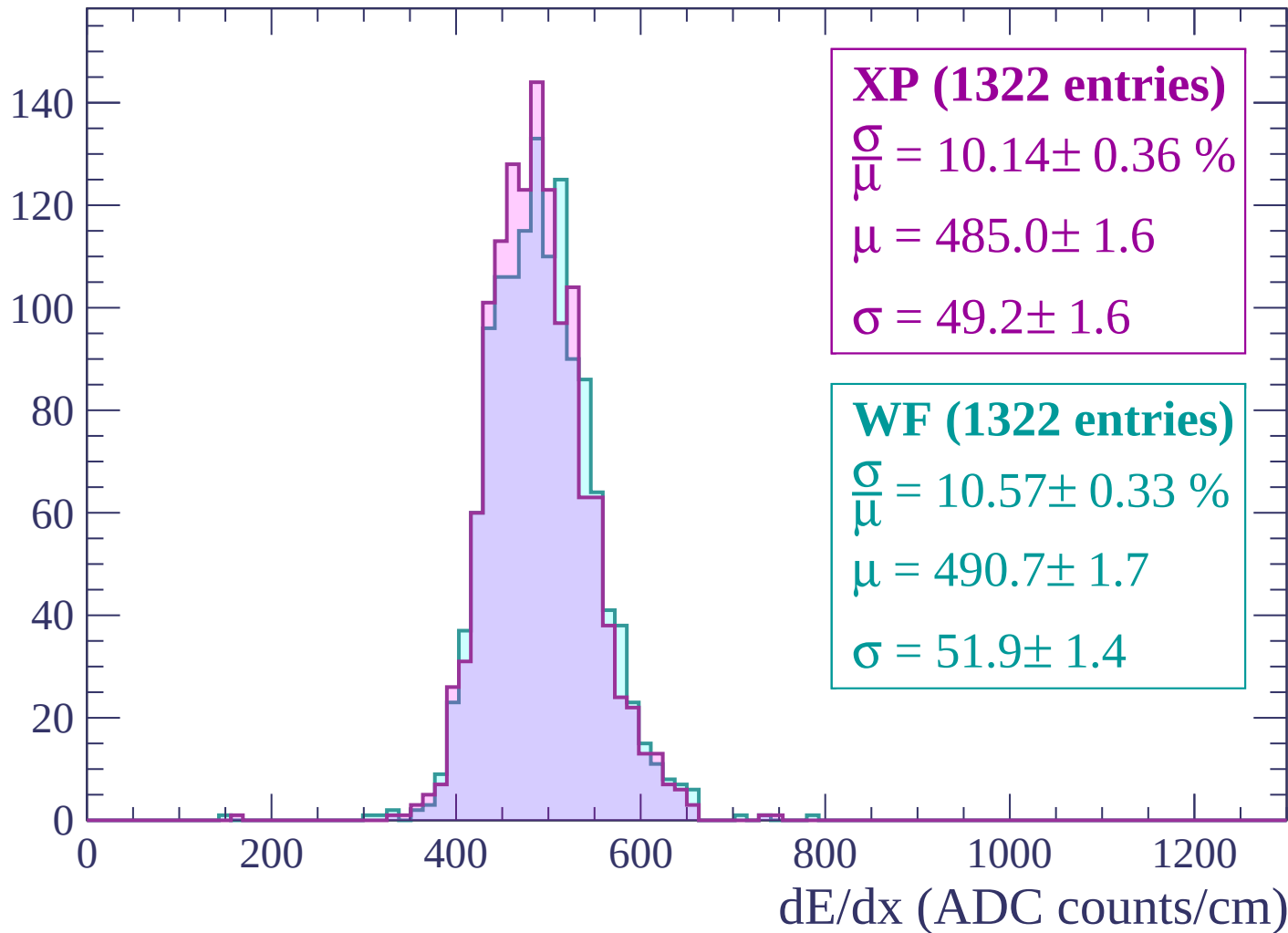
$$\mu = 489.6 \pm 1.5$$

$$\sigma = 49.0 \pm 1.5$$

dE/dx (ADC counts/cm)

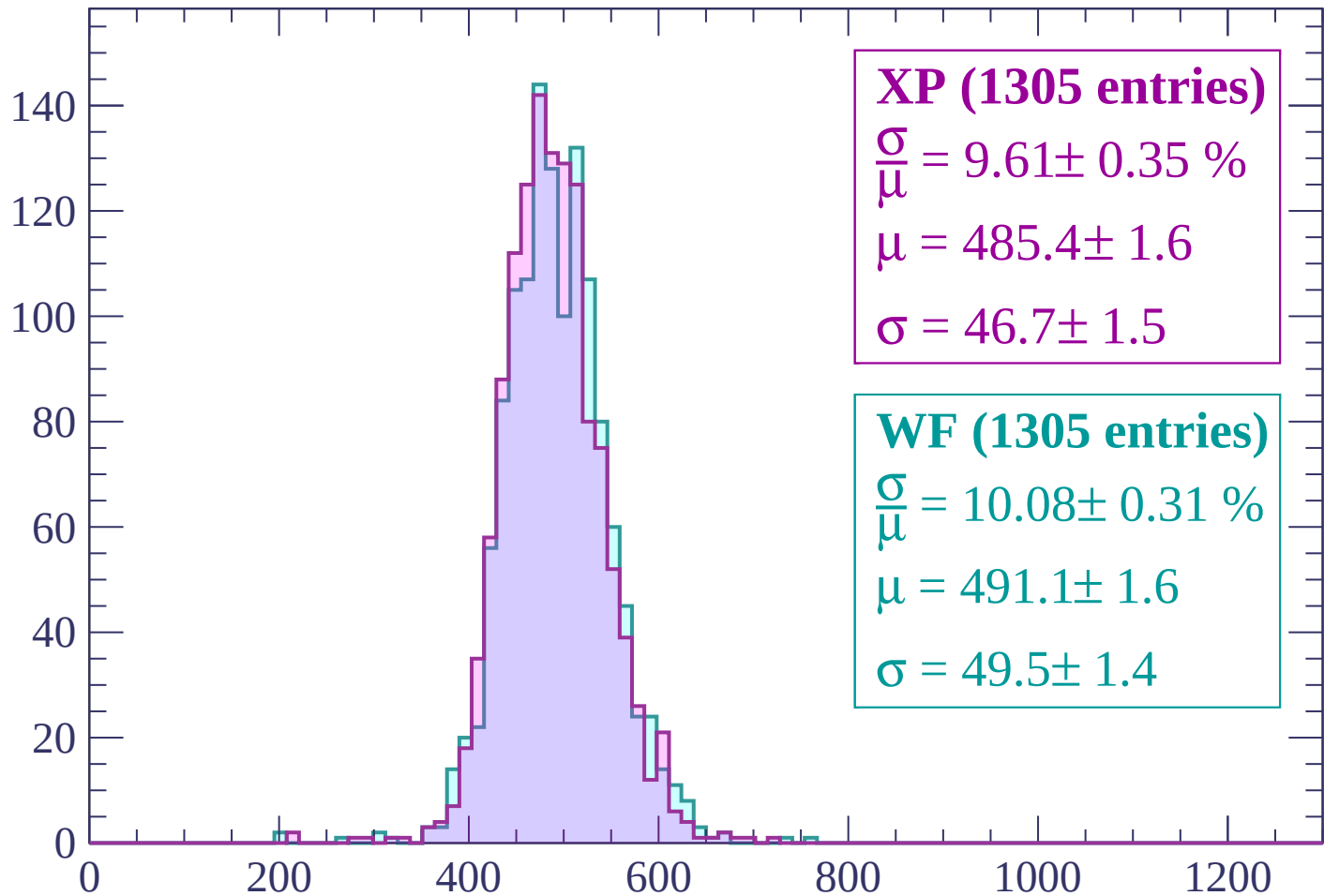
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (1305 entries)

$$\frac{\sigma}{\mu} = 9.61 \pm 0.35 \%$$

$$\mu = 485.4 \pm 1.6$$

$$\sigma = 46.7 \pm 1.5$$

WF (1305 entries)

$$\frac{\sigma}{\mu} = 10.08 \pm 0.31 \%$$

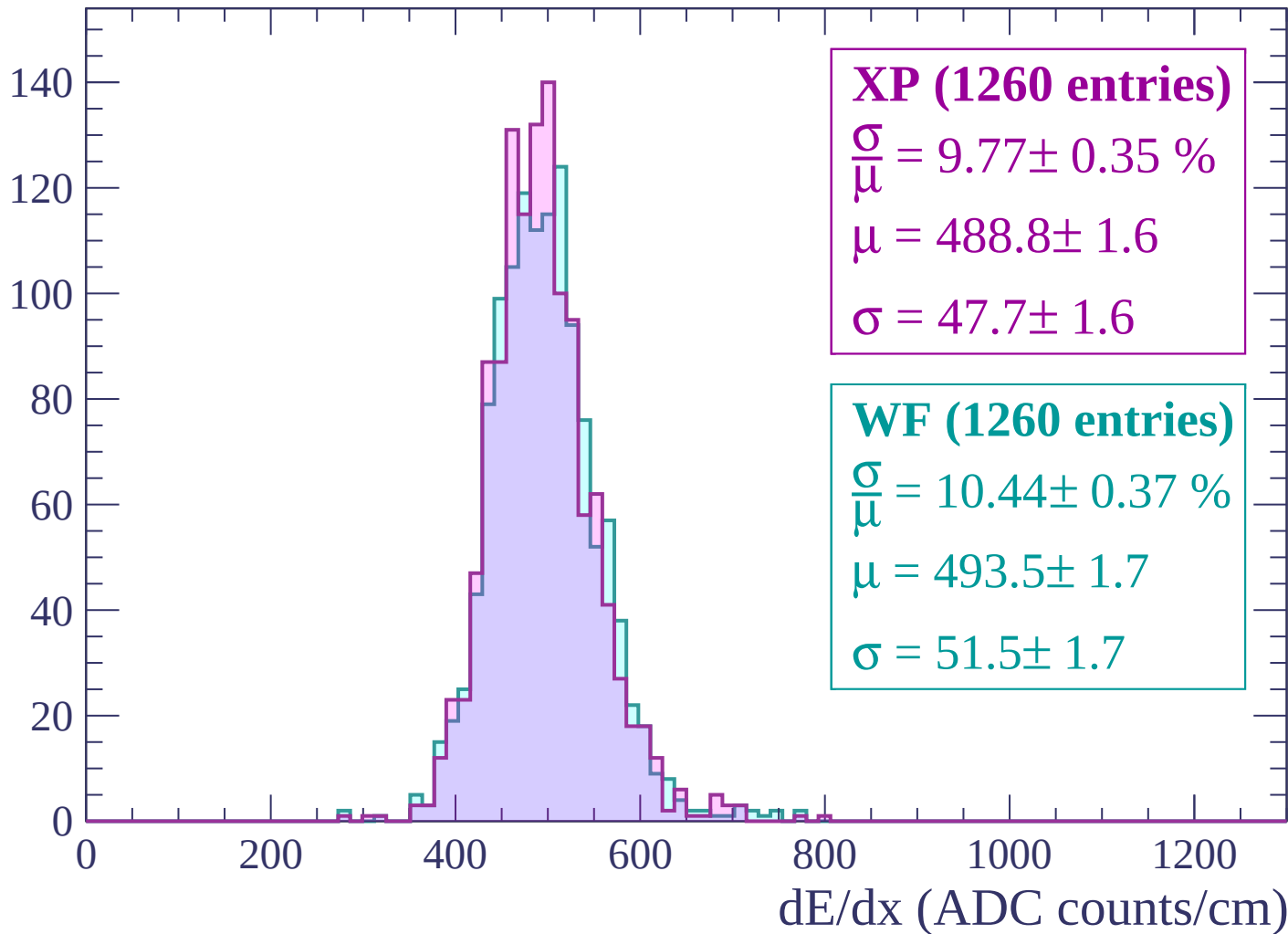
$$\mu = 491.1 \pm 1.6$$

$$\sigma = 49.5 \pm 1.4$$

dE/dx (ADC counts/cm)

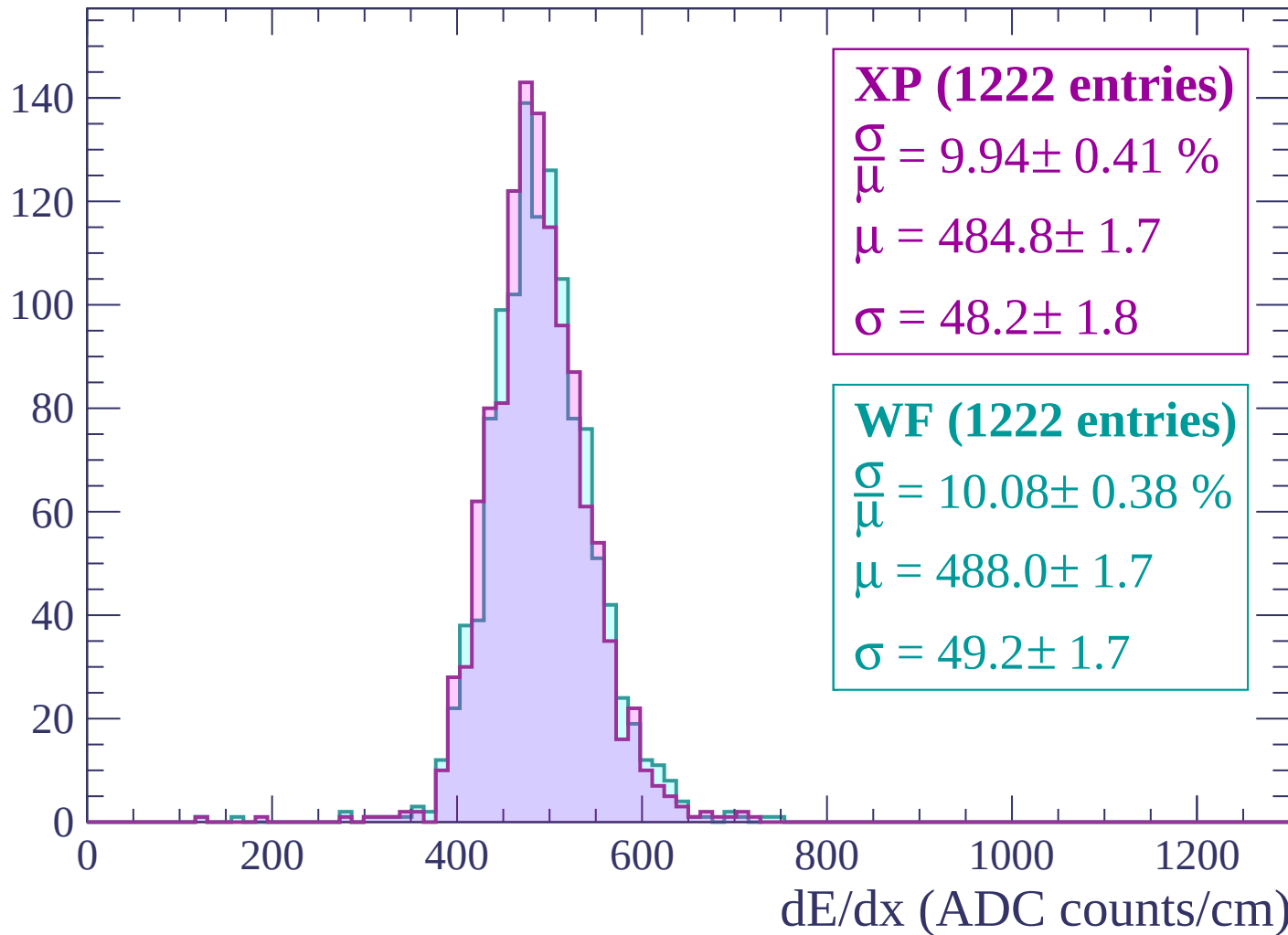
Energy loss | $2760 < p < 2820$

Count



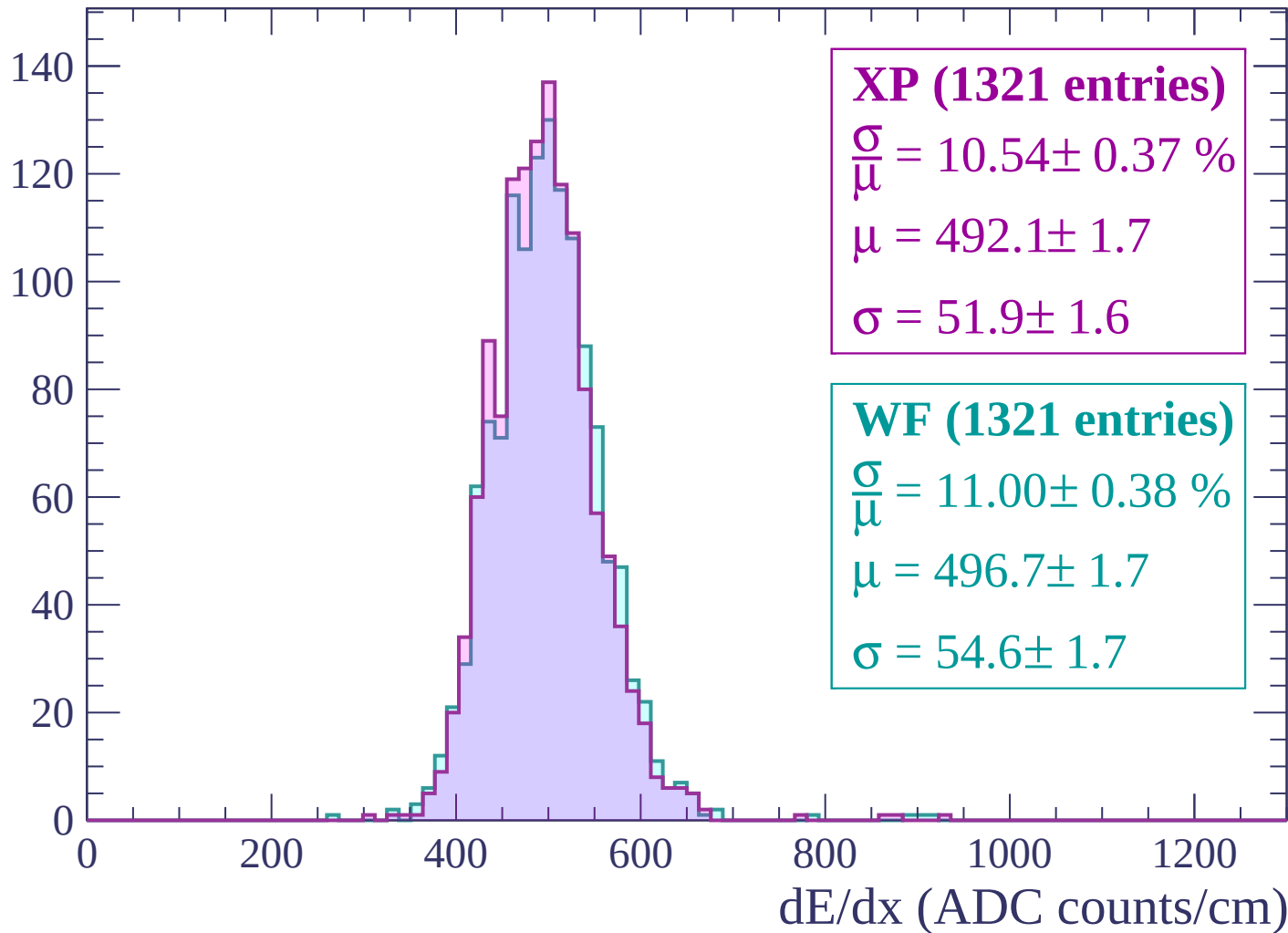
Energy loss | $2820 < p < 2880$

Count



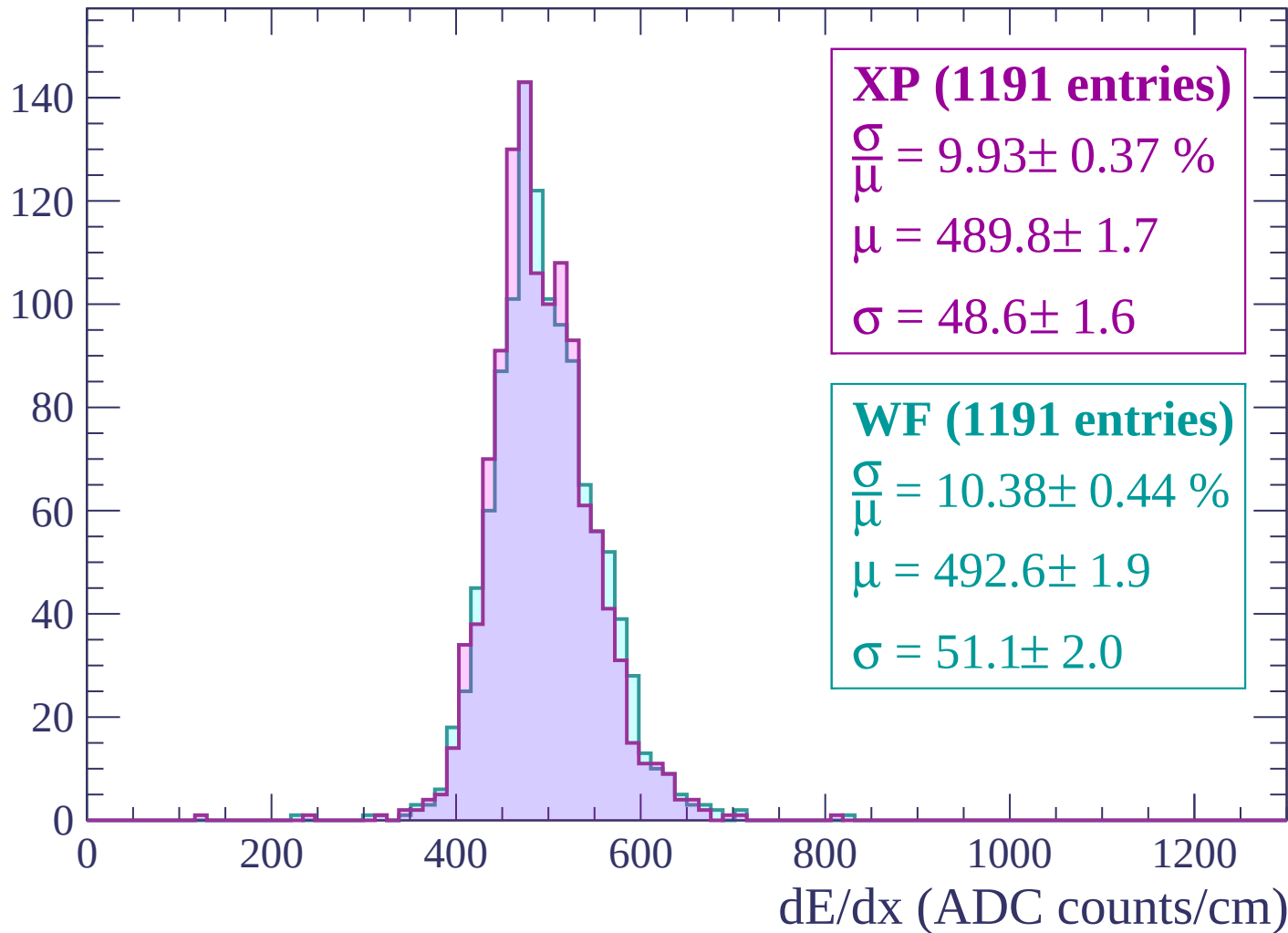
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

